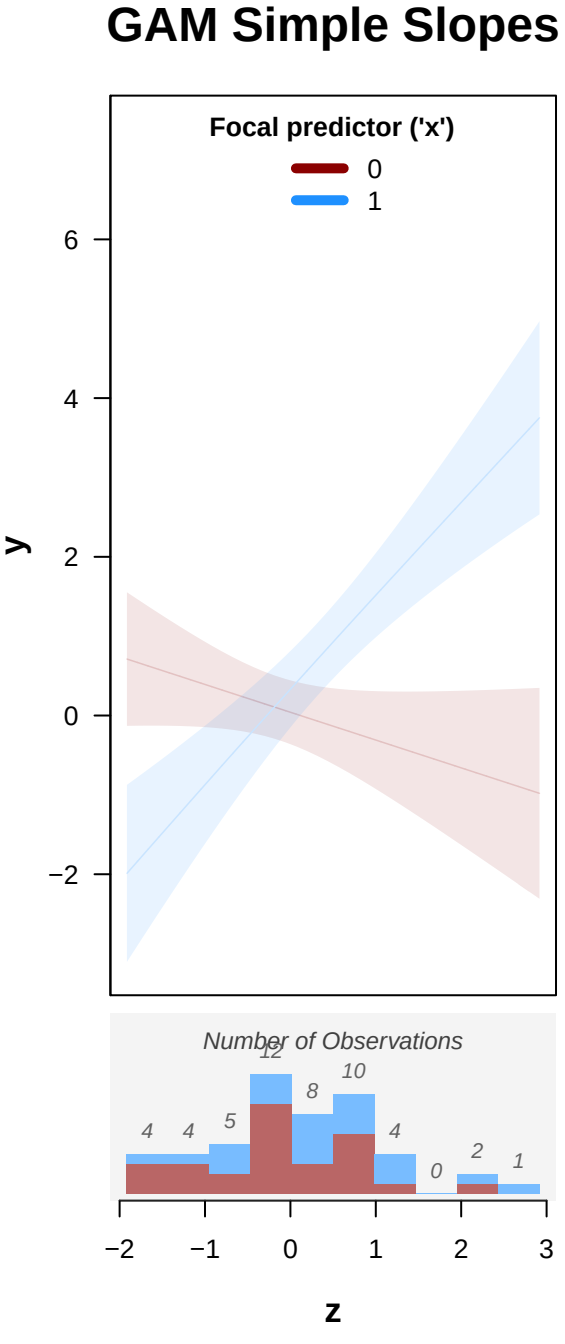
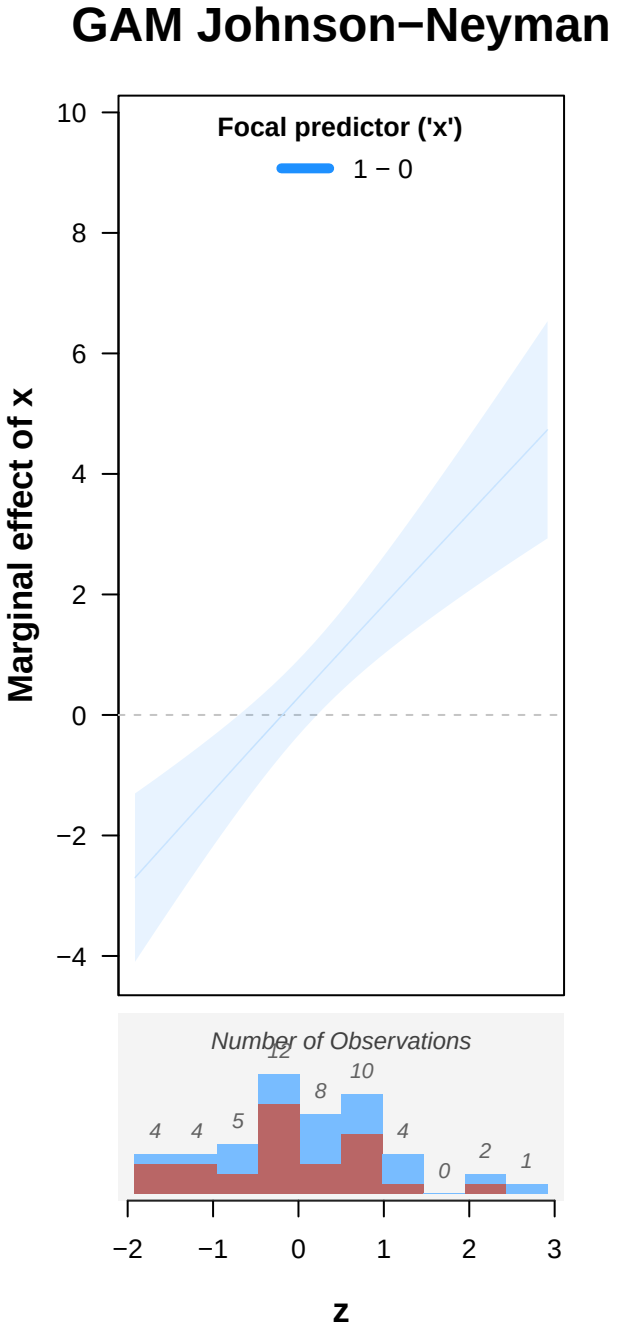
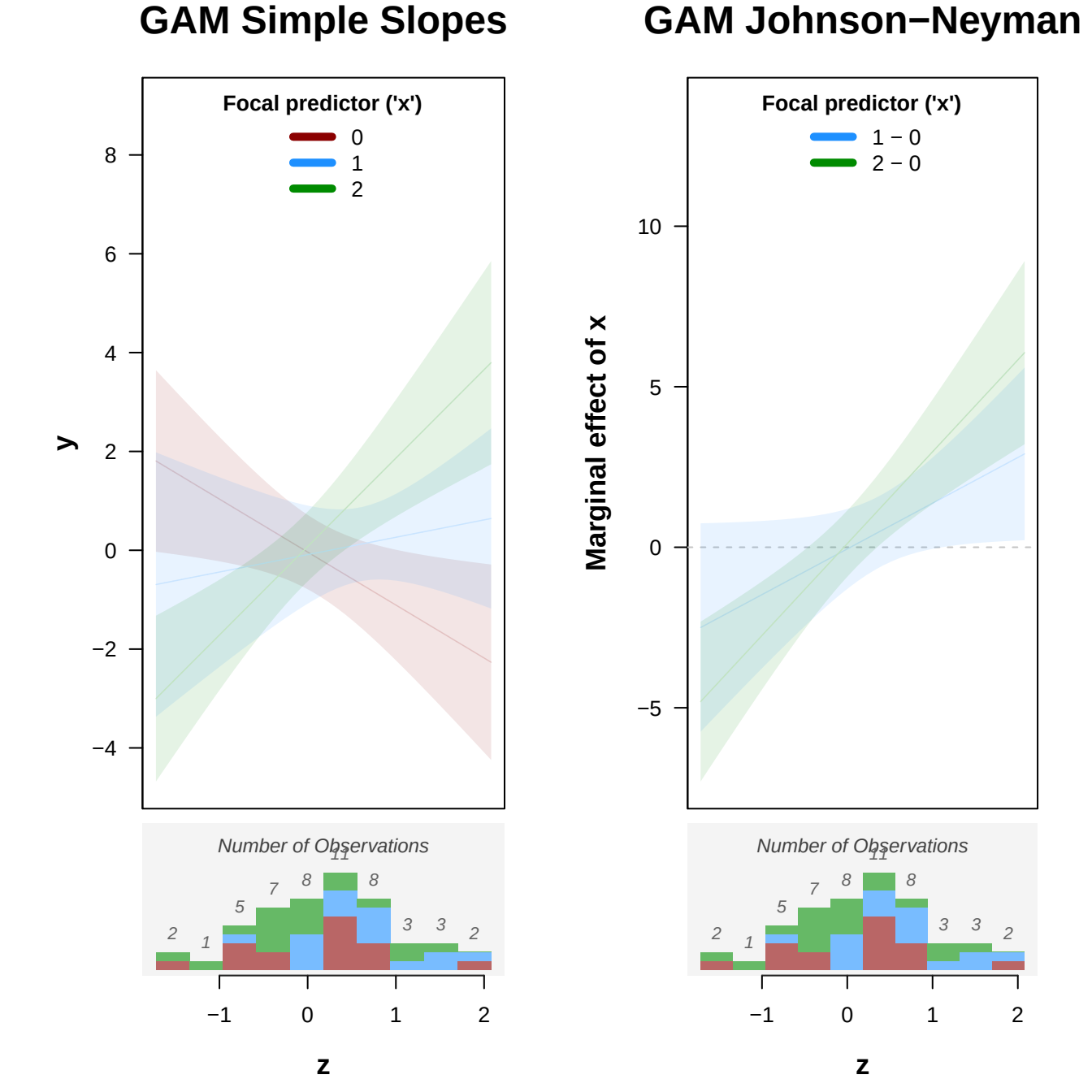
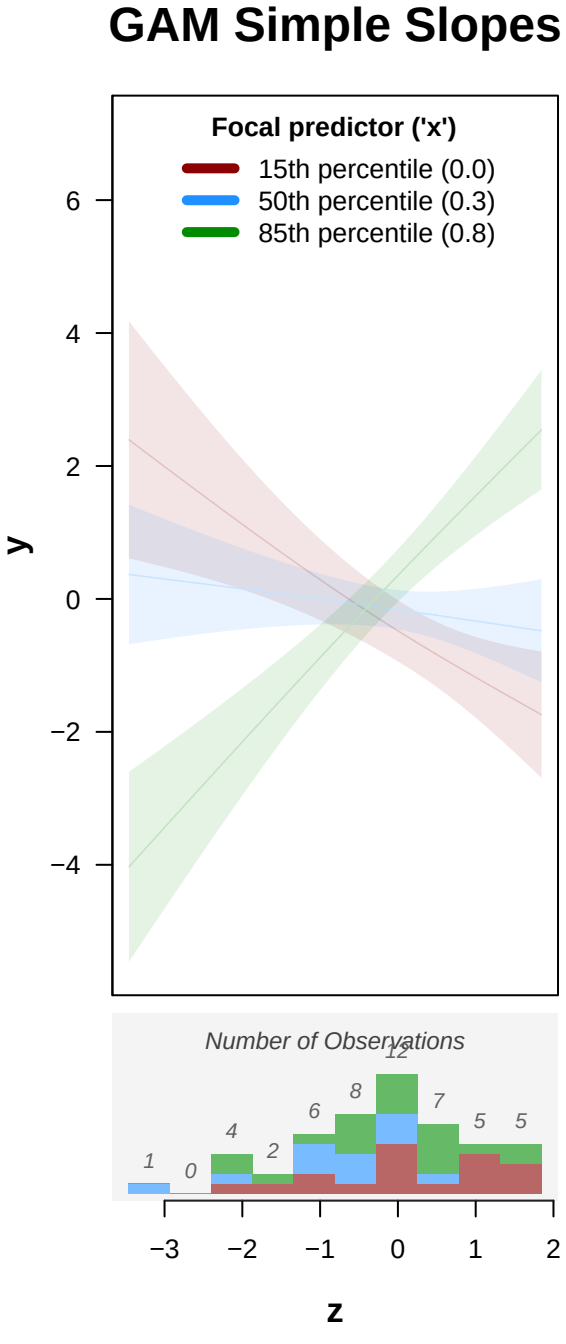
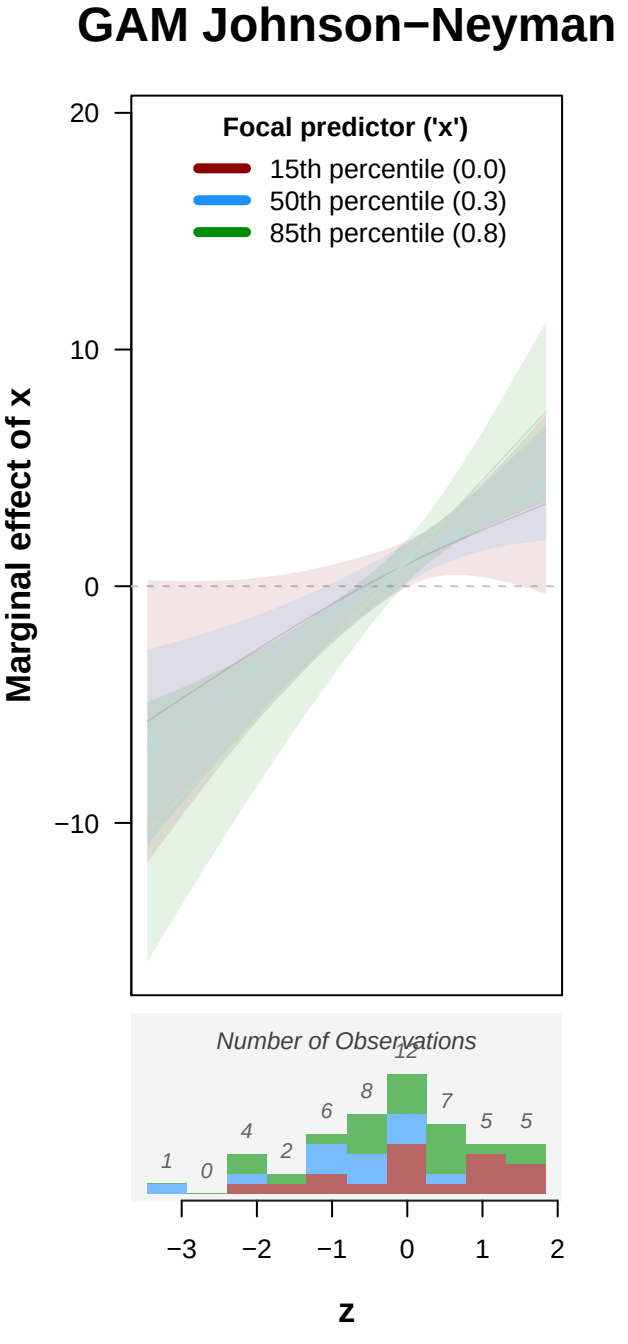
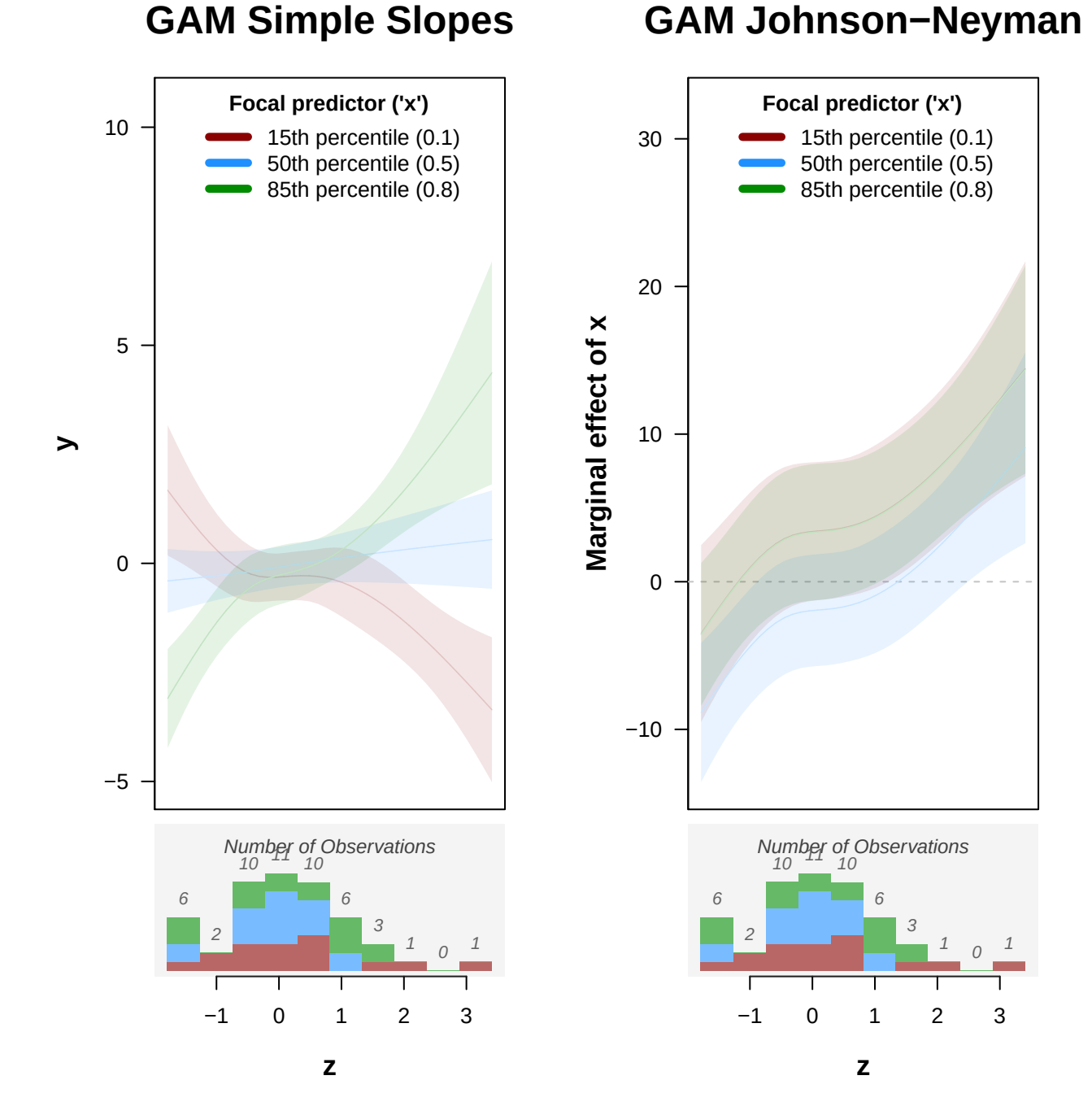

GAM Simple Slopes
Focal predictor ('x')
0
1
y
6
4
2
0
−2
Number of Observations
4 4 5 12 8 10 4 0 2 1
−2 −1 0 1 2 3
z

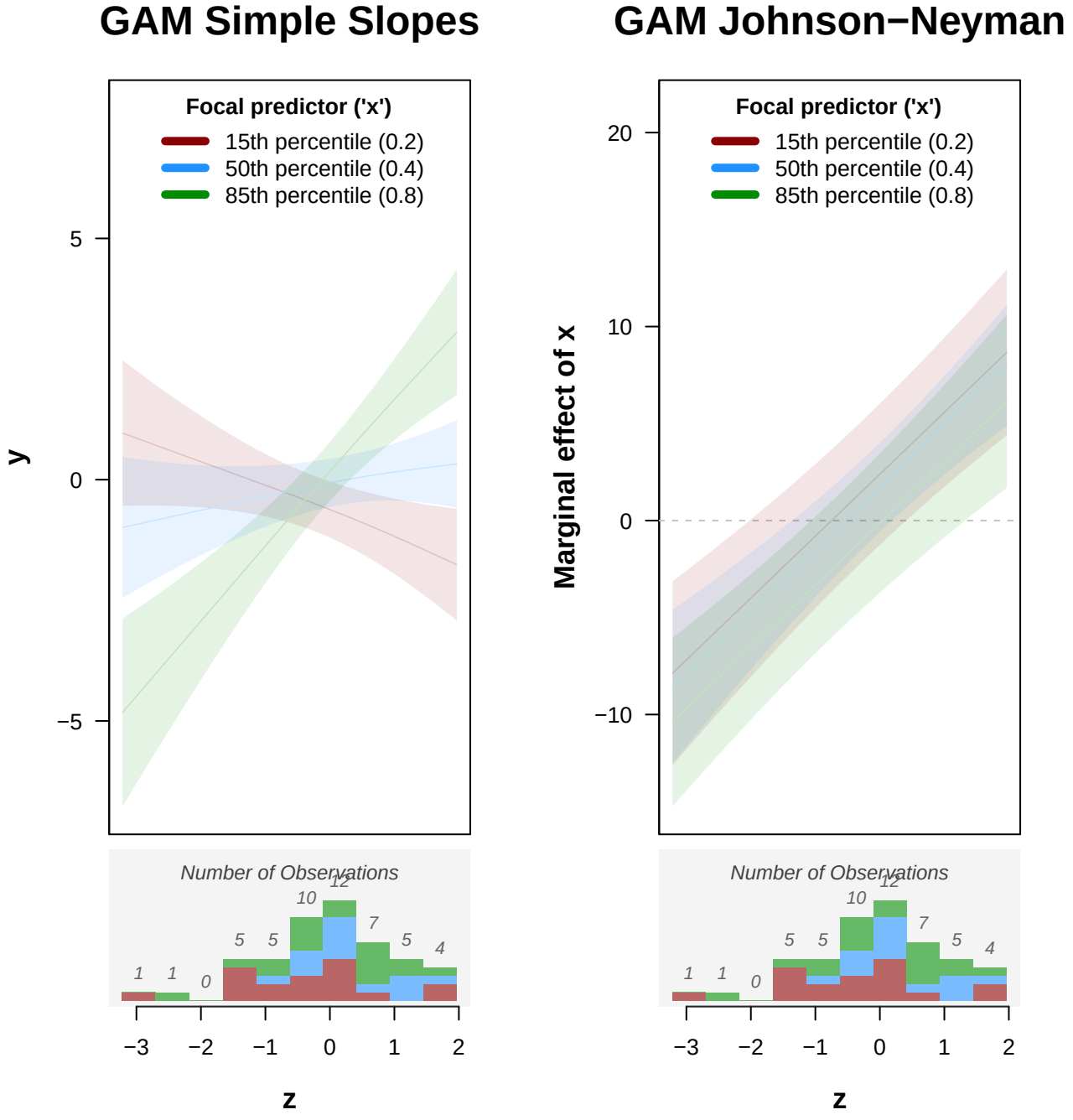

GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
10
8
6
4
2
0
−2
−4
Number of Observations
4 4 5 12 8 10 4 0 2 1
−2 −1 0 1 2 3
z

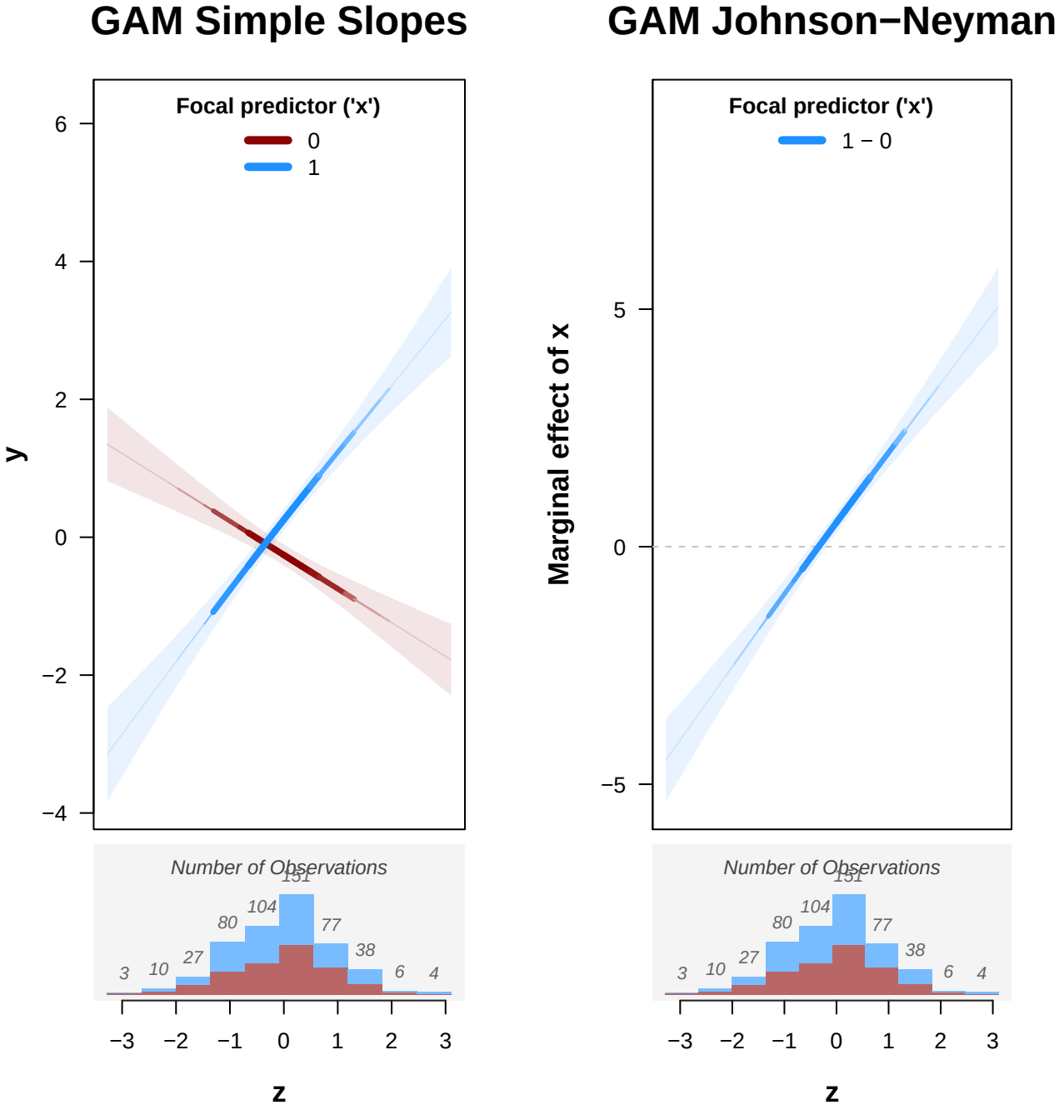

GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
8
6
4
2
0
−2
−4
Number of Observations
2
1
5
7
8
11
8
3
3
2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
10
5
0
−5
Number of Observations
2
1
5
7
8
11
8
3
3
2
−1
0
1
2
z

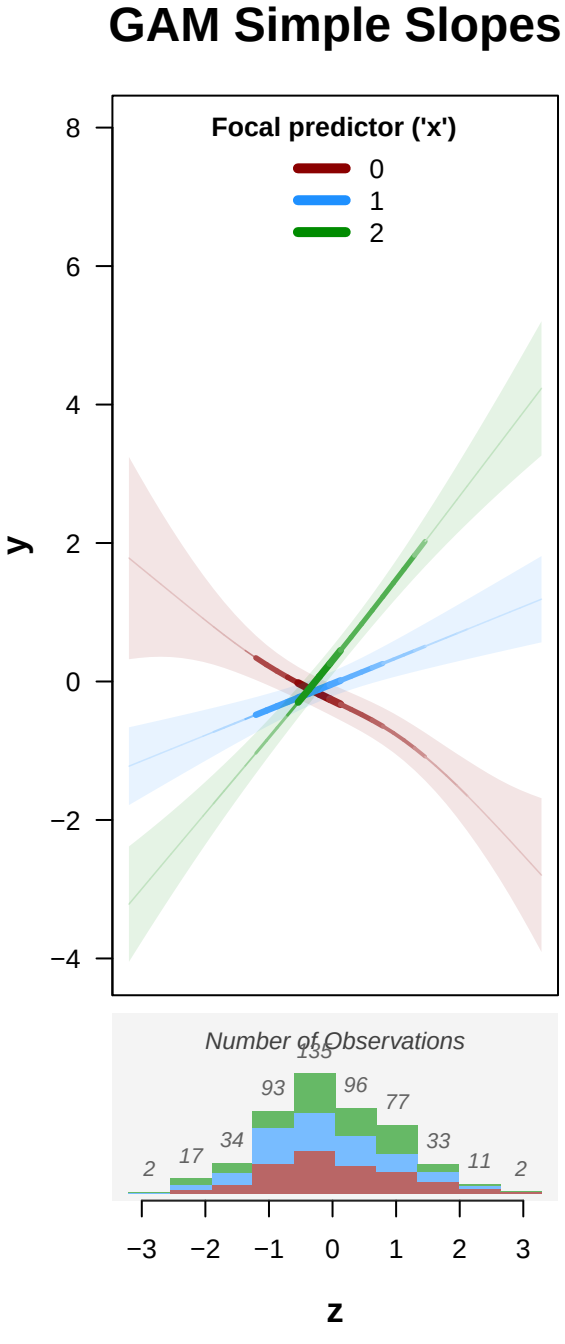

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.3)
85th percentile (0.8)
y
6
4
2
0
−2
−4
Number of Observations
1
0
4
2
6
8
12
7
5
5
−3
−2
−1
0
1
2
z

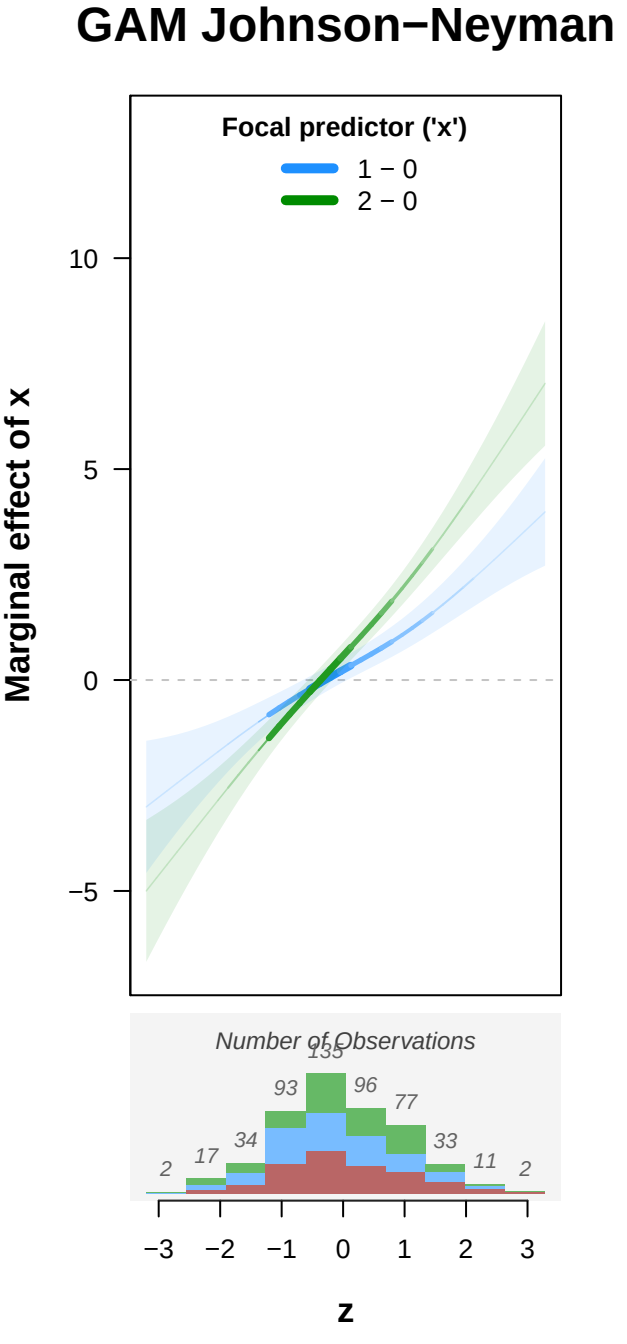

GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.3)
85th percentile (0.8)
Marginal effect of x
20
10
0
−10
Number of Observations
1
0
4
2
6
8
12
7
5
5
−3
−2
−1
0
1
2
z

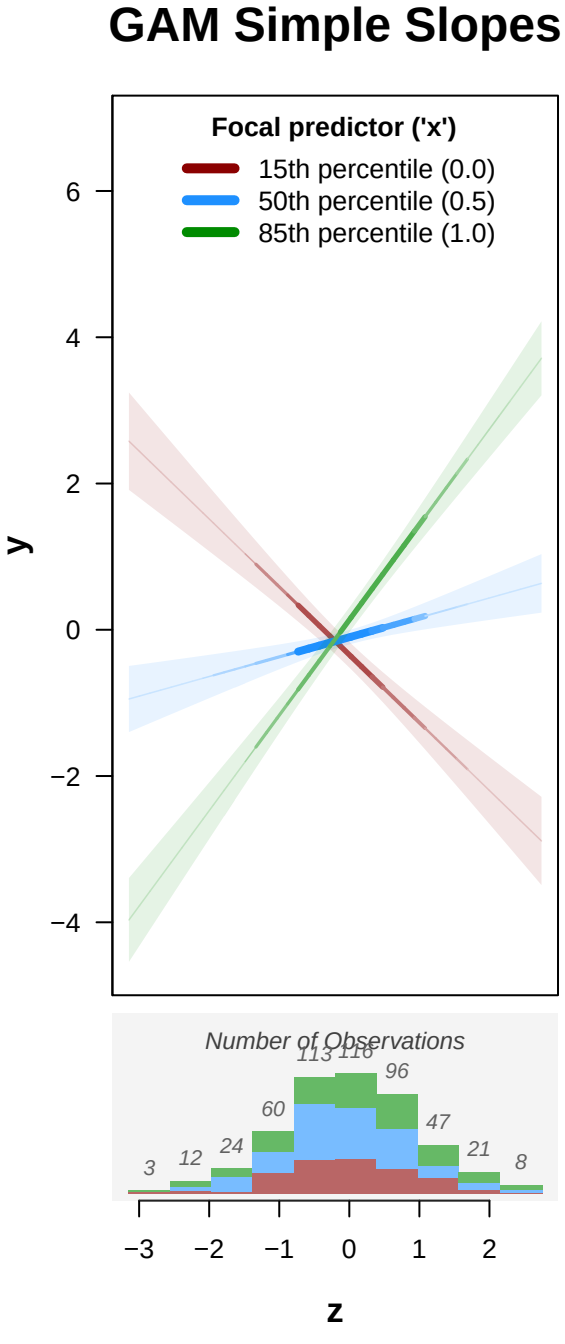

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
y
10
5
0
−5
Number of Observations
6
2
10
11
10
6
3
1
0
1
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
30
20
10
0
−10
Number of Observations
6
2
10
11
10
6
3
1
0
1
−1
0
1
2
3
z

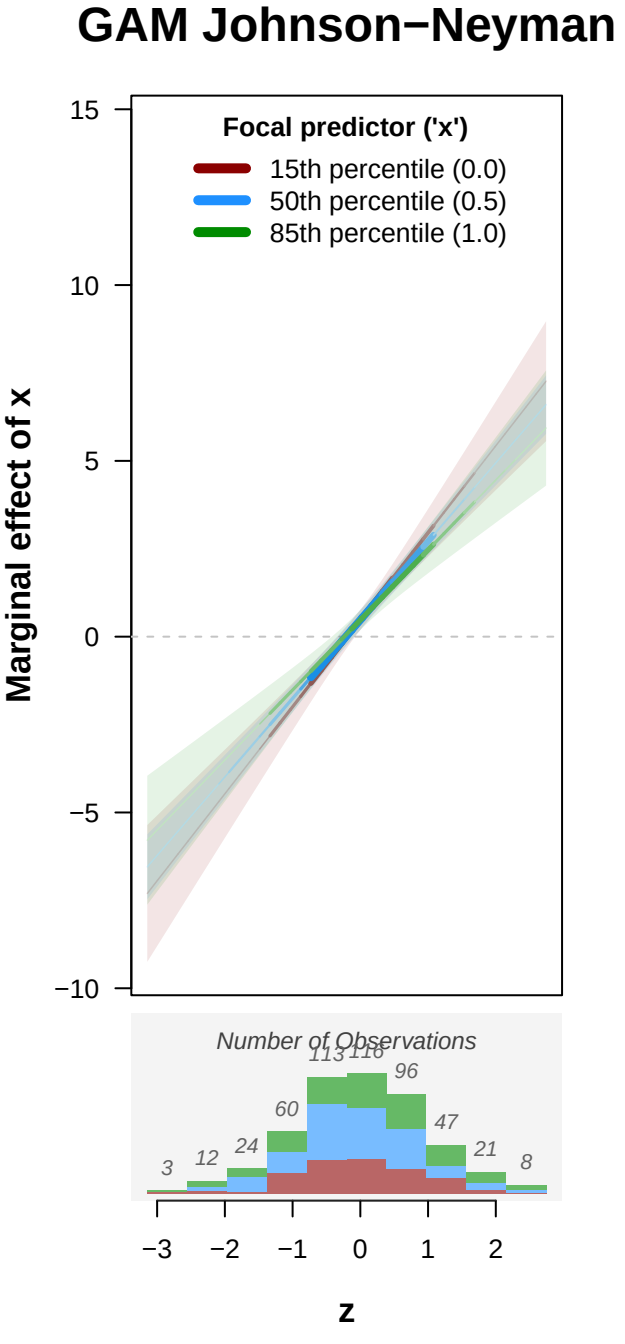

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.4)
85th percentile (0.8)
y
5
0
−5
Number of Observations
1 1 0 5 5 10 12 7 5 4
−3 −2 −1 0 1 2
z
GAM Johnson−Neyman
Marginal effect of x
20
10
0
−10

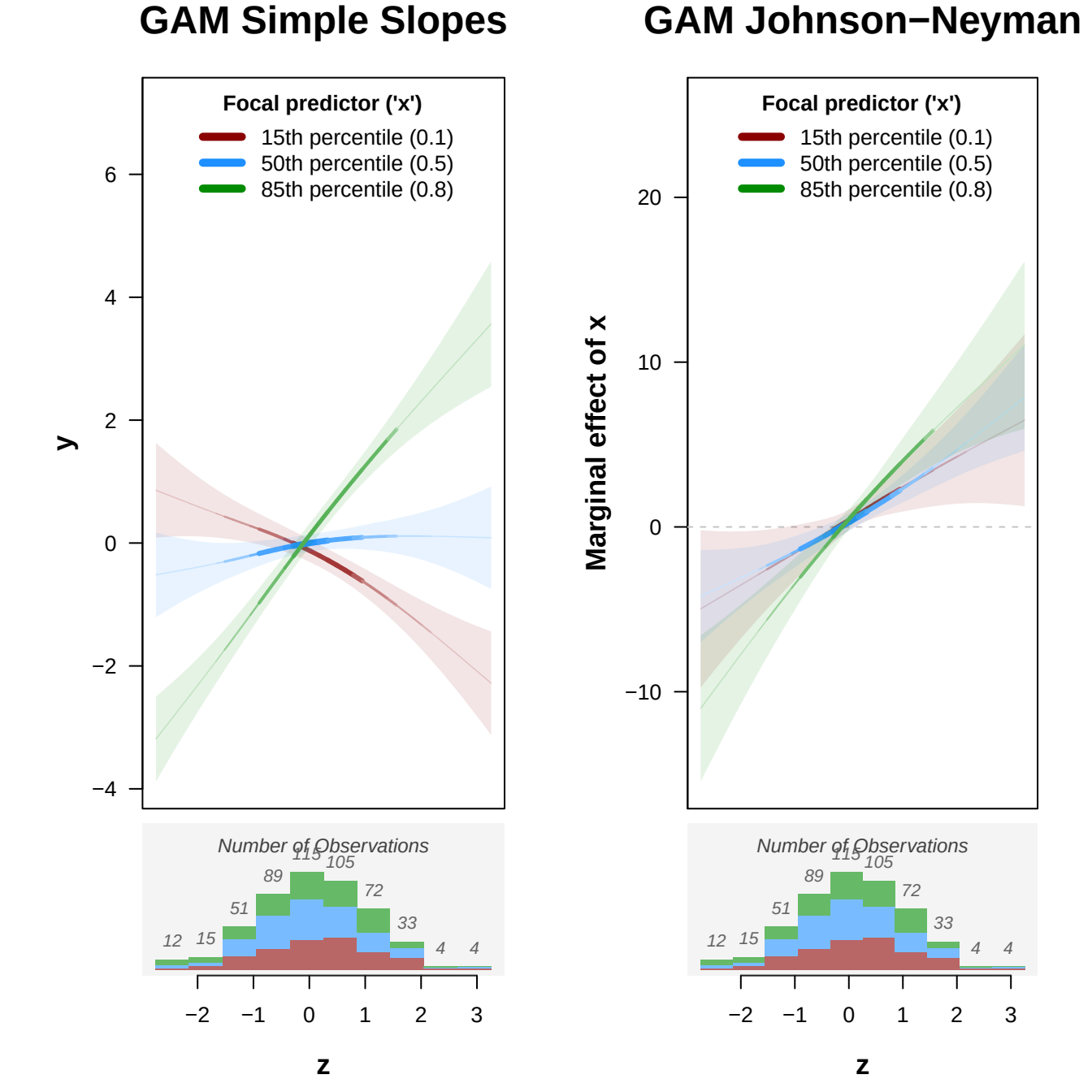

GAM Simple Slopes
Focal predictor ('x')
0
1
y
6
4
2
0
−2
−4
Number of Observations
3
10
27
80
104
151
77
38
6
4
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
5
0
−5
Number of Observations
3
10
27
80
104
151
77
38
6
4
−3
−2
−1
0
1
2
3
z

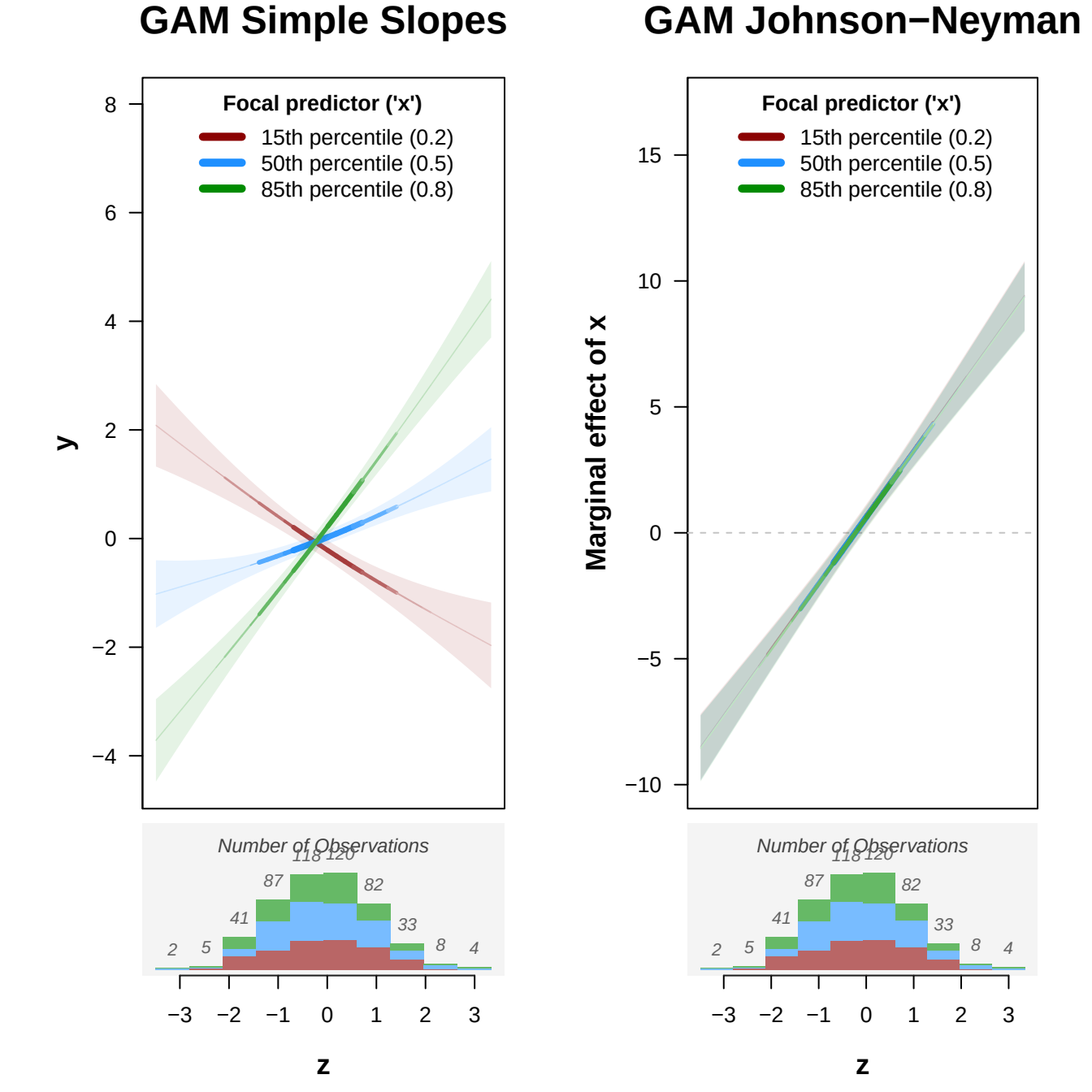

GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
8
6
4
2
0
−2
−4
Number of Observations
135
93
96
77
33
34
17
11
2
2
−3
−2
−1
0
1
2
3
z

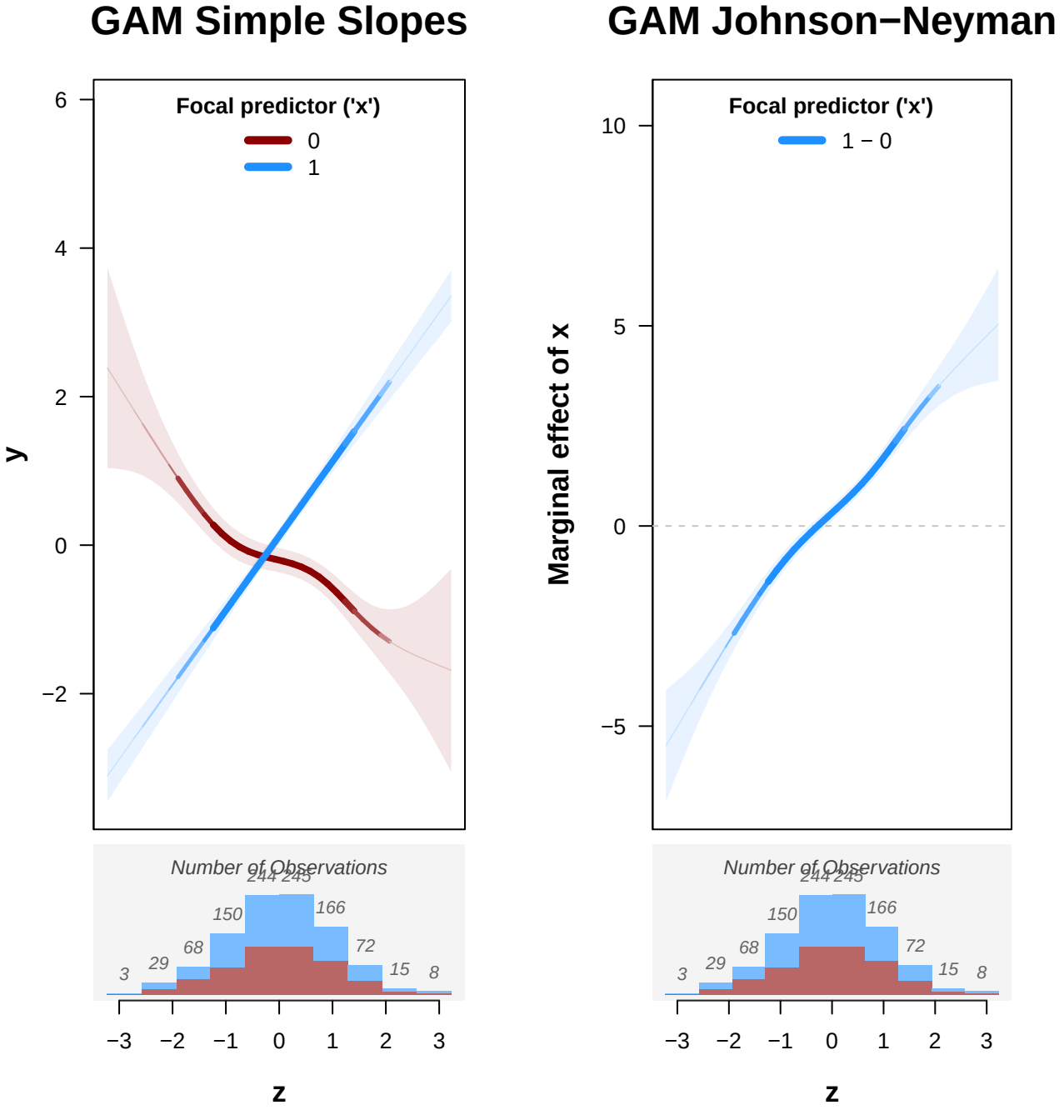

GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
10
5
0
−5
Number of Observations
135
93
96
77
33
34
17
11
2
2
−3
−2
−1
0
1
2
3
z

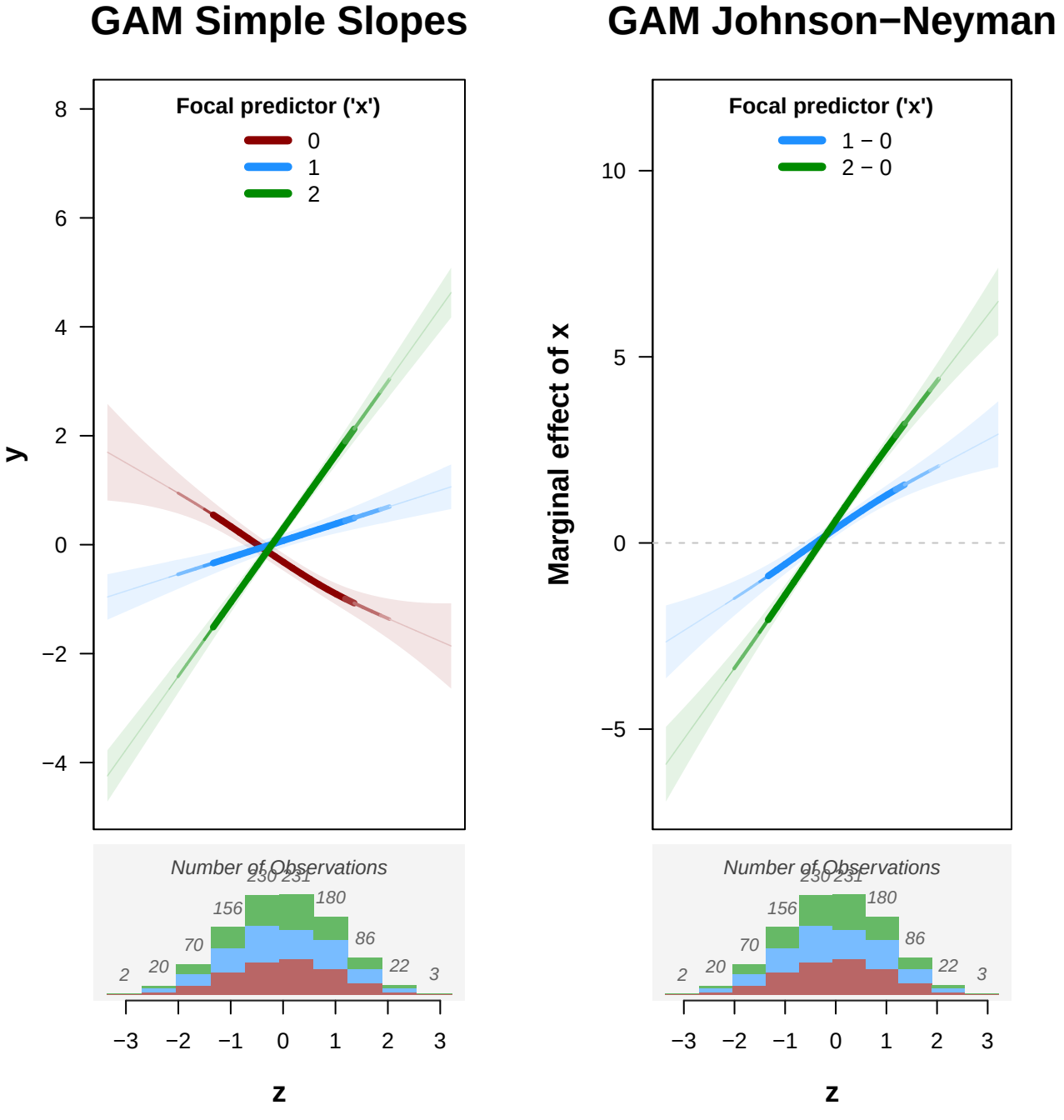

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.5)
85th percentile (1.0)
y
6
4
2
0
−2
−4
Number of Observations
3
12
24
60
113
116
96
47
21
8
−3
−2
−1
0
1
2
z

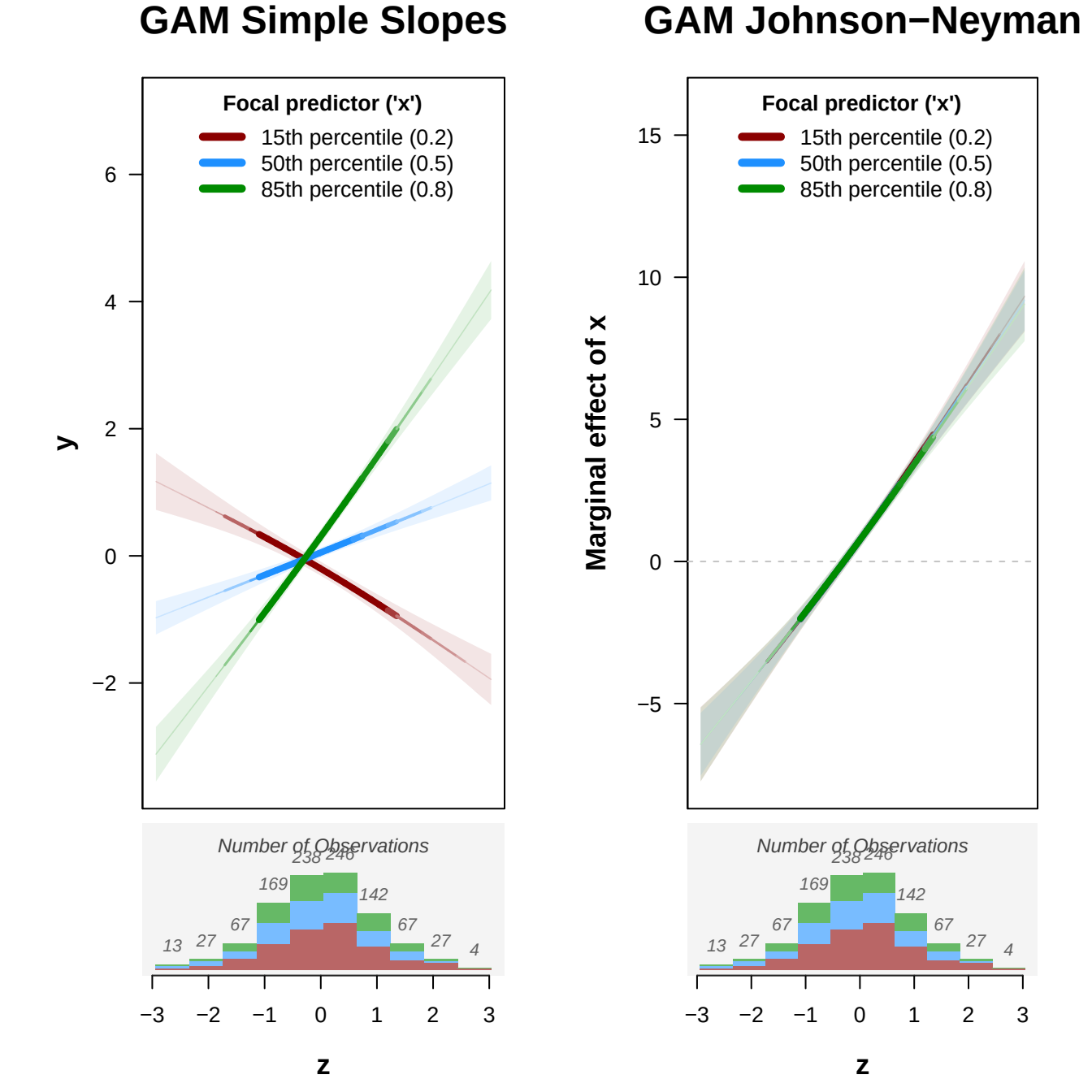

GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.5)
85th percentile (1.0)
Marginal effect of x
15
10
5
0
−5
−10
Number of Observations
3
12
24
60
113
116
96
47
21
8
−3
−2
−1
0
1
2
z

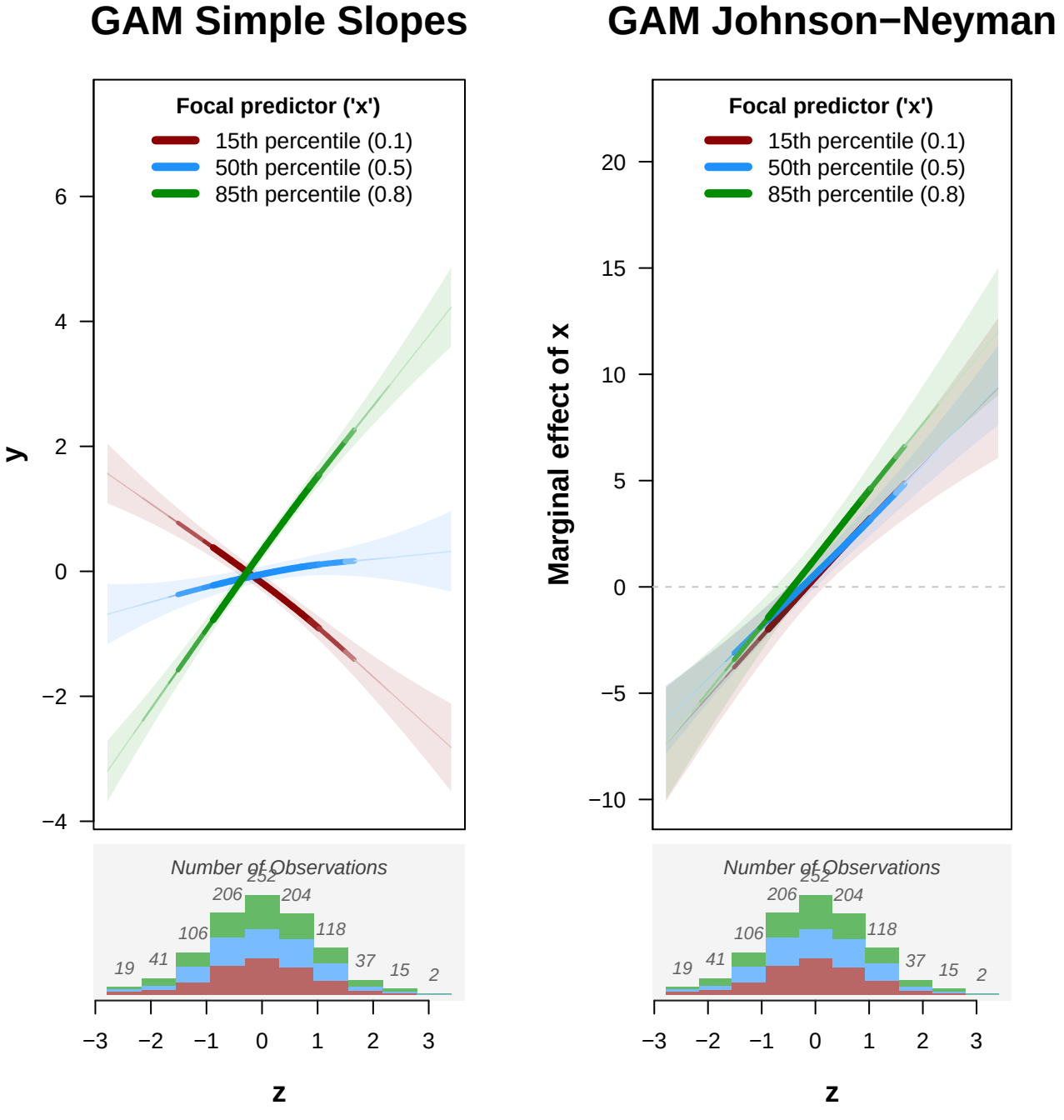

GAM Simple Slopes
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
y
Marginal effect of x
z
Number of Observations
12
15
51
89
115
105
72
33
4
4

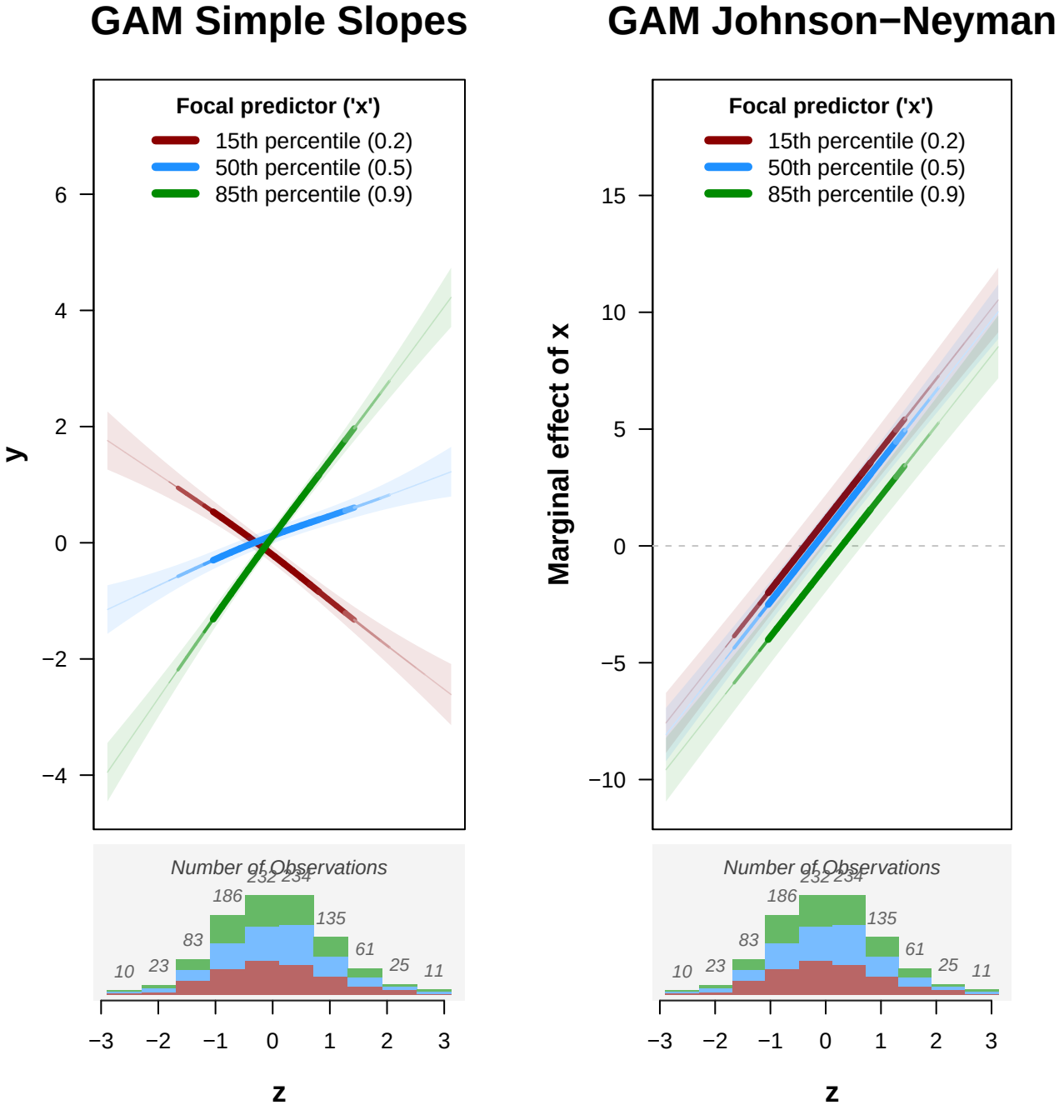

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
y
8
6
4
2
0
−2
−4
Number of Observations
118 120
87
82
41
33
8
4
2
5
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
15
10
5
0
−5
−10
Number of Observations
118 120
87
82
41
33
8
4
2
5
−3
−2
−1
0
1
2
3
z

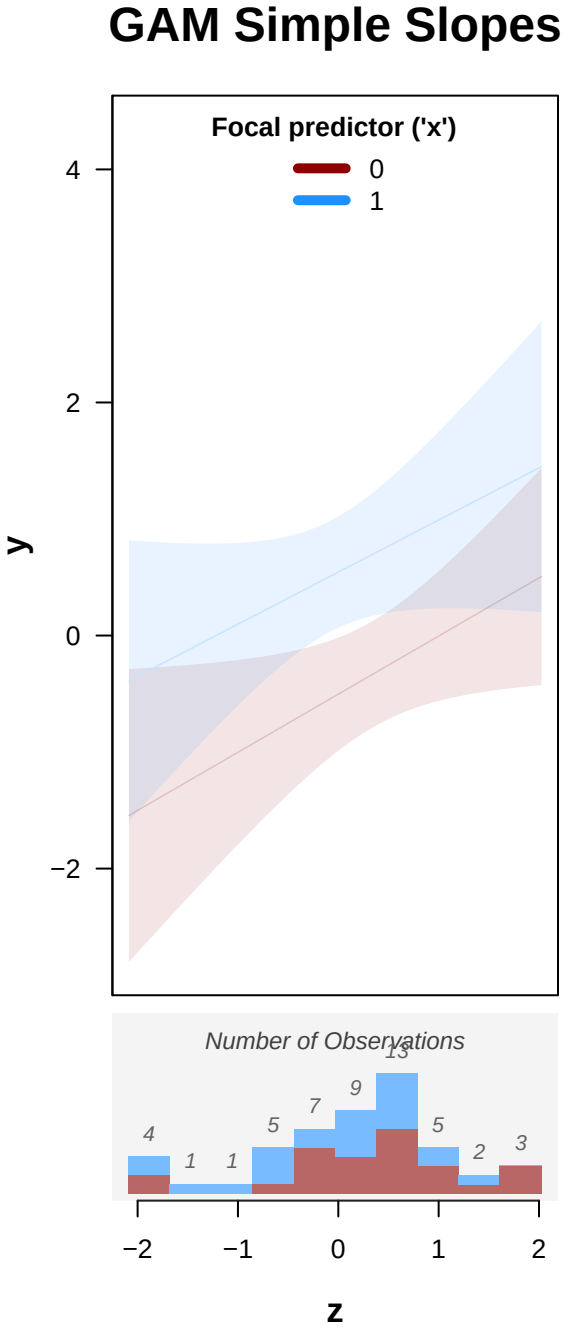

GAM Simple Slopes
Focal predictor ('x')
0
1
y
6
4
2
0
−2
Number of Observations
244 245
150
166
68
72
29
15
3
8
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
10
5
0
−5
Number of Observations
244 245
150
166
68
72
29
15
3
8
−3
−2
−1
0
1
2
3
z

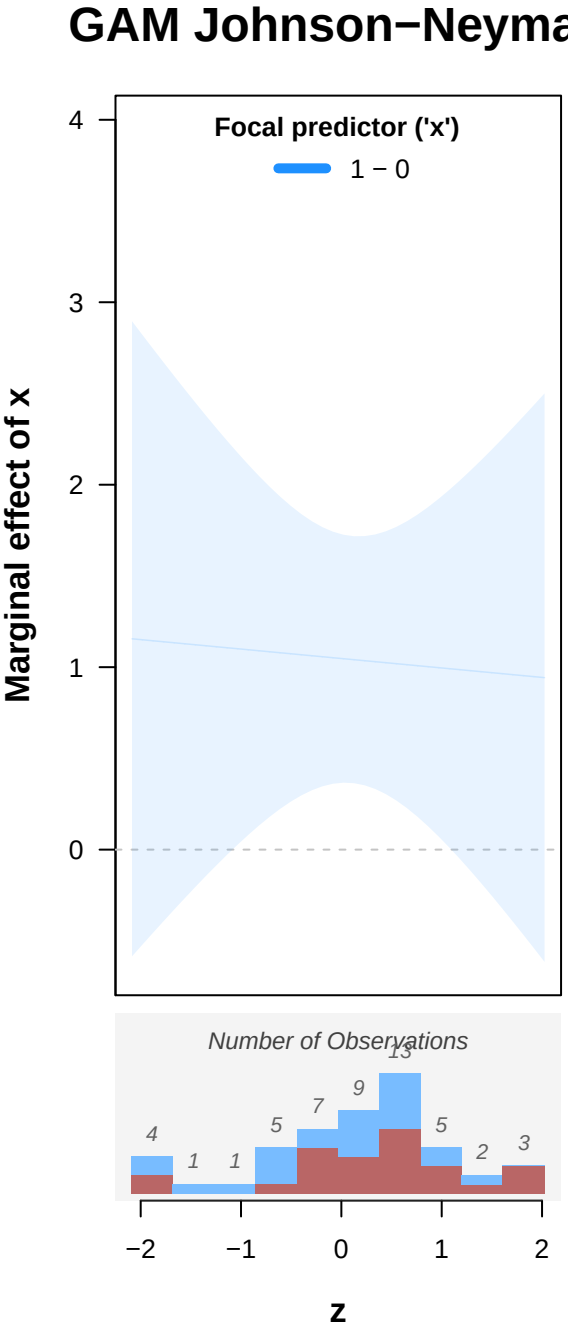

GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
8
6
4
2
0
−2
−4
Number of Observations
2
20
70
156
230 231
180
86
22
3
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
10
5
0
−5
Number of Observations
2
20
70
156
230 231
180
86
22
3
−3
−2
−1
0
1
2
3
z

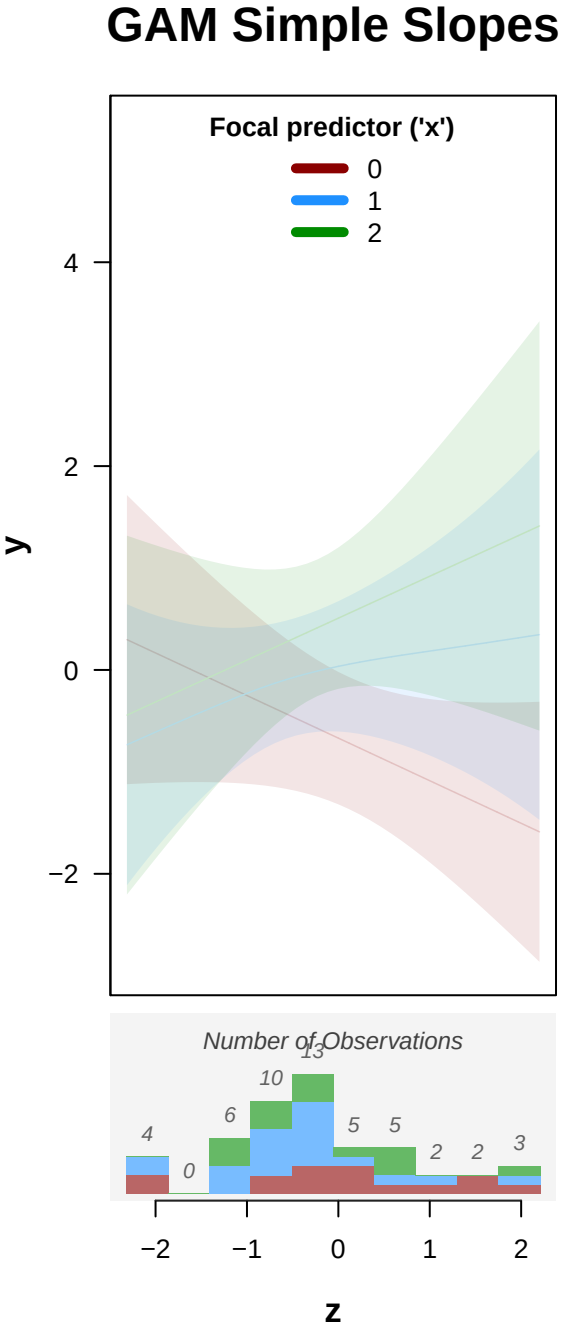

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
y
6
4
2
0
−2
Number of Observations
13
27
67
169
238
246
142
67
27
4
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
15
10
5
0
−5
Number of Observations
13
27
67
169
238
246
142
67
27
4
−3
−2
−1
0
1
2
3
z

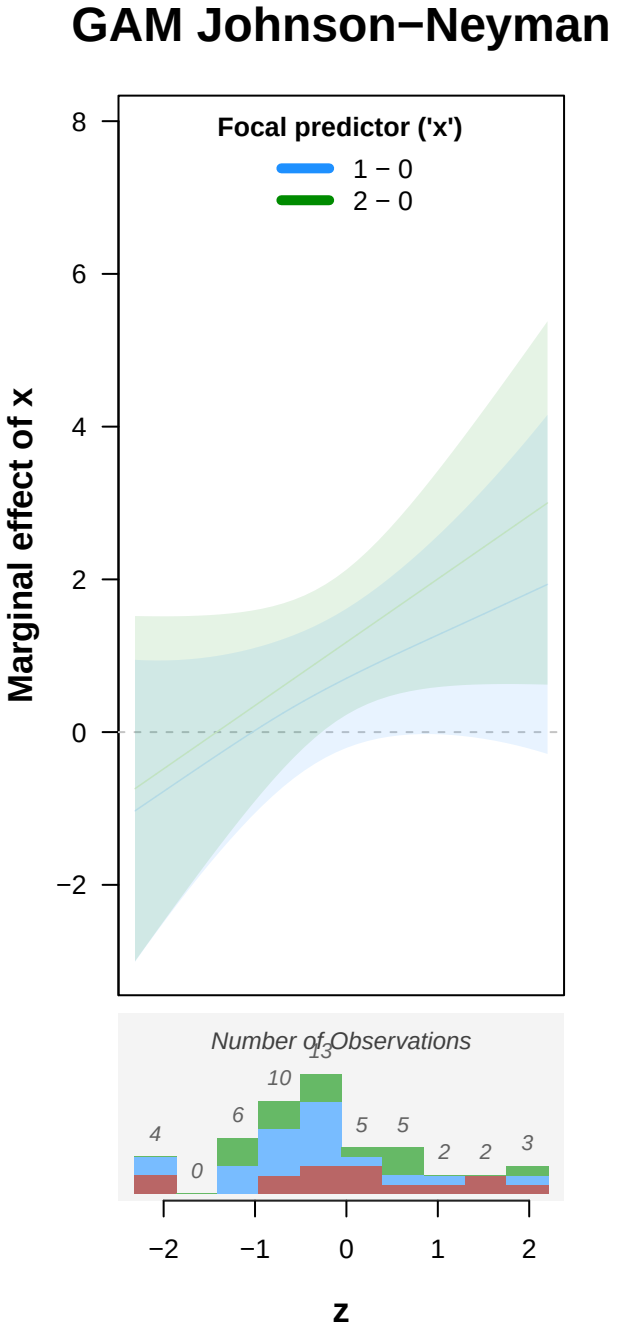

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
y
6
4
2
0
−2
−4
Number of Observations
252
206
204
106
118
19
41
37
15
2
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
20
15
10
5
0
−5
−10
Number of Observations
252
206
204
106
118
19
41
37
15
2
−3
−2
−1
0
1
2
3
z

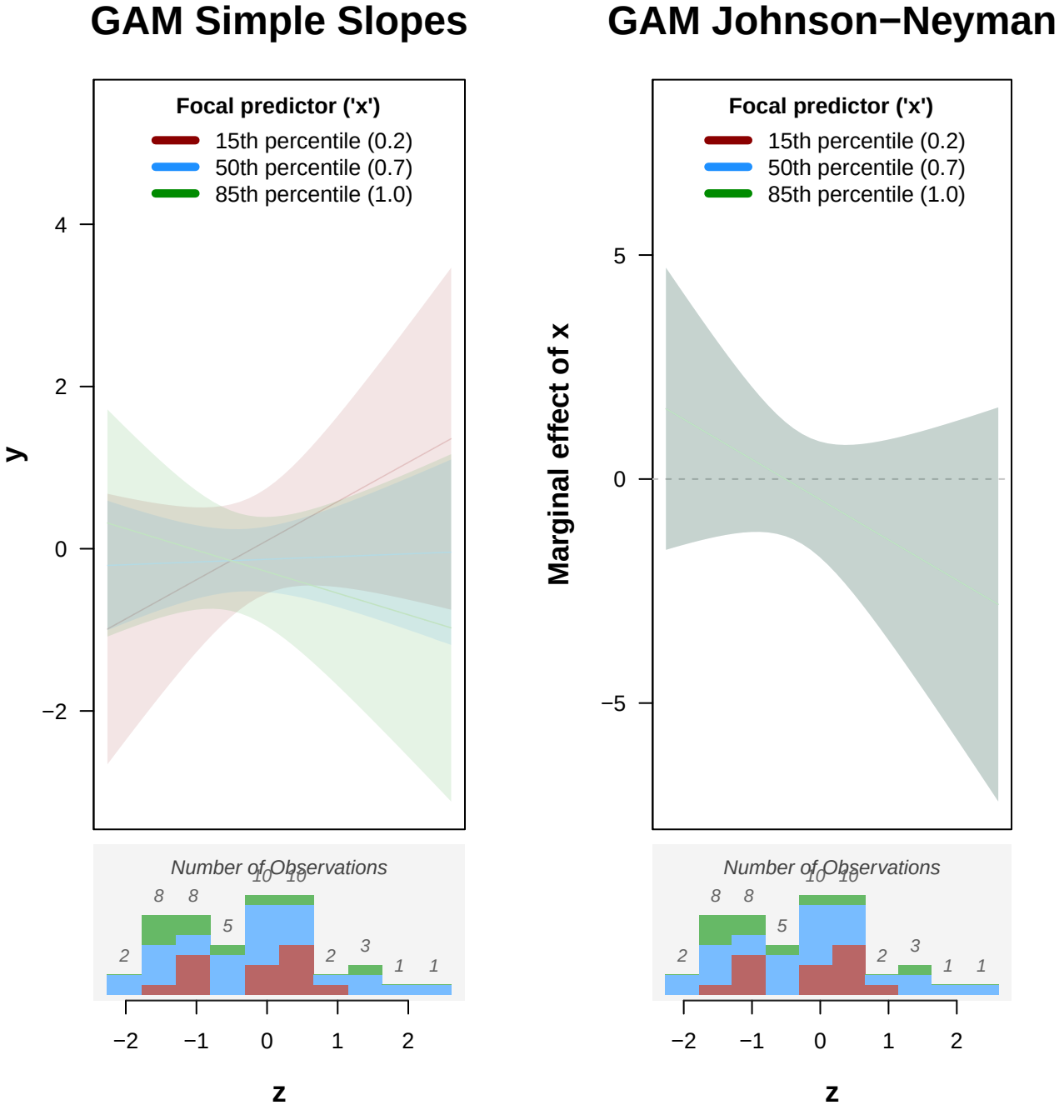

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
y
6
4
2
0
−2
−4
Number of Observations
232 234
186
135
83
61
25
11
10
23
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
Marginal effect of x
15
10
5
0
−5
−10
Number of Observations
232 234
186
135
83
61
25
11
10
23
−3
−2
−1
0
1
2
3
z

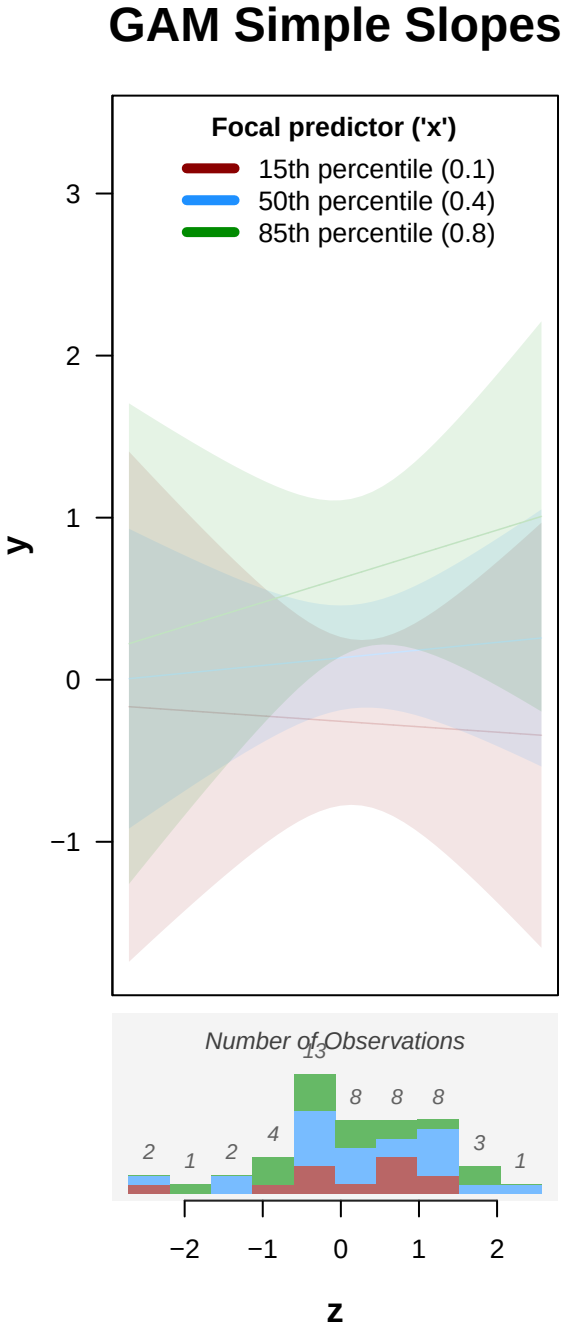

GAM Simple Slopes
Focal predictor ('x')
0
1
y
4
2
0
−2
Number of Observations
4
1
1
5
7
9
13
5
2
3
−2
−1
0
1
2
z

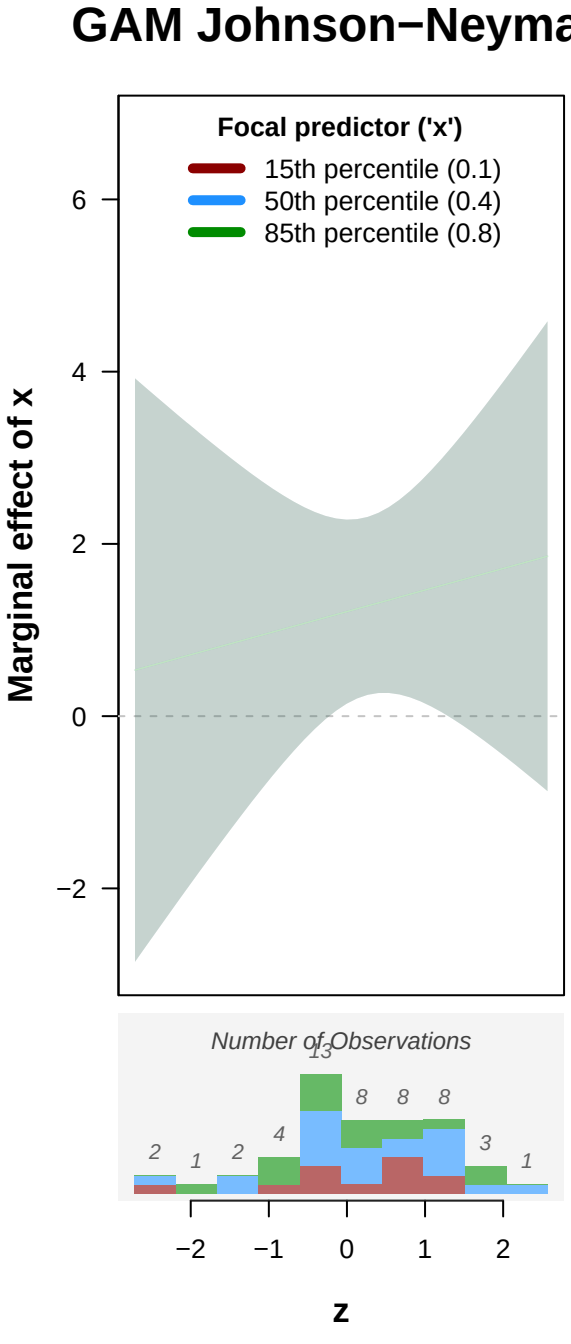

GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
4
3
2
1
0
Number of Observations
4
1
1
5
7
9
13
5
2
3
−2
−1
0
1
2
z

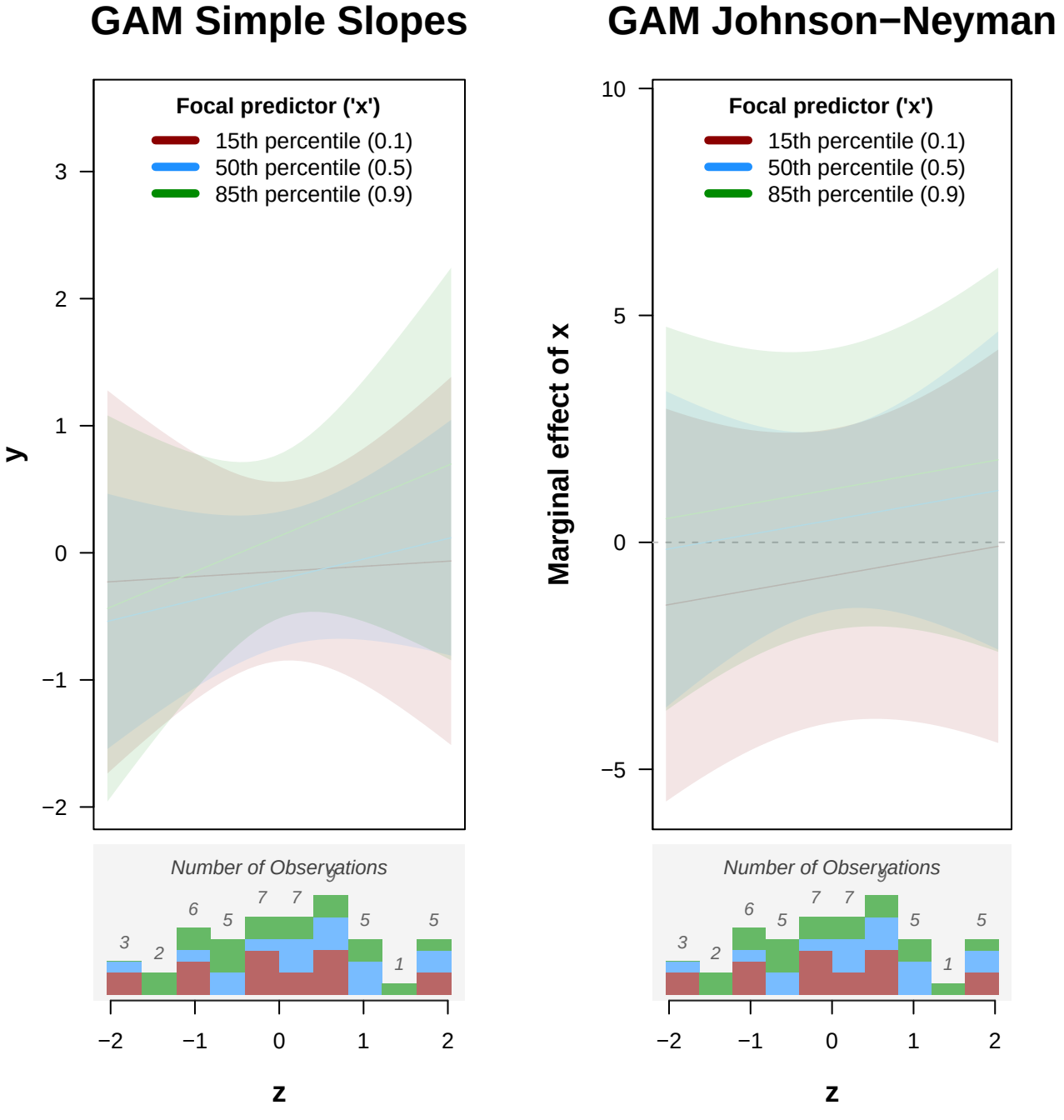

GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
4
2
0
−2
Number of Observations
13
10
6
4
0
5
5
2
2
3
−2
−1
0
1
2
z

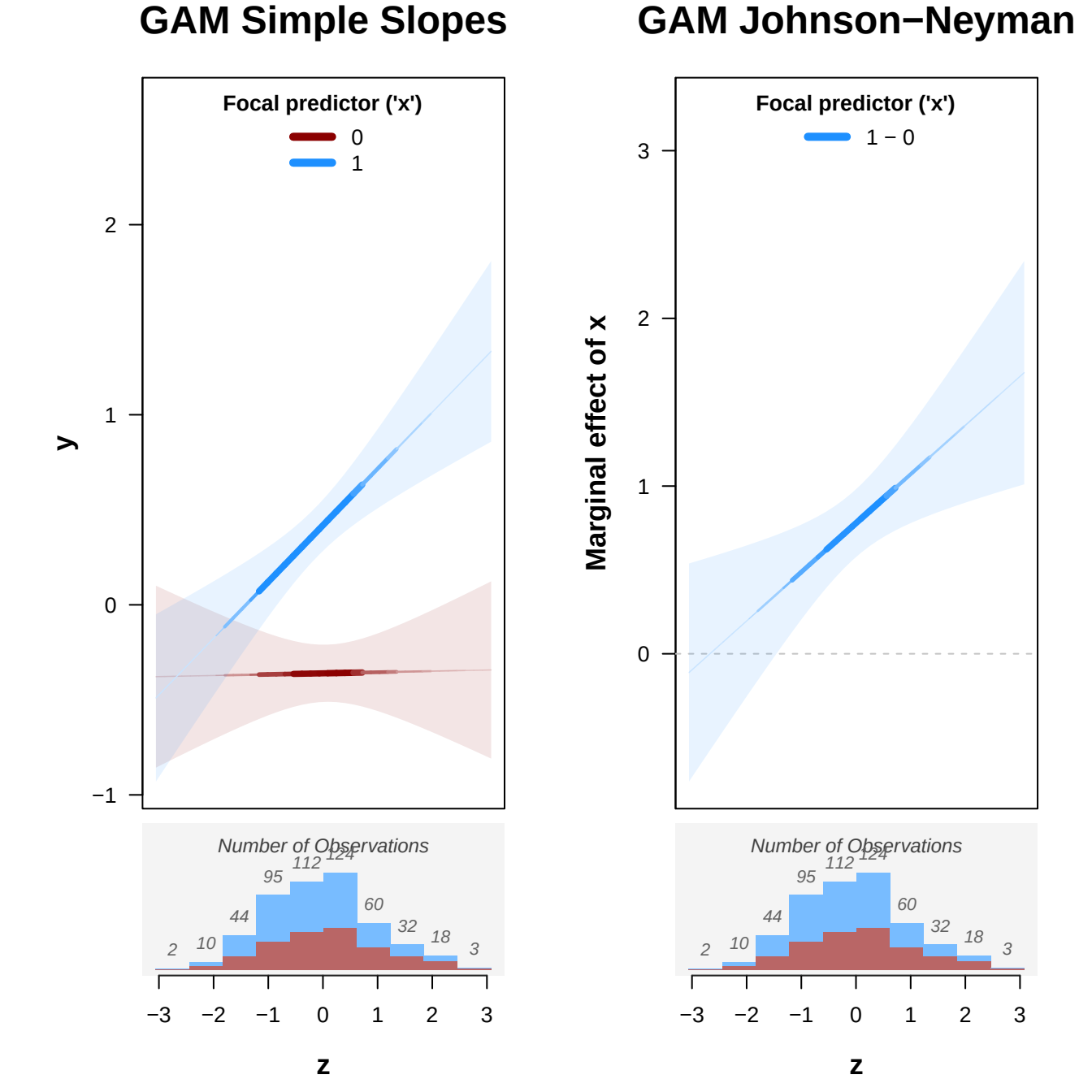

GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
8
6
4
2
0
−2
Number of Observations
13
10
6
4
0
5
5
2
2
3
−2
−1
0
1
2
z

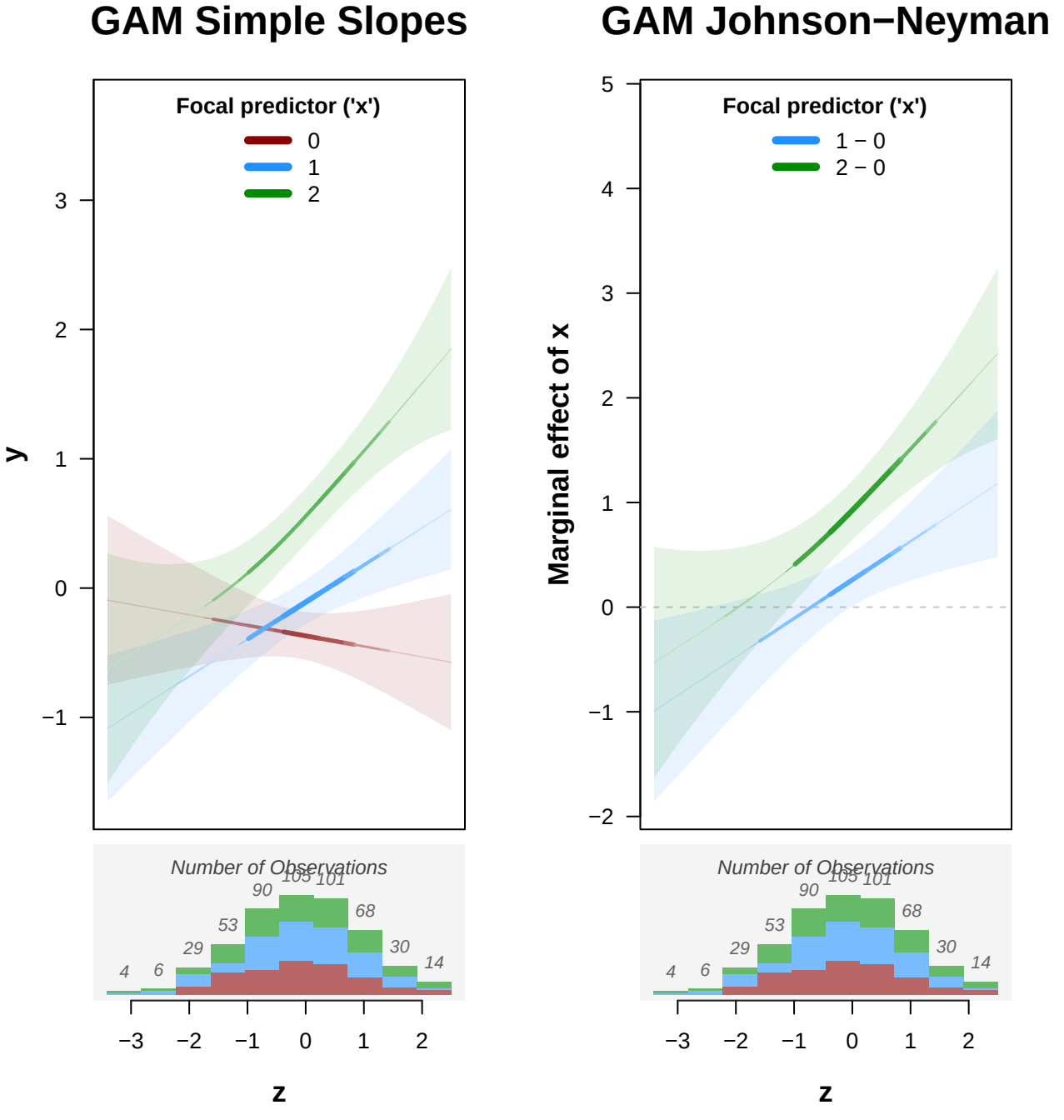

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.7)
85th percentile (1.0)
y
4
2
0
−2
Number of Observations
8 8
10 10
5
2
3
1 1
−2 −1 0 1 2
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.7)
85th percentile (1.0)
Marginal effect of x
5
0
−5
Number of Observations
8 8
10 10
5
2
3
1 1
−2 −1 0 1 2
z

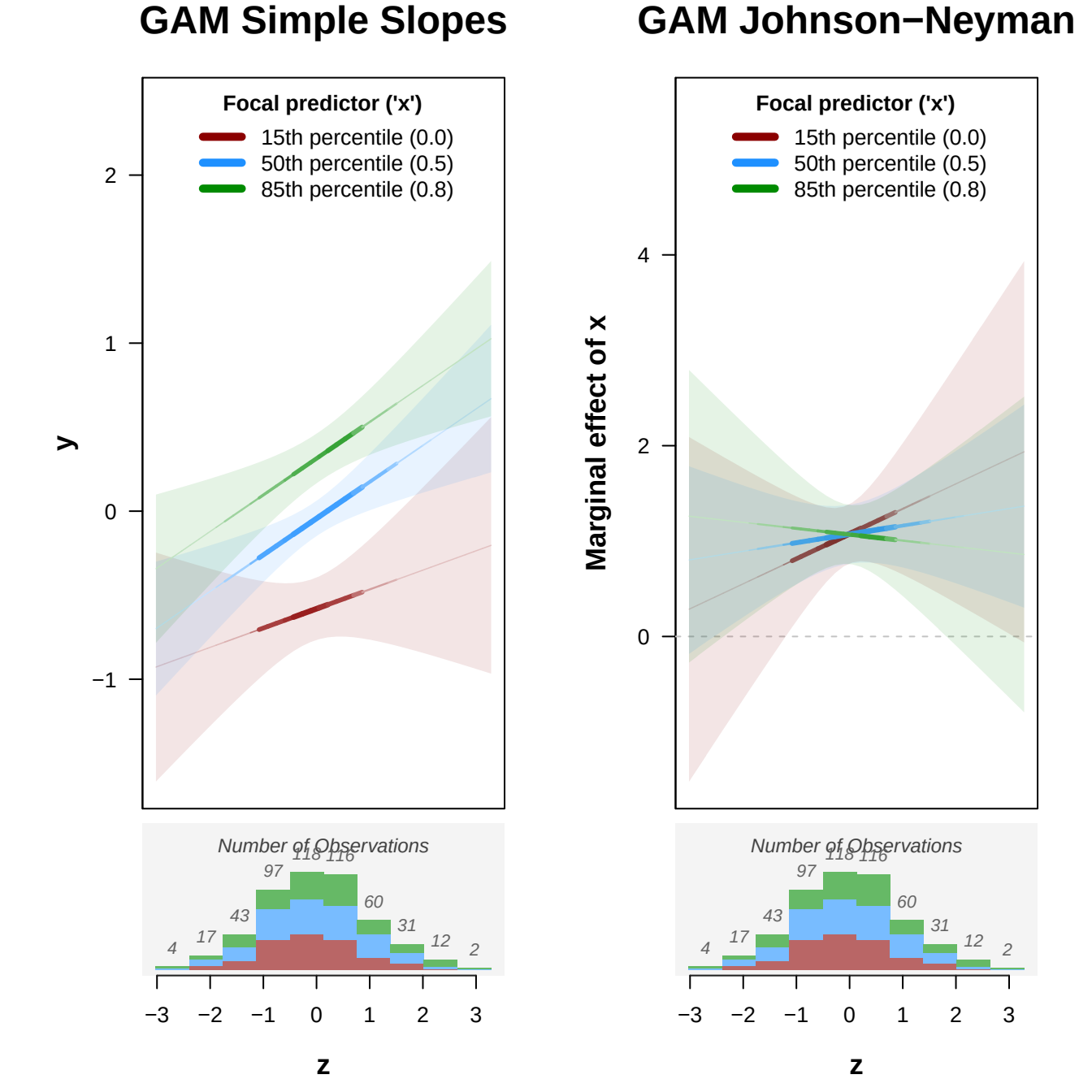

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.4)
85th percentile (0.8)
y
3
2
1
0
−1
Number of Observations
13
8
8
8
4
2
1
2
3
1
−2
−1
0
1
2
z

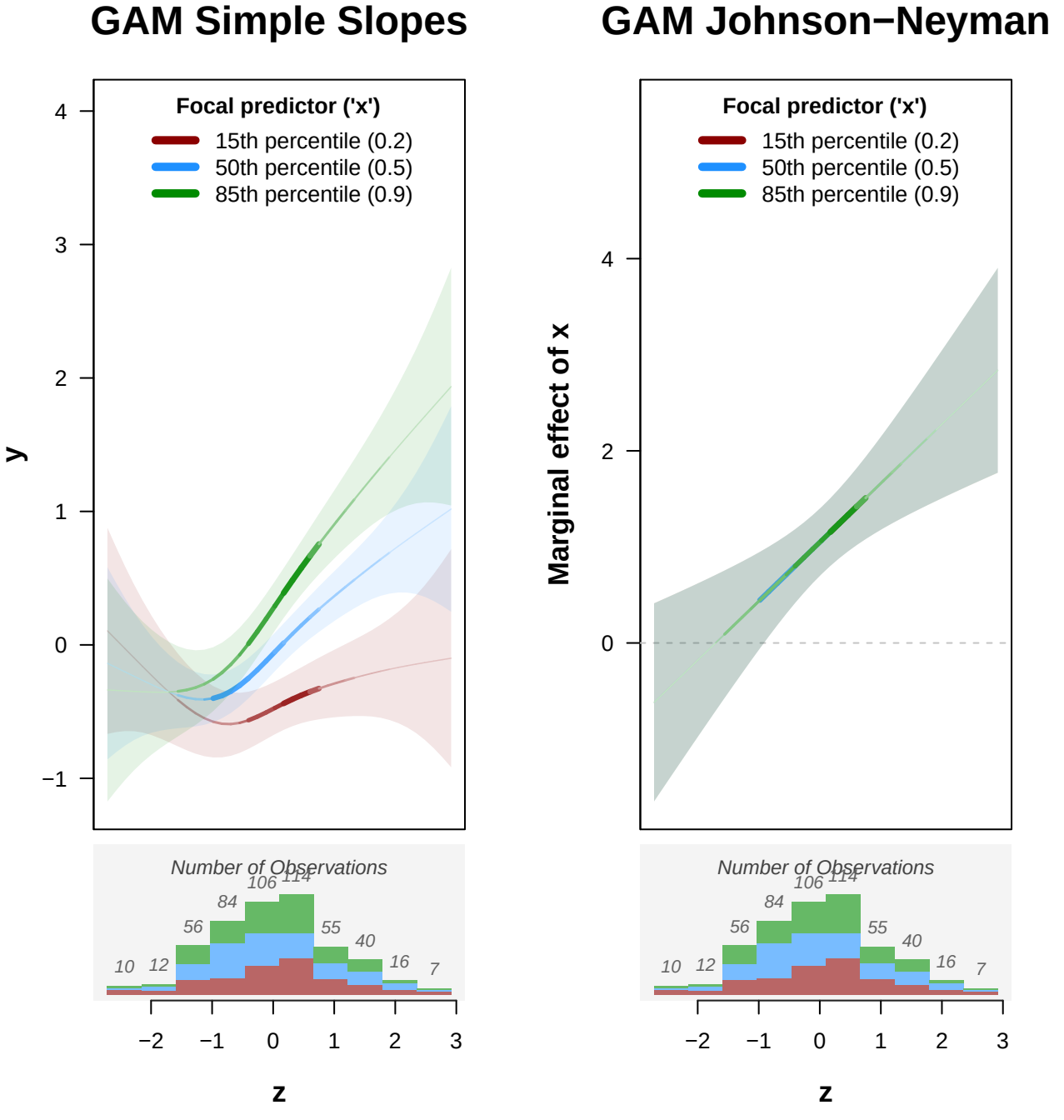

GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.4)
85th percentile (0.8)
Marginal effect of x
6
4
2
0
−2
Number of Observations
13
8
8
8
4
2
1
2
3
1
−2
−1
0
1
2
z

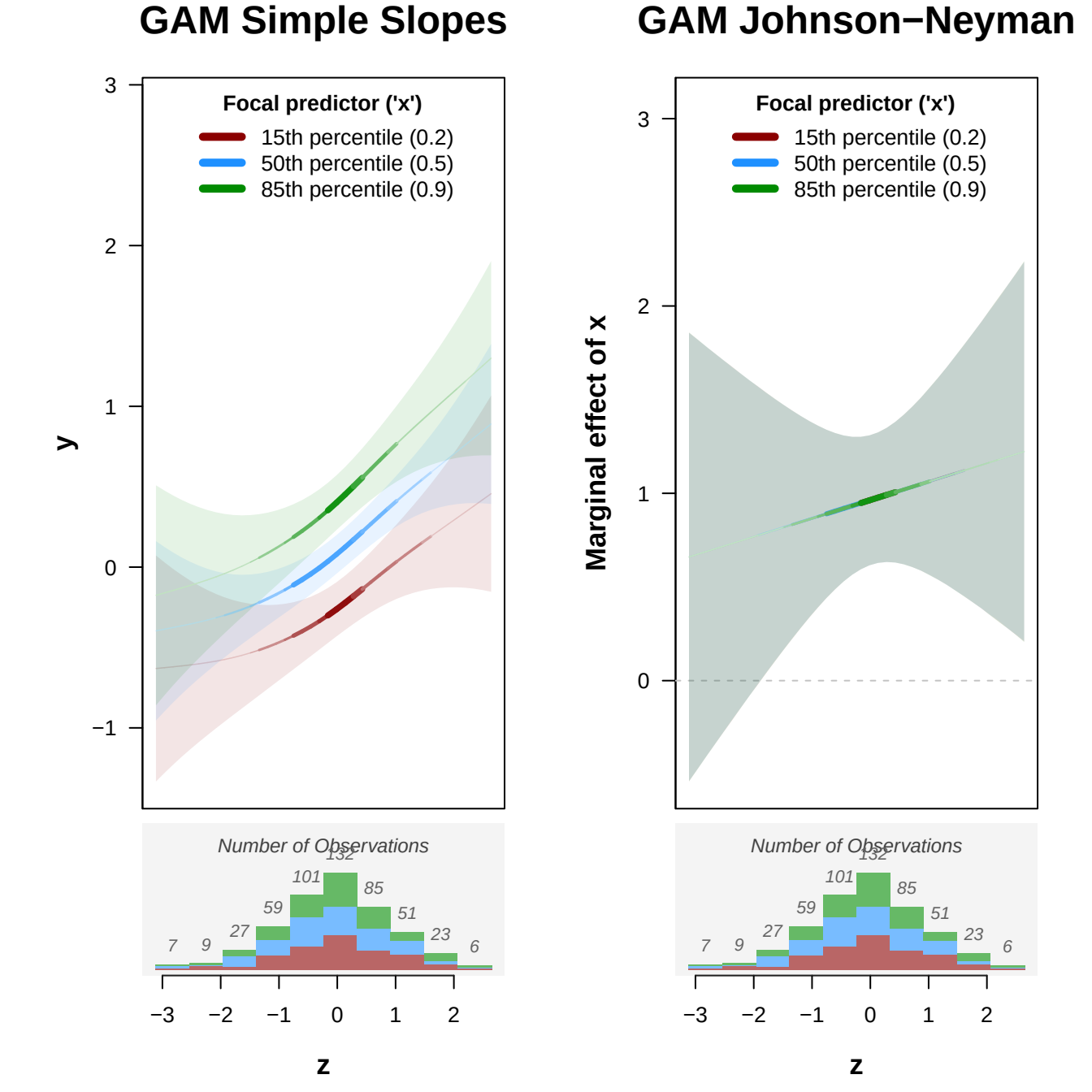

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.9)
y
3
2
1
0
−1
−2
Number of Observations
3
2
6
5
7
7
9
5
1
5
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.9)
Marginal effect of x
10
5
0
−5
Number of Observations
3
2
6
5
7
7
9
5
1
5
−2
−1
0
1
2
z

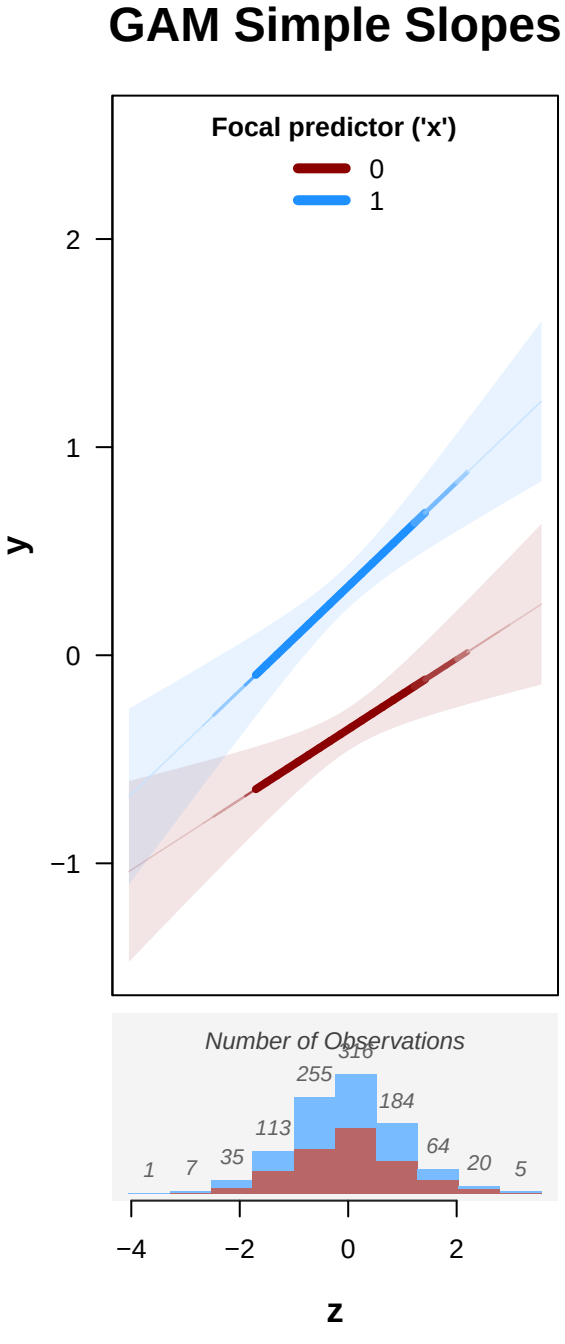

GAM Simple Slopes
Focal predictor ('x')
0
1
y
2
1
0
−1
Number of Observations
2
10
44
95
112
124
60
32
18
3
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
3
2
1
0
Number of Observations
2
10
44
95
112
124
60
32
18
3
−3
−2
−1
0
1
2
3
z

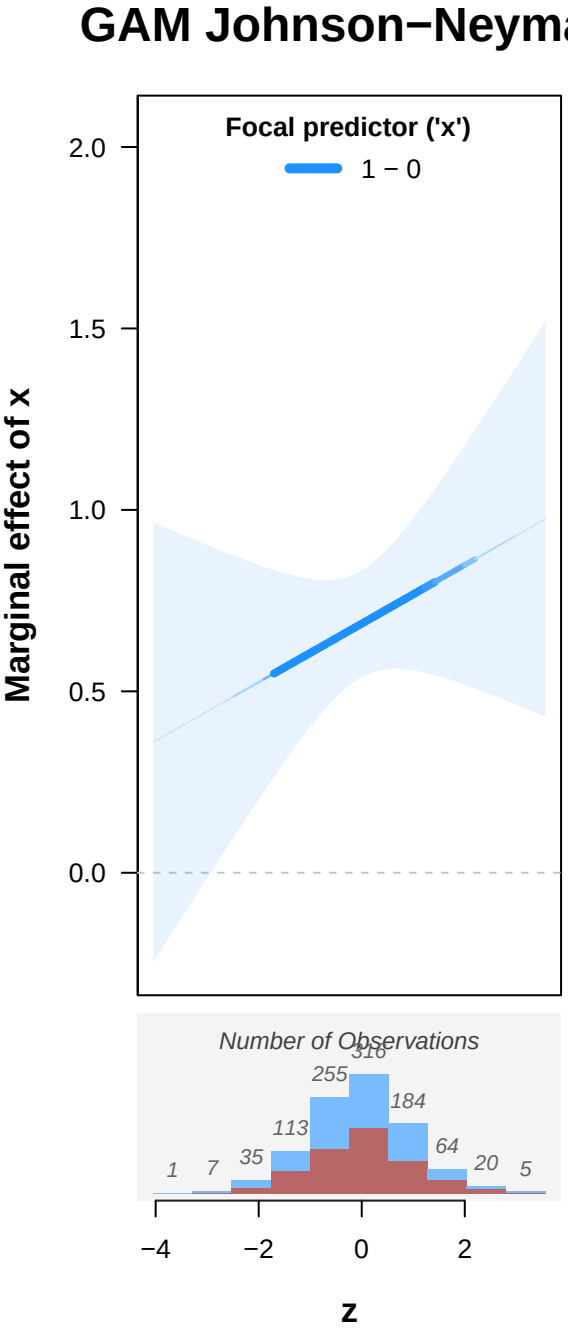

GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
3
2
1
0
−1
Number of Observations
4
6
29
53
90
103
101
68
30
14
−3
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
5
4
3
2
1
0
−1
−2
Number of Observations
4
6
29
53
90
103
101
68
30
14
−3
−2
−1
0
1
2
z

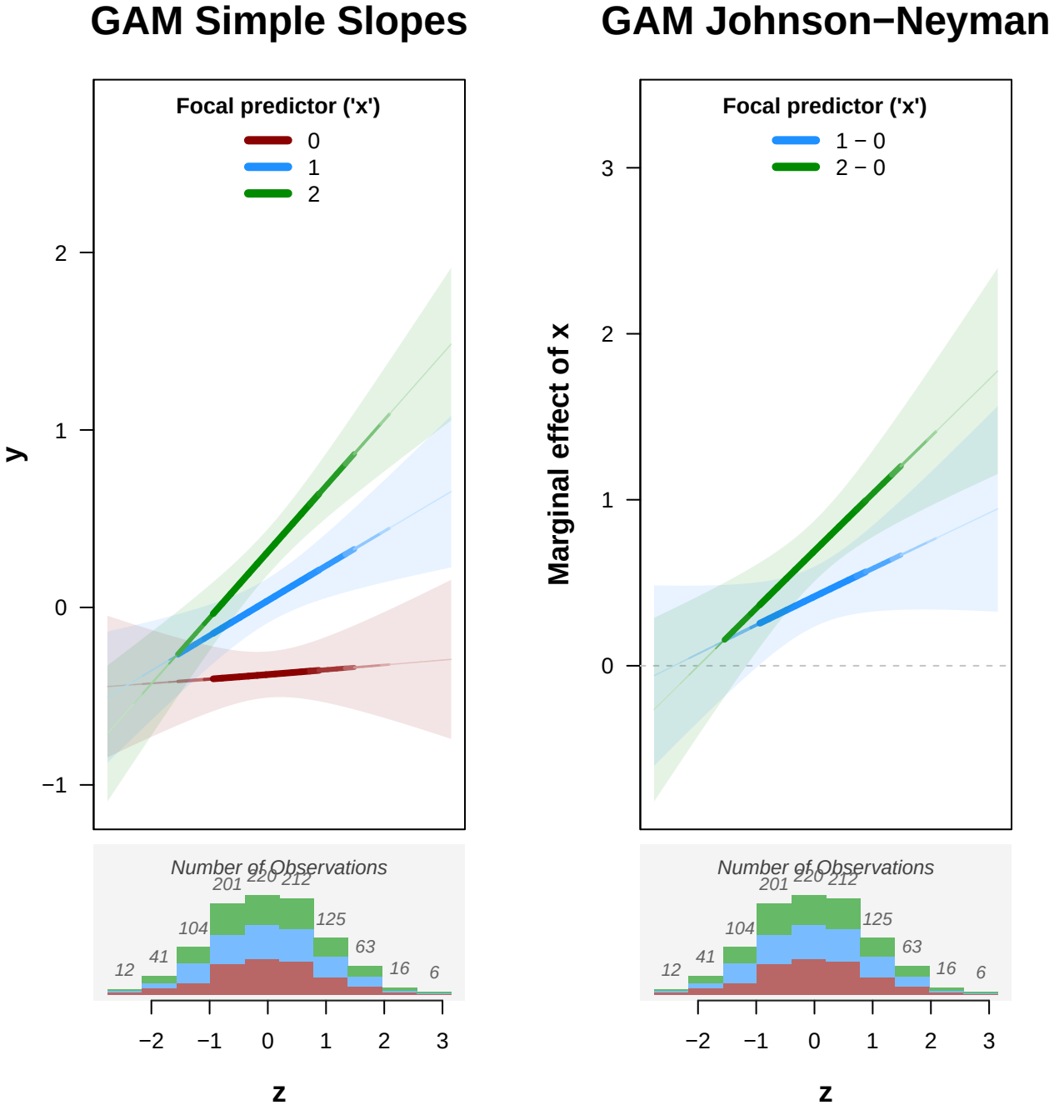

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.5)
85th percentile (0.8)
y
2
1
0
−1
Number of Observations
4
17
43
97
118
116
60
31
12
2
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
4
2
0
Number of Observations
4
17
43
97
118
116
60
31
12
2
−3
−2
−1
0
1
2
3
z

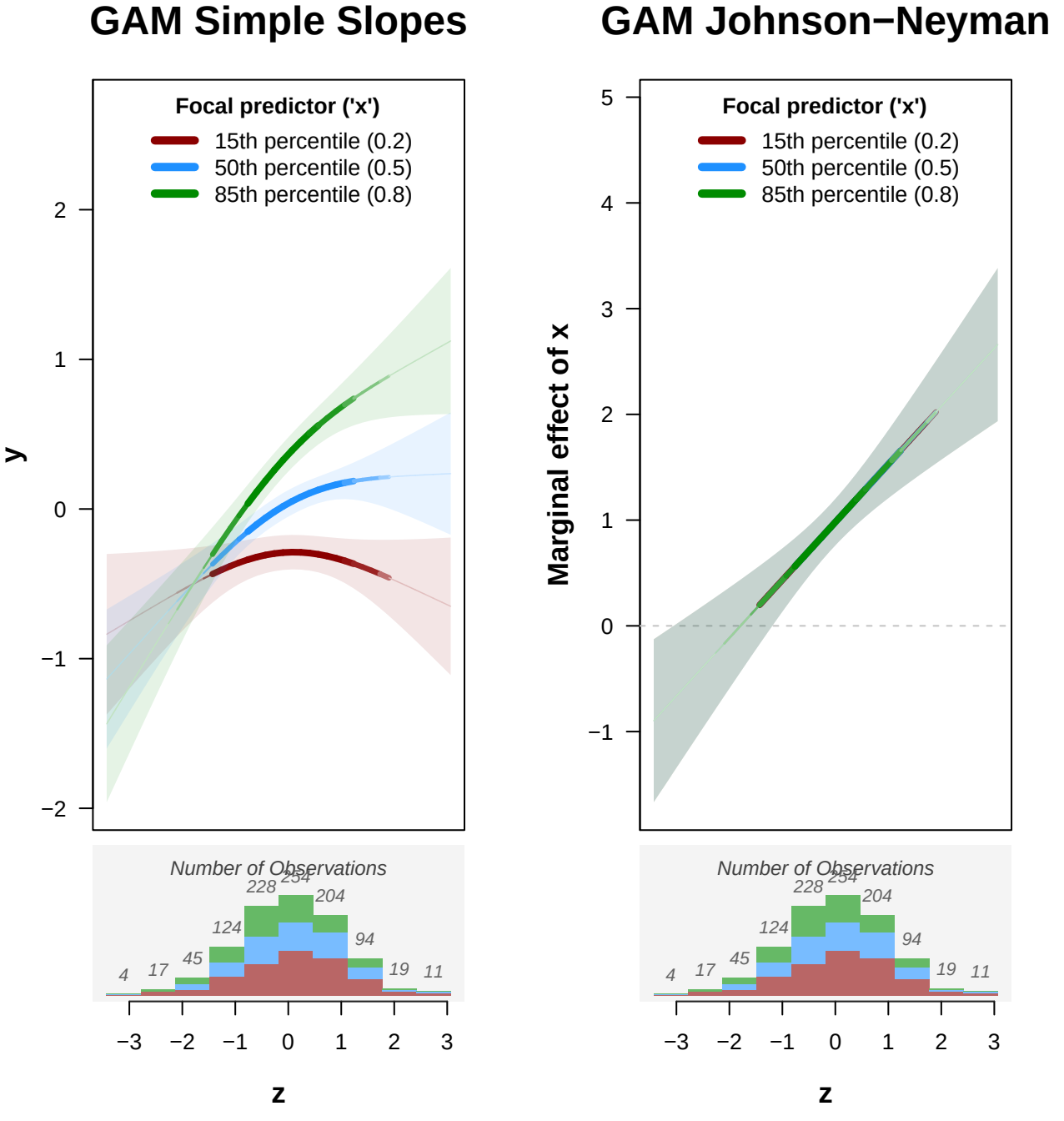

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
y
Number of Observations
10 12 56 84 106 114 55 40 16 7
z
GAM Johnson−Neyman
Marginal effect of x

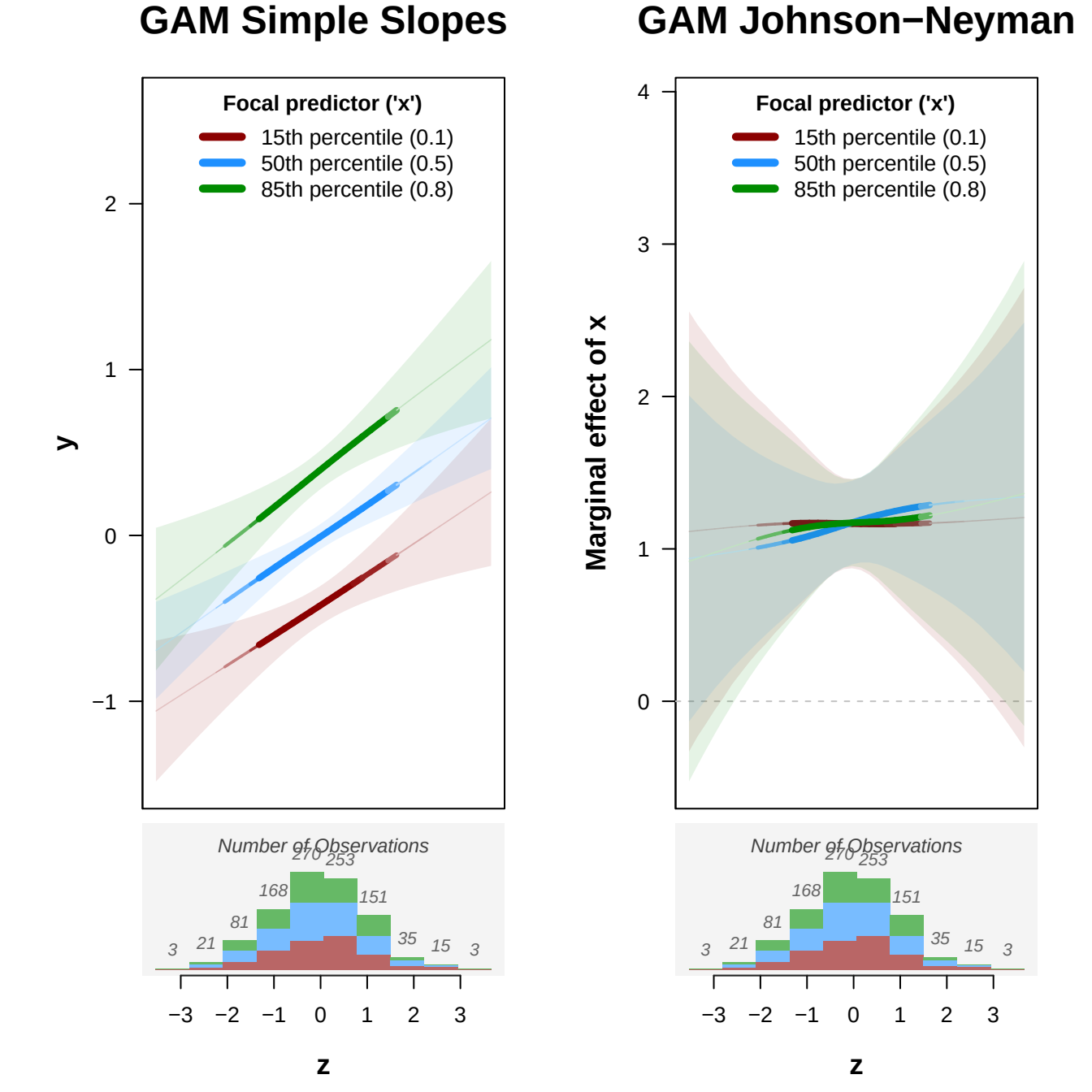

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
y
3
2
1
0
−1
Number of Observations
132
101
85
59
51
27
23
7
9
6
−3
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
Marginal effect of x
3
2
1
0
Number of Observations
132
101
85
59
51
27
23
7
9
6
−3
−2
−1
0
1
2
z

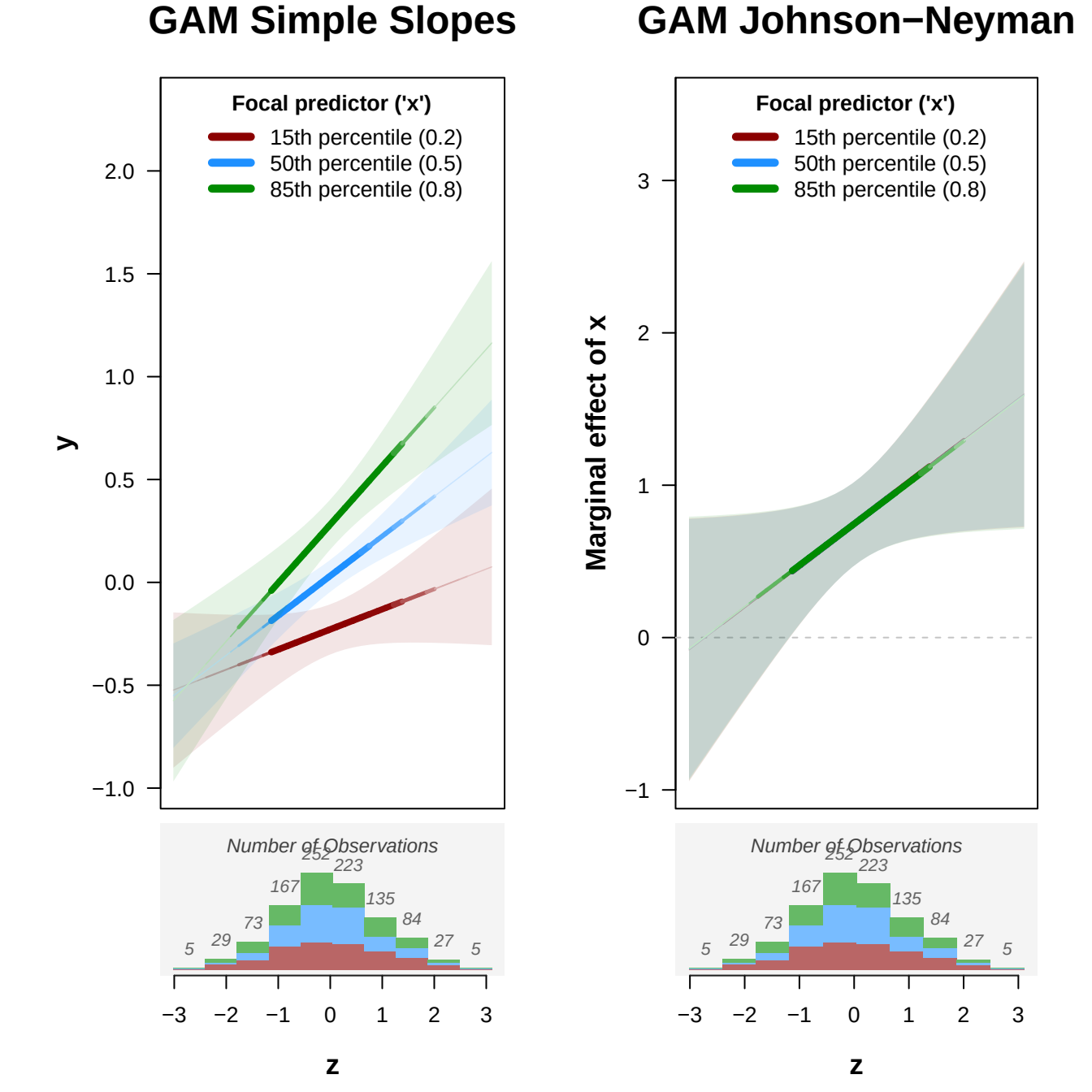

GAM Simple Slopes
Focal predictor ('x')
0
1
y
2
1
0
−1
Number of Observations
316
255
184
113
64
35
20
7
5
1
−4
−2
0
2
z

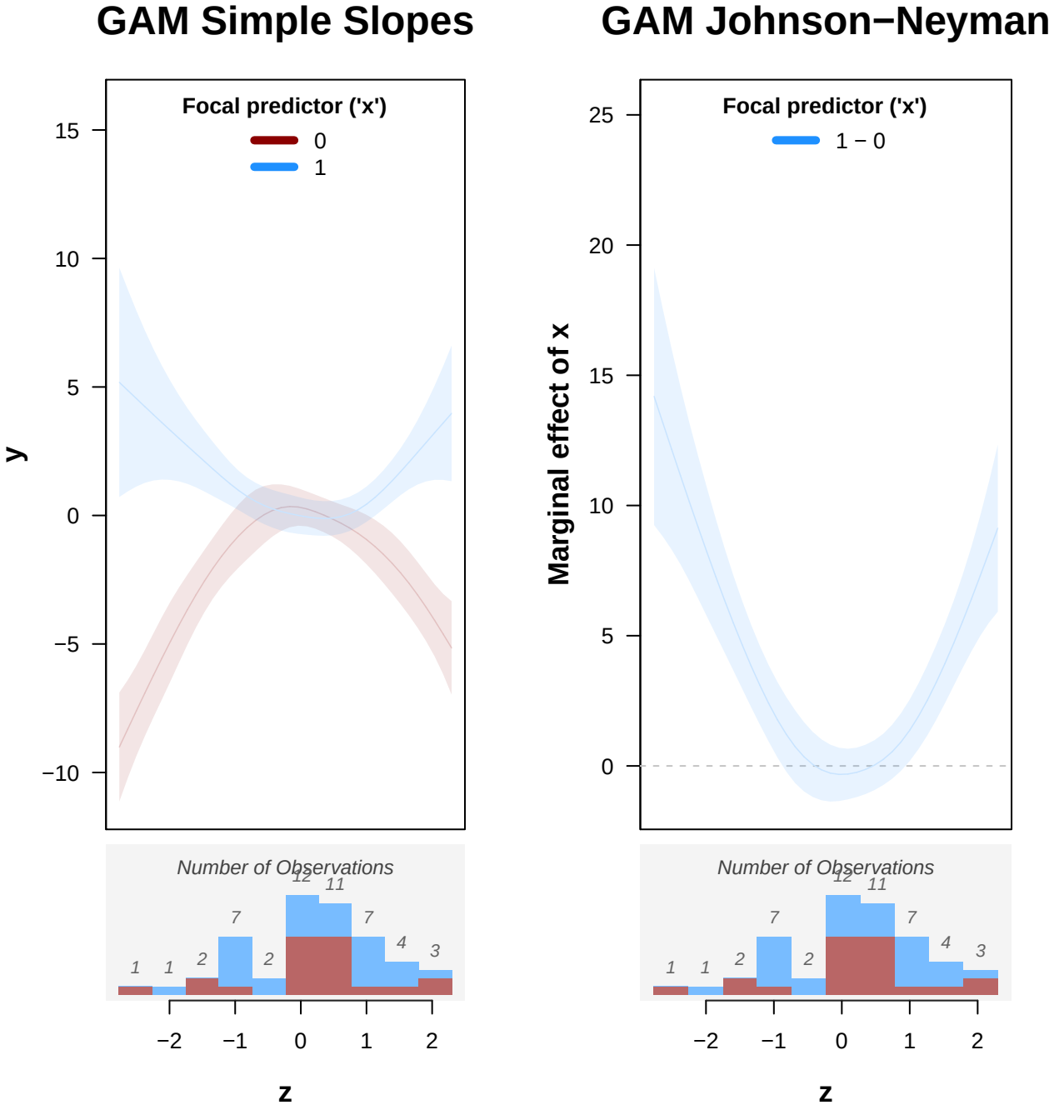

GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
2.0
1.5
1.0
0.5
0.0
Number of Observations
316
255
184
113
64
35
20
7
5
1
−4
−2
0
2
z

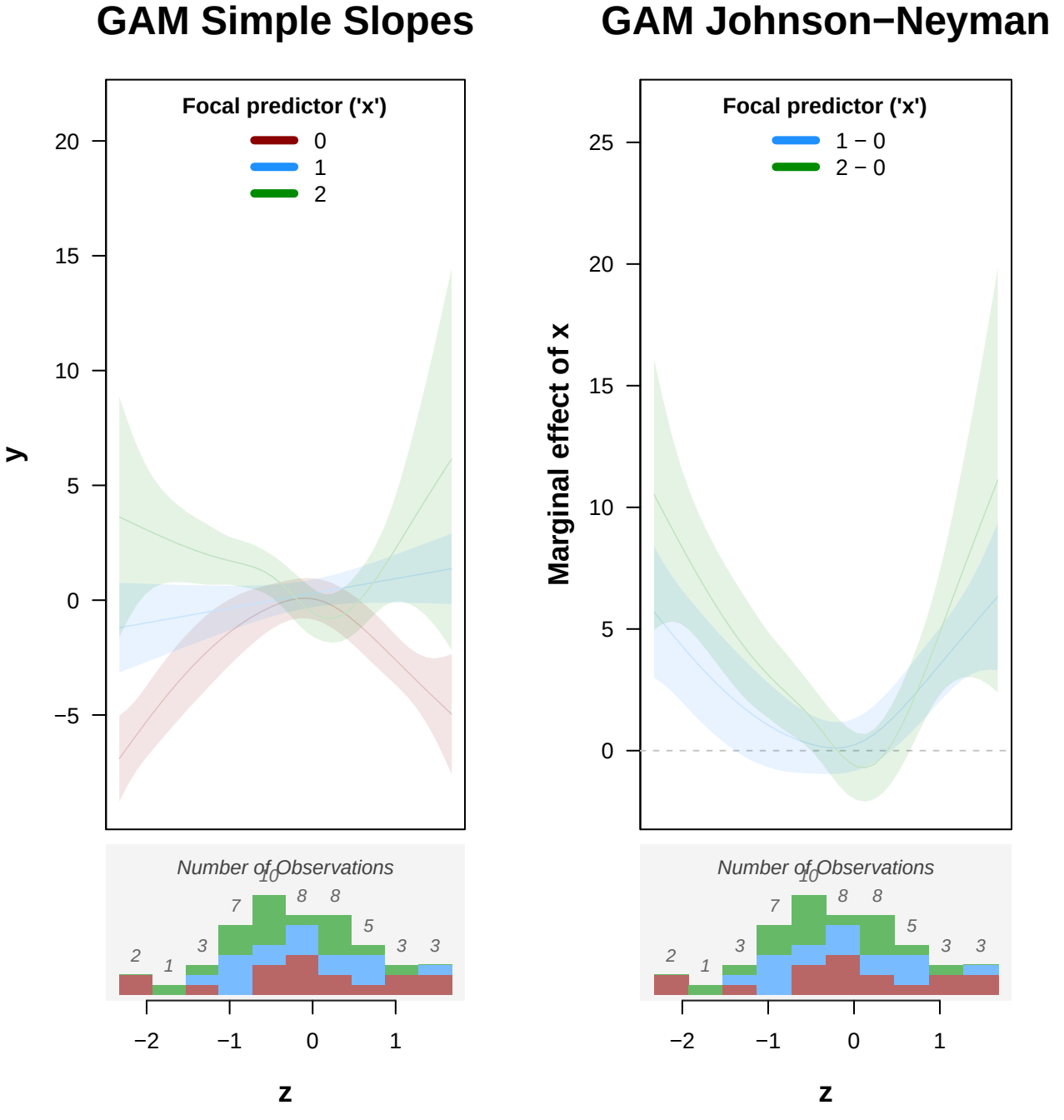

GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
2
1
0
−1
Number of Observations
201
220
212
104
125
63
41
16
12
6
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
3
2
1
0
Number of Observations
201
220
212
104
125
63
41
16
12
6
−2
−1
0
1
2
3
z

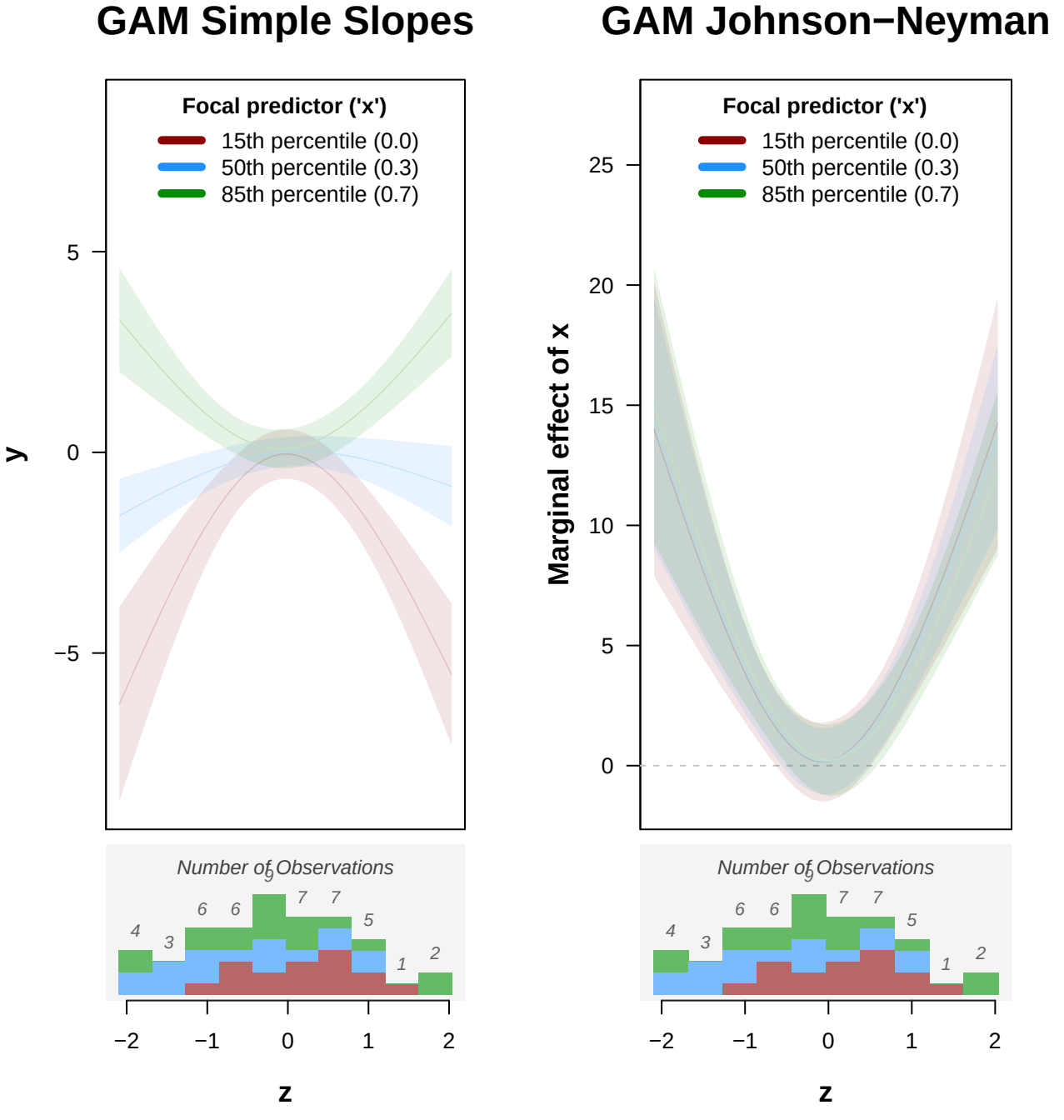

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
y
2
1
0
−1
−2
Number of Observations
4
17
45
124
228
254
204
94
19
11
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
5
4
3
2
1
0
−1
Number of Observations
4
17
45
124
228
254
204
94
19
11
−3
−2
−1
0
1
2
3
z

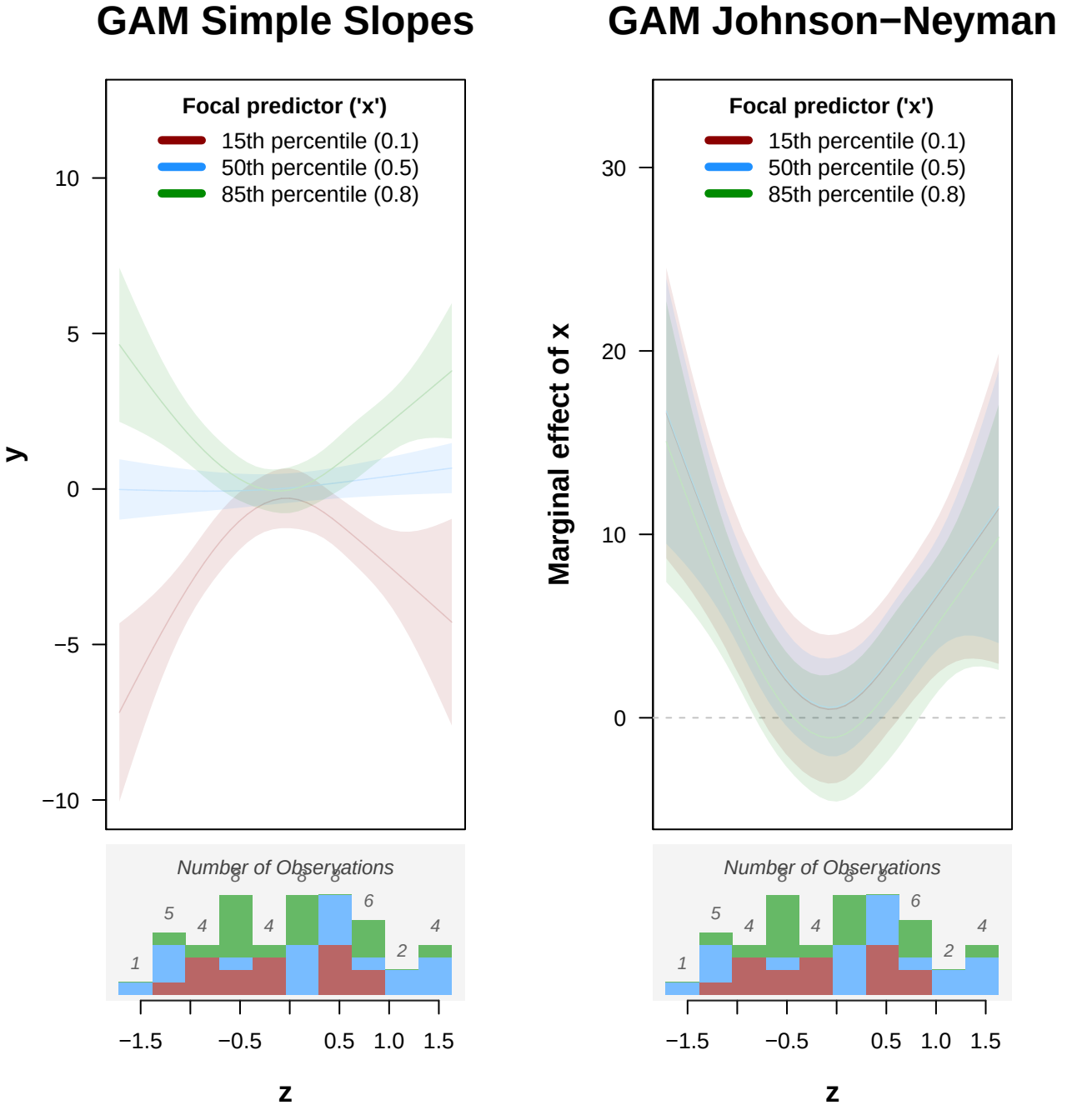

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
y
2
1
0
−1
Number of Observations
270 253
168
151
81
35
15
3
21
3
−3 −2 −1 0 1 2 3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
4
3
2
1
0
Number of Observations
270 253
168
151
81
35
15
3
21
3
−3 −2 −1 0 1 2 3
z

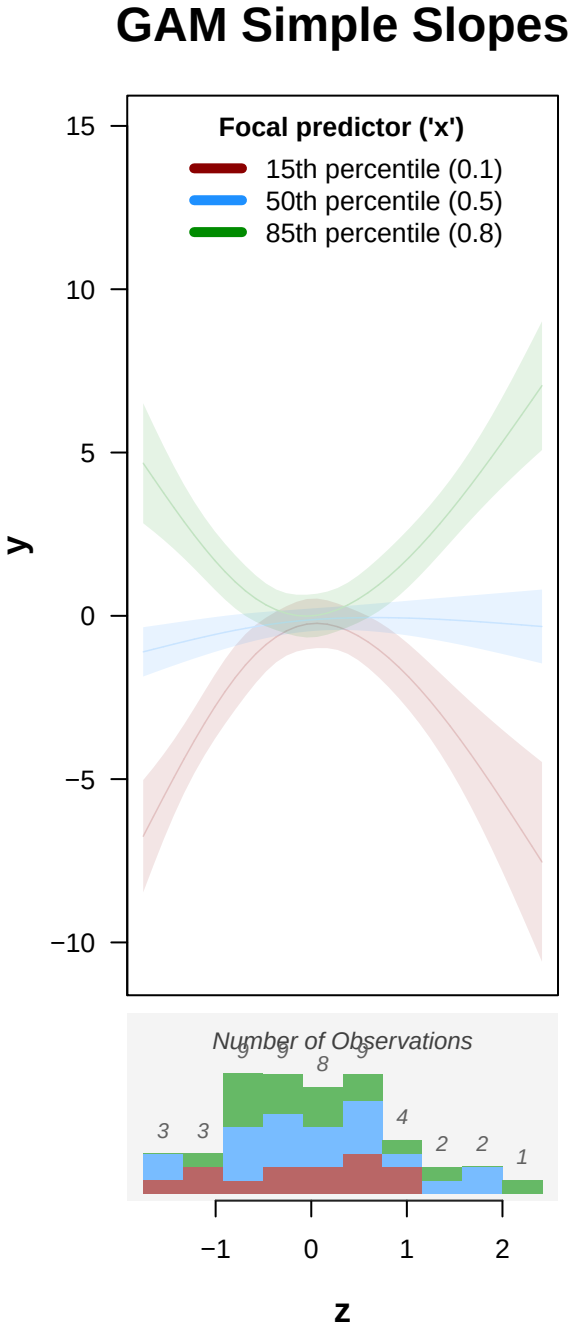

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
y
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
Number of Observations
252
223
167
135
84
73
29
27
5
5
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
3
2
1
0
−1
Number of Observations
252
223
167
135
84
73
29
27
5
5
−3
−2
−1
0
1
2
3
z

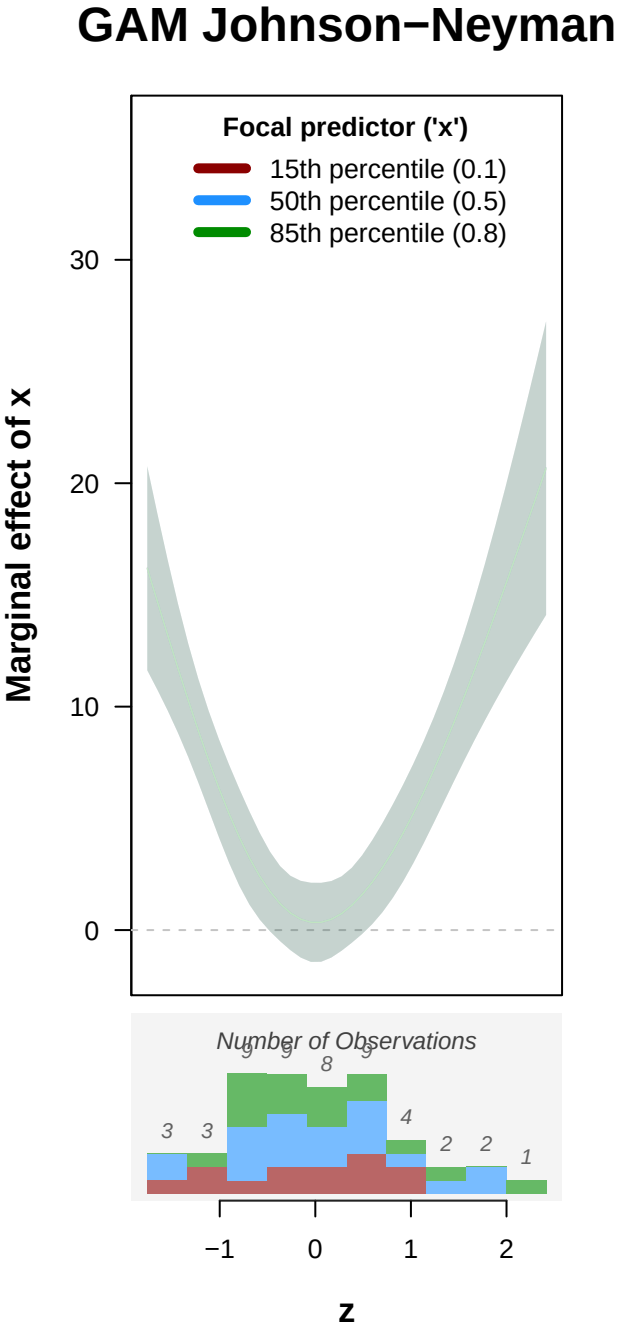

GAM Simple Slopes
Focal predictor ('x')
0
1
y
15
10
5
0
−5
−10
Number of Observations
1
1
2
7
2
12
11
7
4
3
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
25
20
15
10
5
0
Number of Observations
1
1
2
7
2
12
11
7
4
3
−2
−1
0
1
2
z

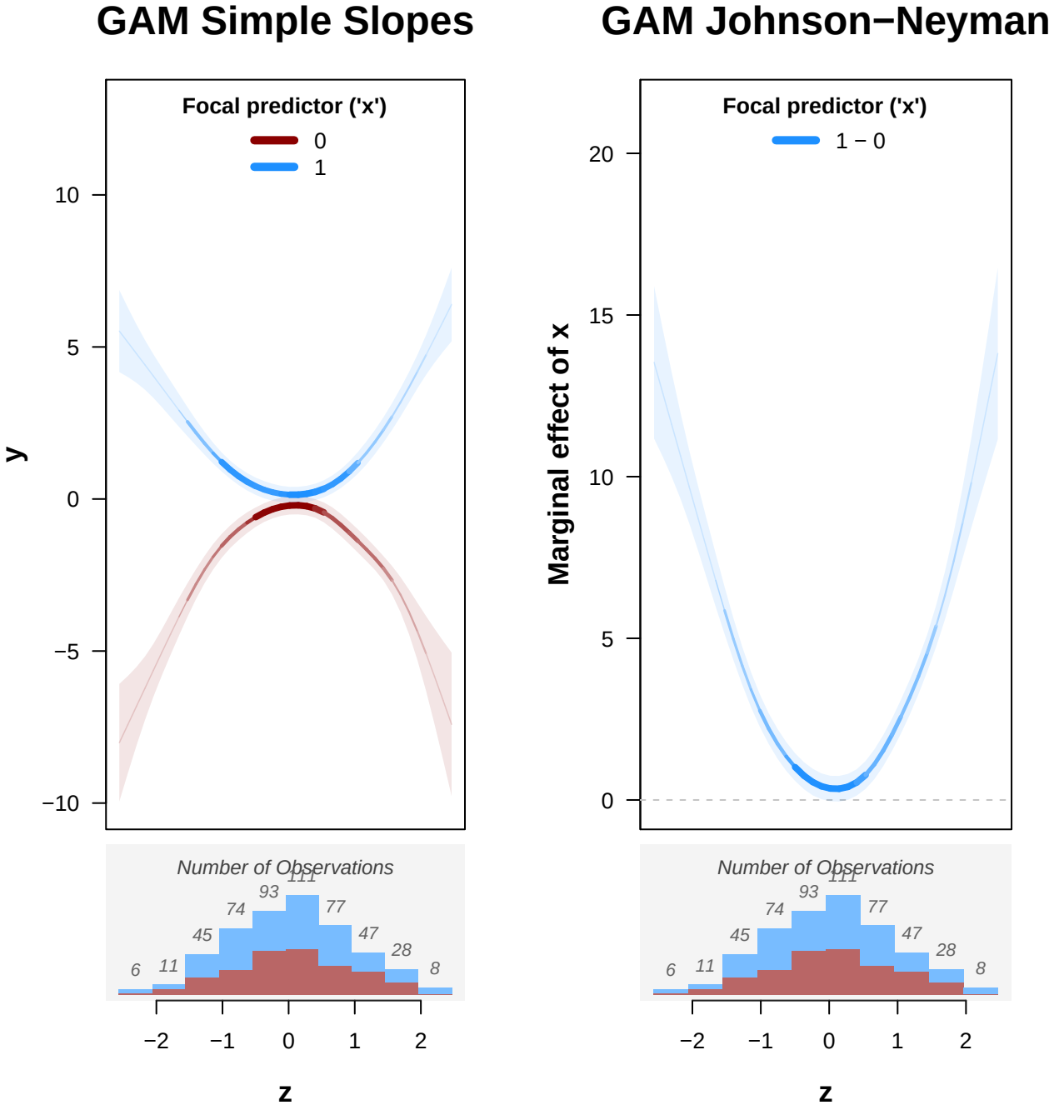

GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
20
15
10
5
0
−5
Number of Observations
10
8
8
7
5
3
3
2
1
3
−2
−1
0
1
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
25
20
15
10
5
0
Number of Observations
10
8
8
7
5
3
3
2
1
3
−2
−1
0
1
z

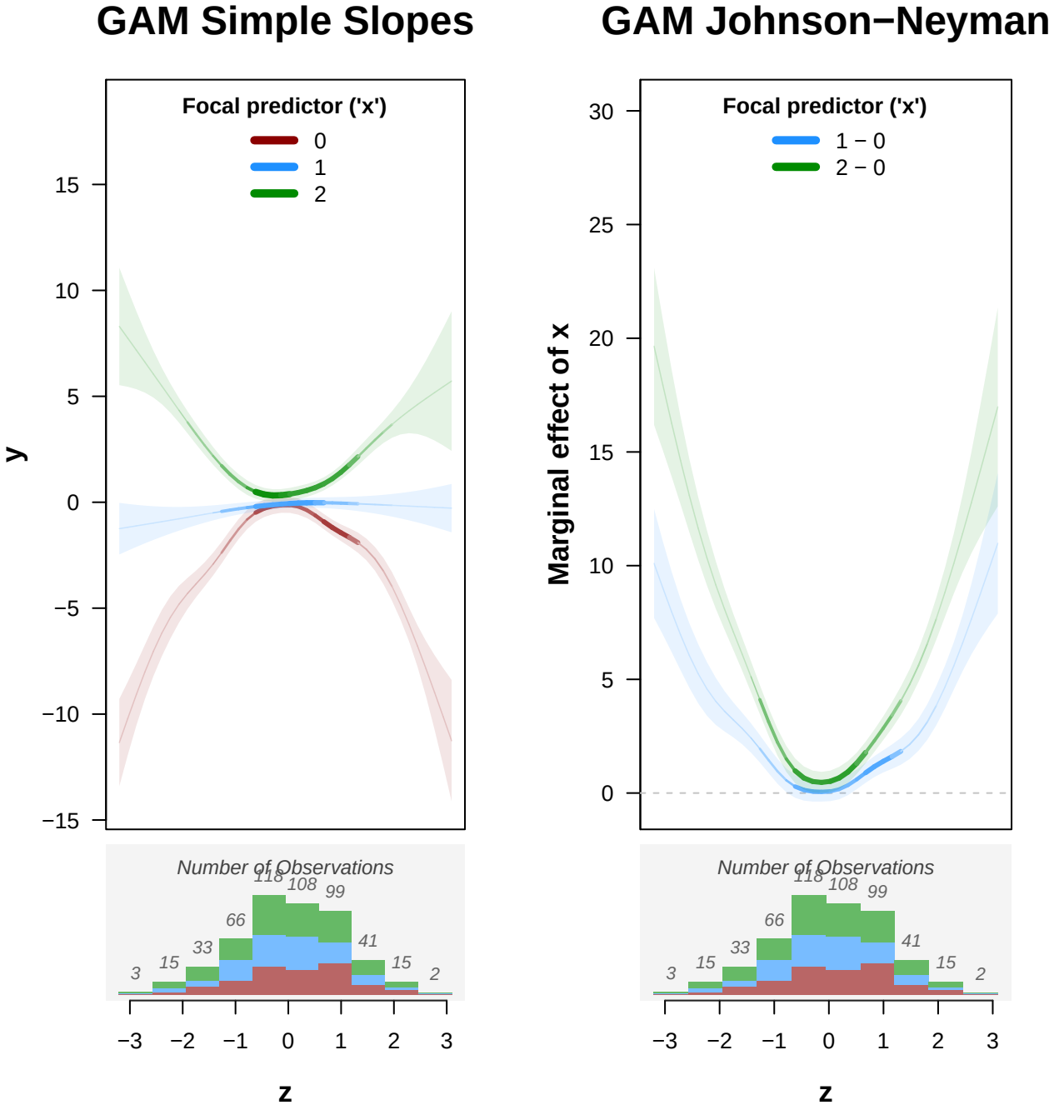

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.3)
85th percentile (0.7)
y
5
0
−5
Number of Observations
9
4
3
6
6
7
7
5
1
2
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.3)
85th percentile (0.7)
Marginal effect of x
25
20
15
10
5
0
Number of Observations
9
4
3
6
6
7
7
5
1
2
−2
−1
0
1
2
z

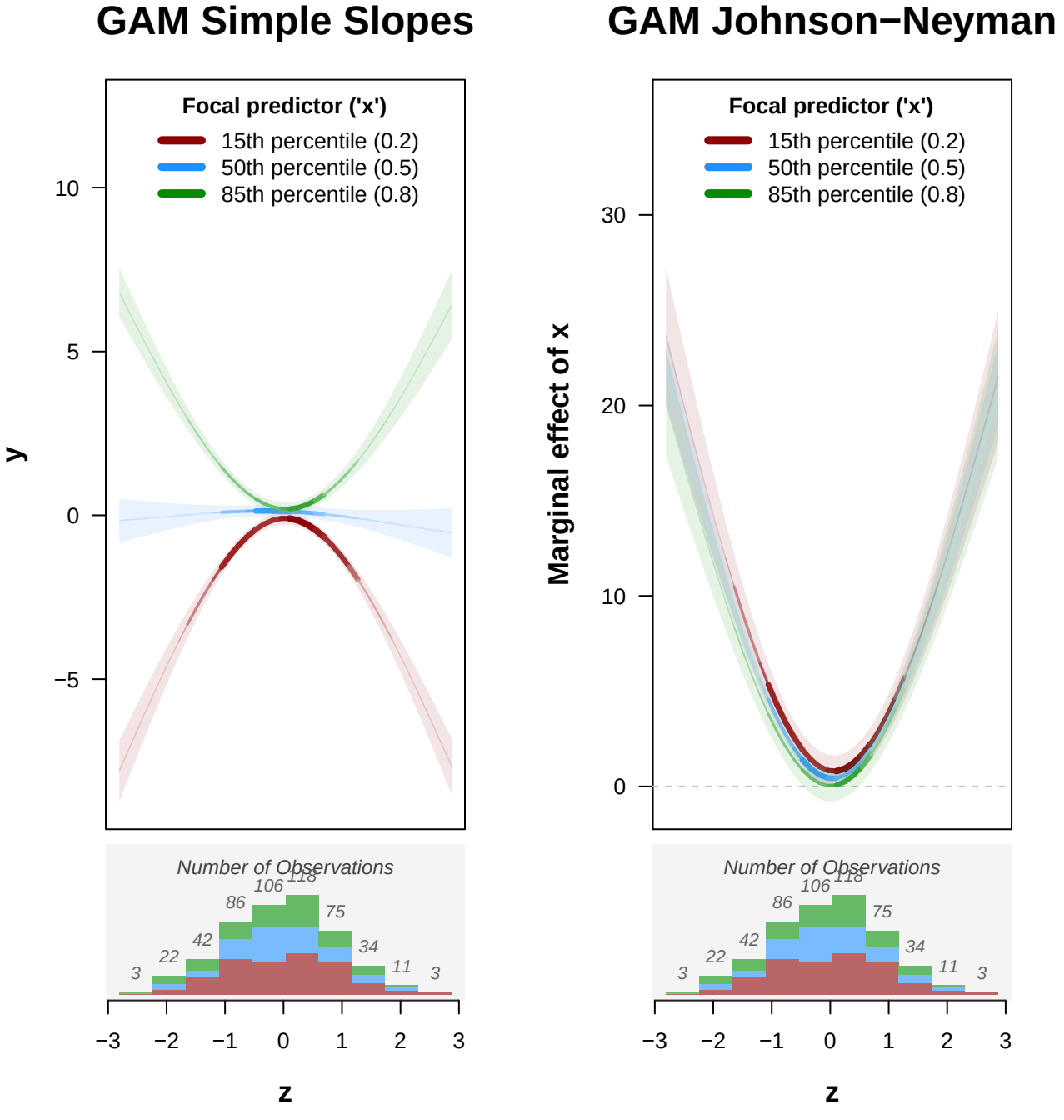

GAM Simple Slopes
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
y
Marginal effect of x
z
Number of Observations
10
5
0
−5
−10
30
20
10
0
−1.5
−0.5
0.5
1.0
1.5
1
5
4
8
4
8
8
6
2
4

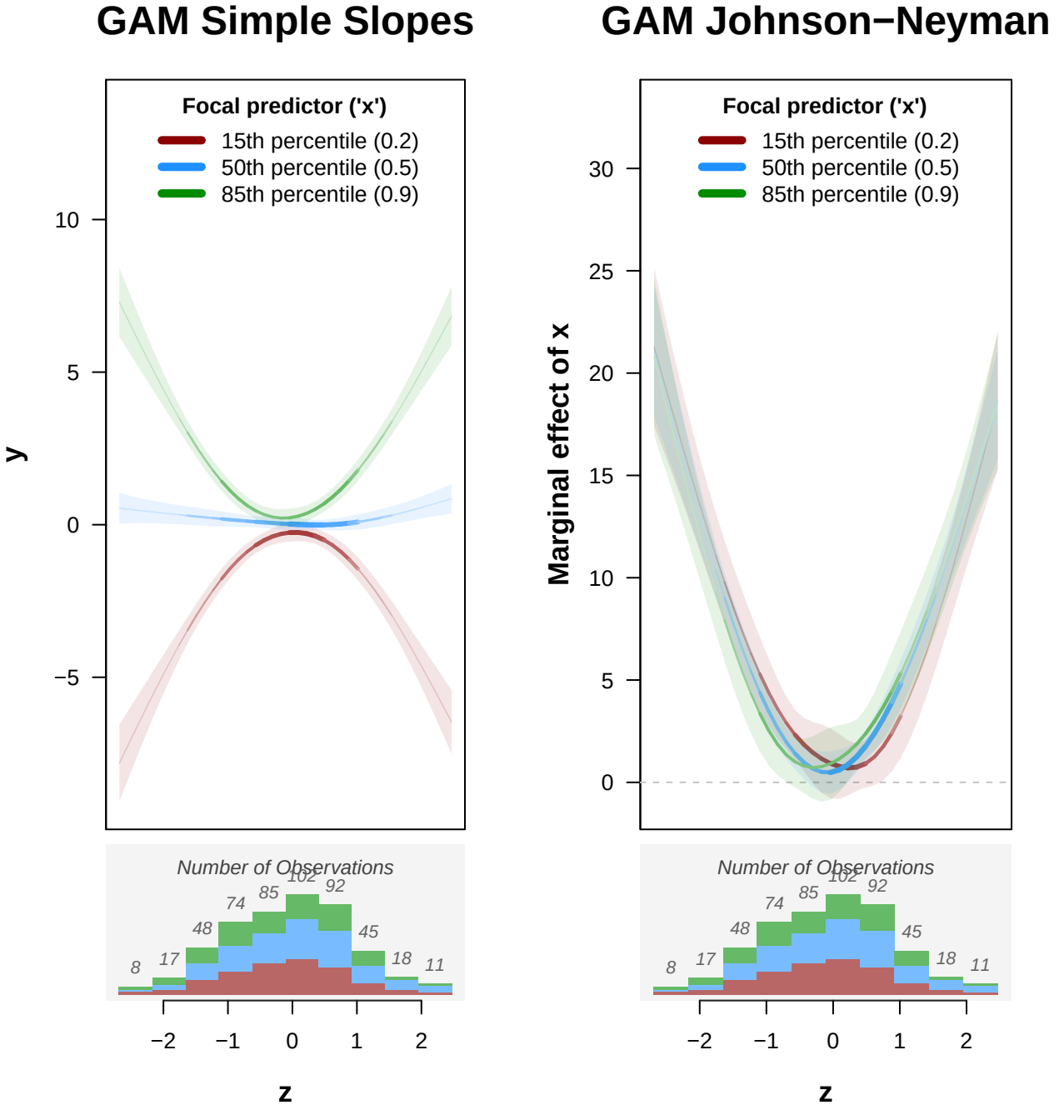

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
y
15
10
5
0
−5
−10
Number of Observations
3 3 9 9 8 9 4 2 2 1
−1 0 1 2
z

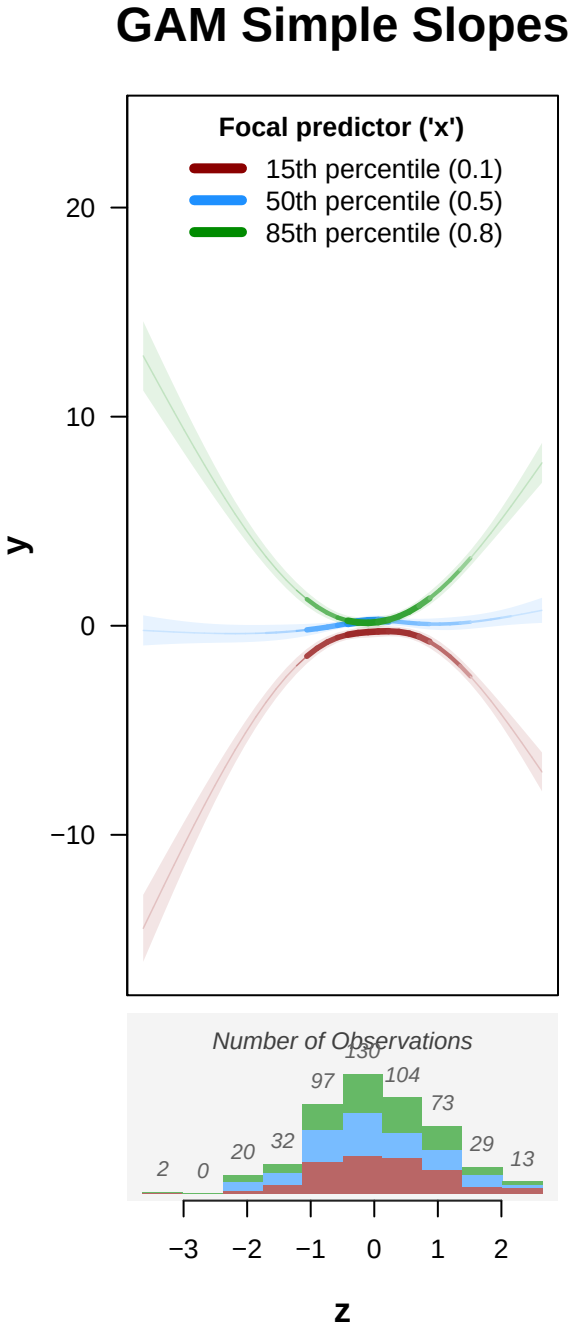

GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
30
20
10
0
Number of Observations
3 3 9 9 8 9 4 2 2 1
−1 0 1 2
z

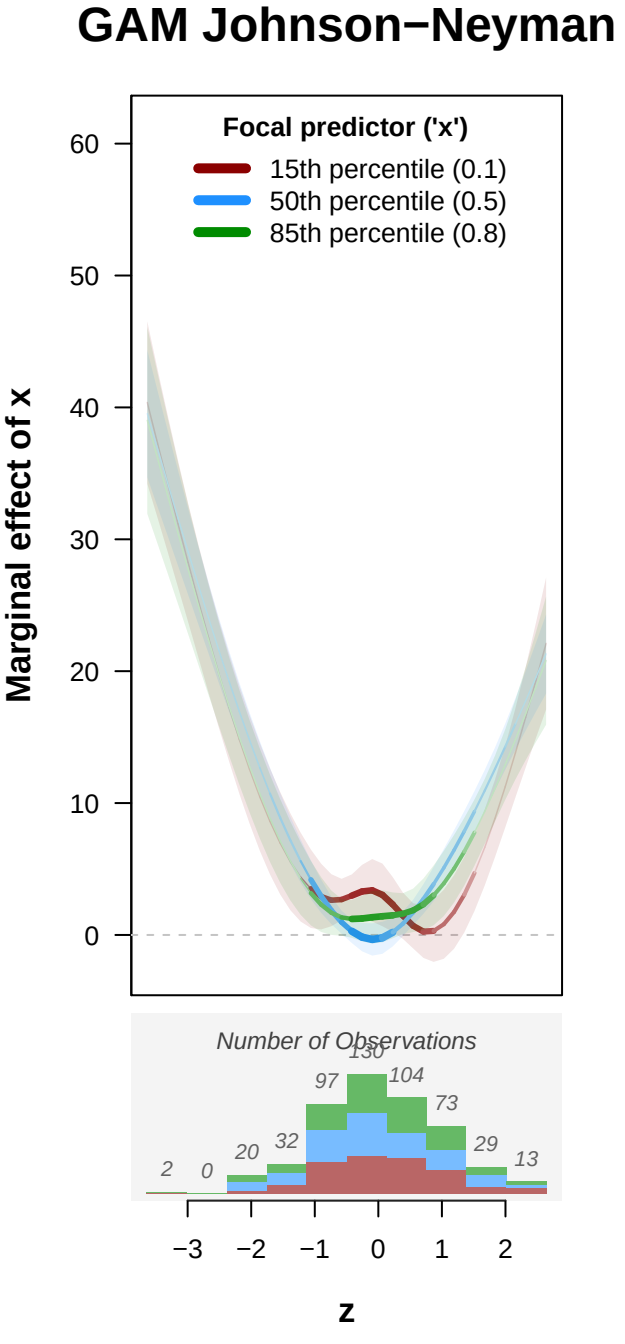

GAM Simple Slopes
Focal predictor ('x')
0
1
y
10
5
0
−5
−10
Number of Observations
6
11
45
74
93
111
77
47
28
8
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
20
15
10
5
0
Number of Observations
6
11
45
74
93
111
77
47
28
8
−2
−1
0
1
2
z

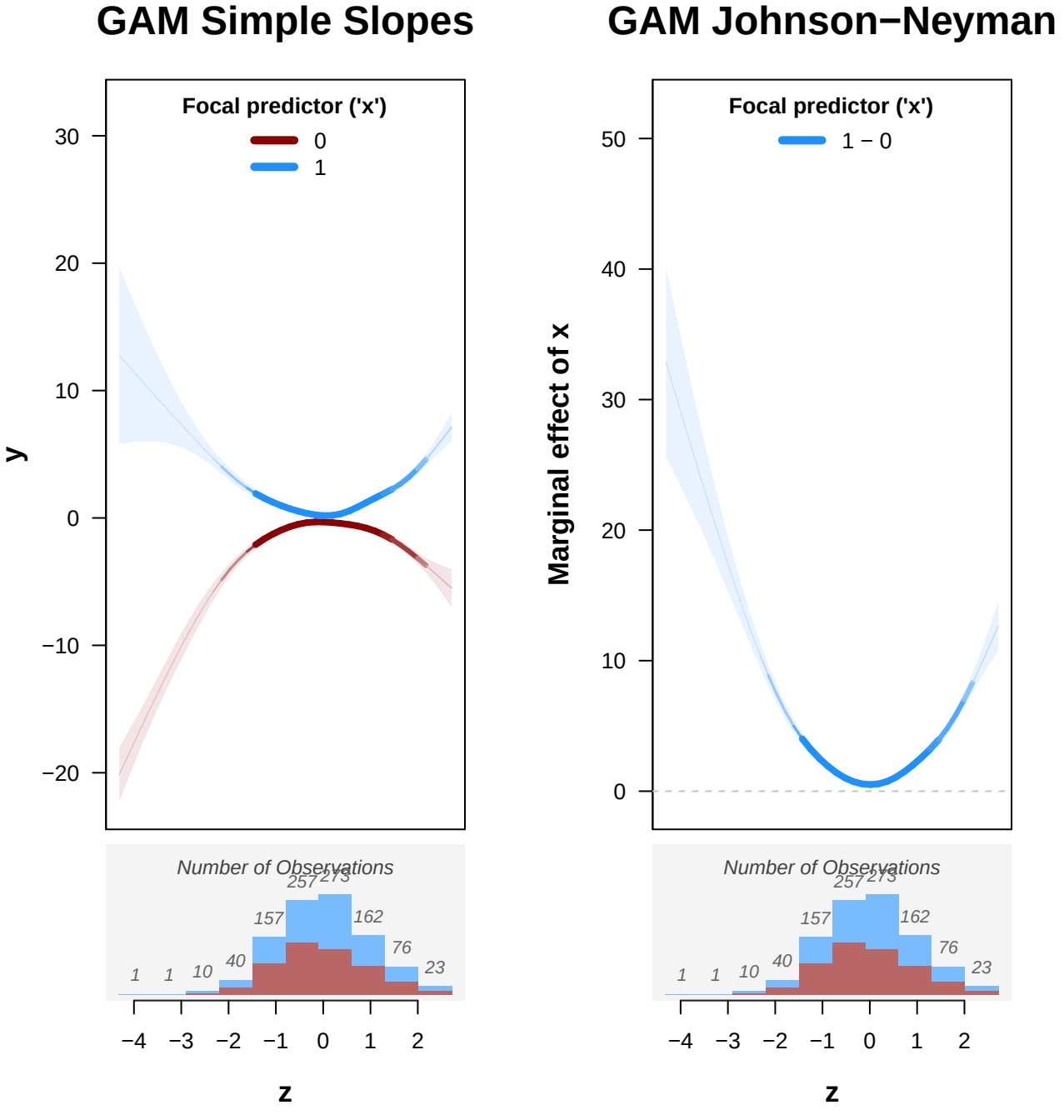

GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
15
10
5
0
−5
−10
−15
Number of Observations
3
15
33
66
118
108
99
41
15
2
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
30
25
20
15
10
5
0
Number of Observations
3
15
33
66
118
108
99
41
15
2
−3
−2
−1
0
1
2
3
z

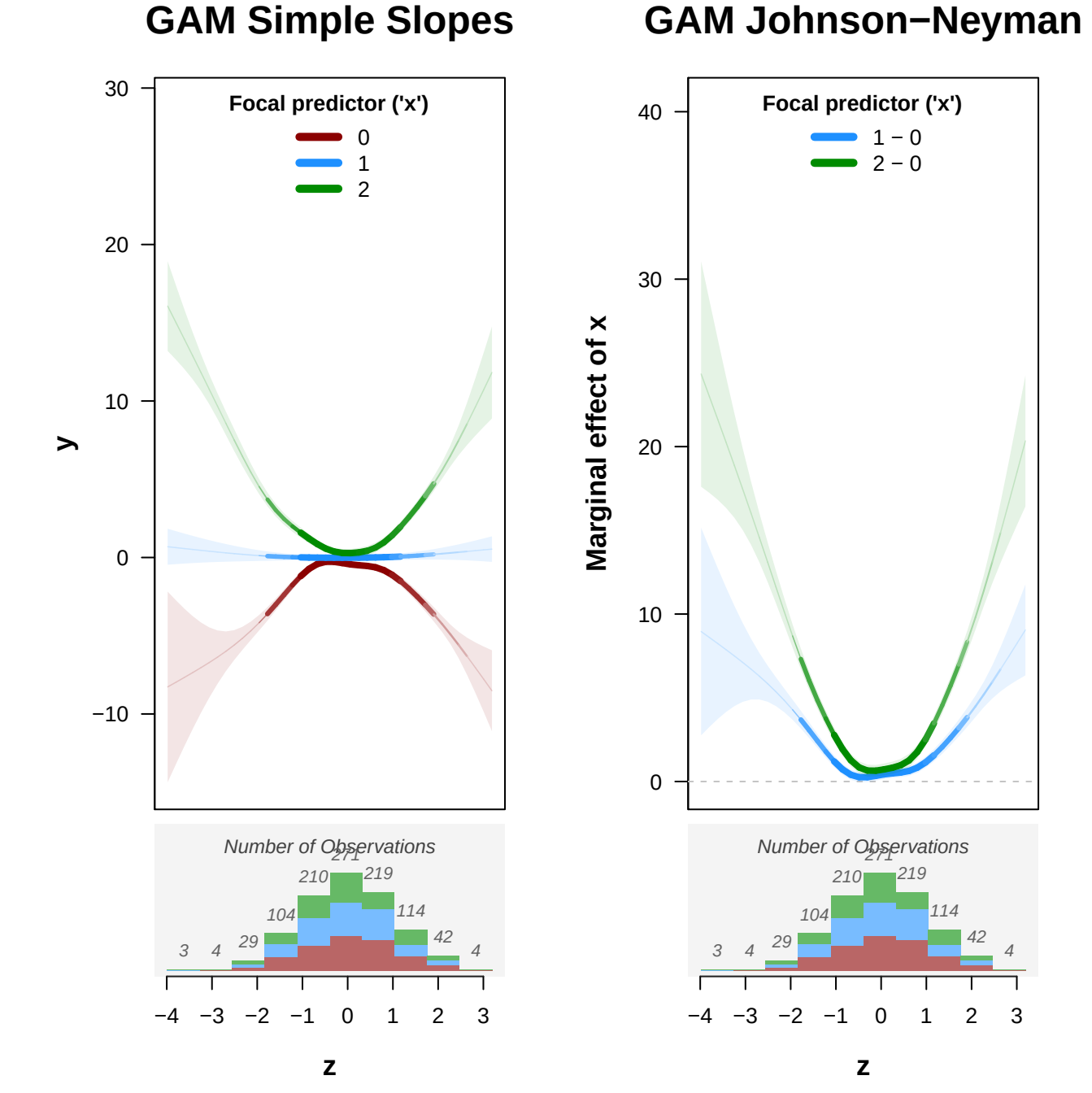

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
y
10
5
0
−5
Number of Observations
3
22
42
86
106
118
75
34
11
3
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
30
20
10
0
Number of Observations
3
22
42
86
106
118
75
34
11
3
−3
−2
−1
0
1
2
3
z

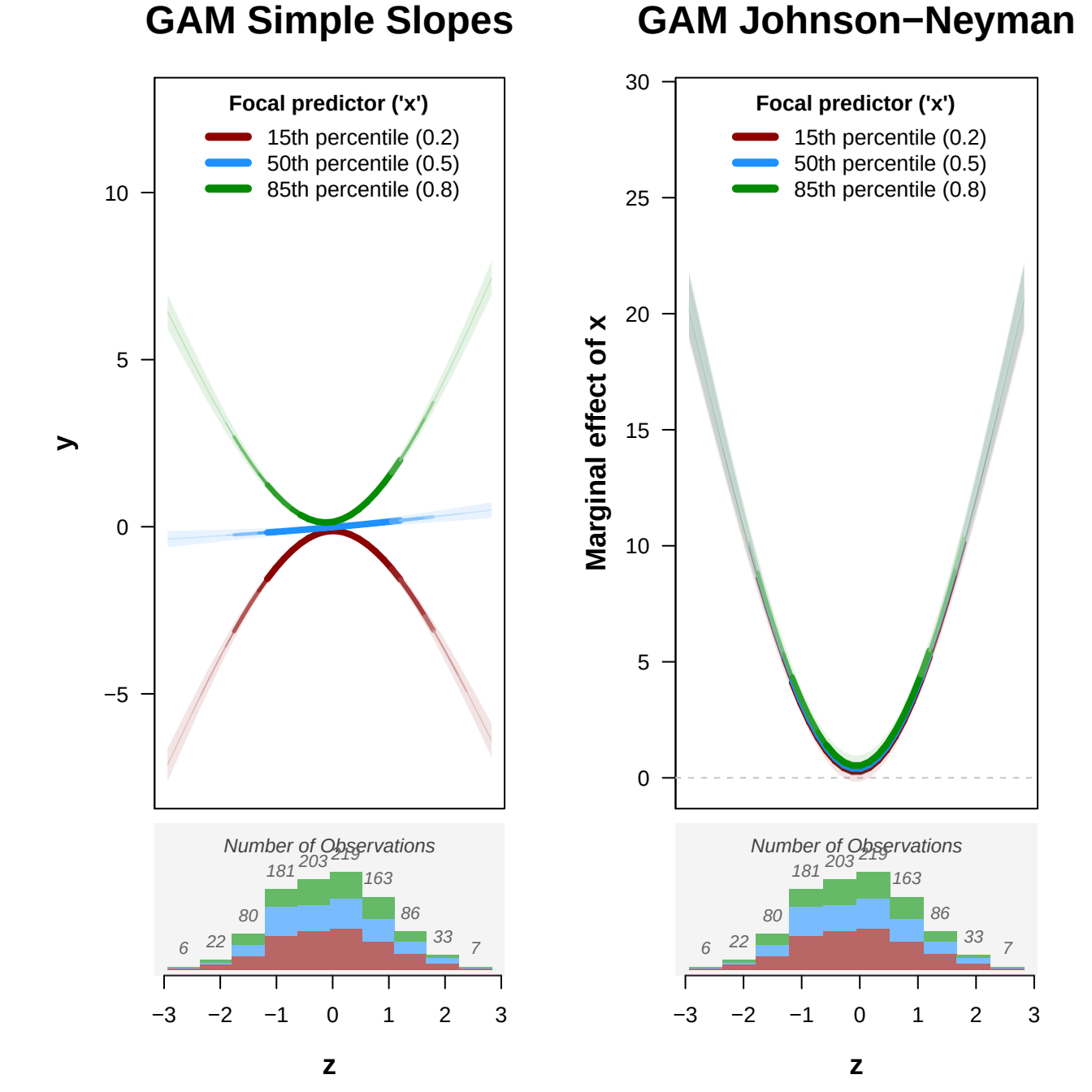

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
y
10
5
0
−5
Number of Observations
8
17
48
74
85
102
92
45
18
11
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
Marginal effect of x
30
25
20
15
10
5
0
Number of Observations
8
17
48
74
85
102
92
45
18
11
−2
−1
0
1
2
z

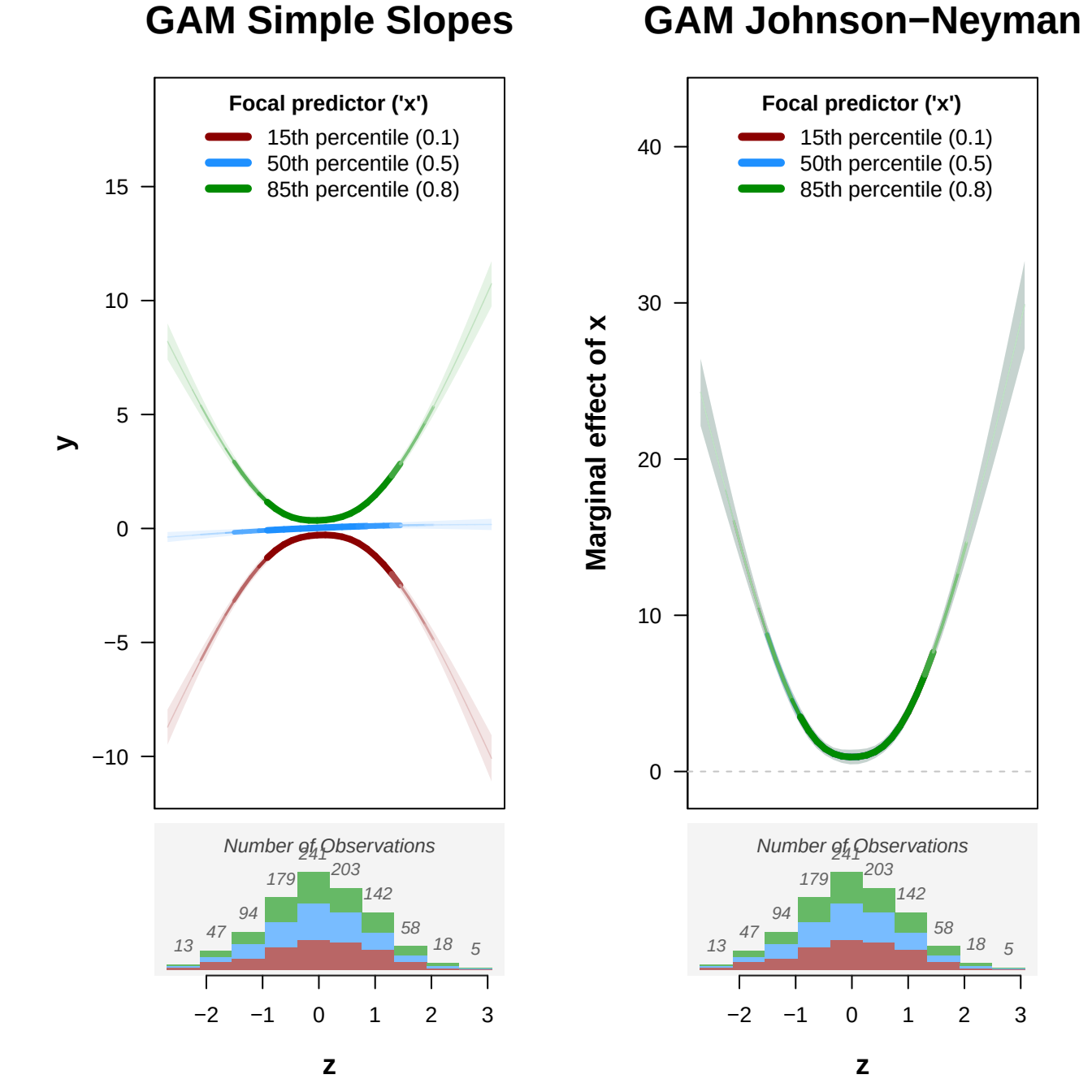

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
y
20
10
0
−10
Number of Observations
2
0
20
32
97
130
104
73
29
13
−3
−2
−1
0
1
2
z

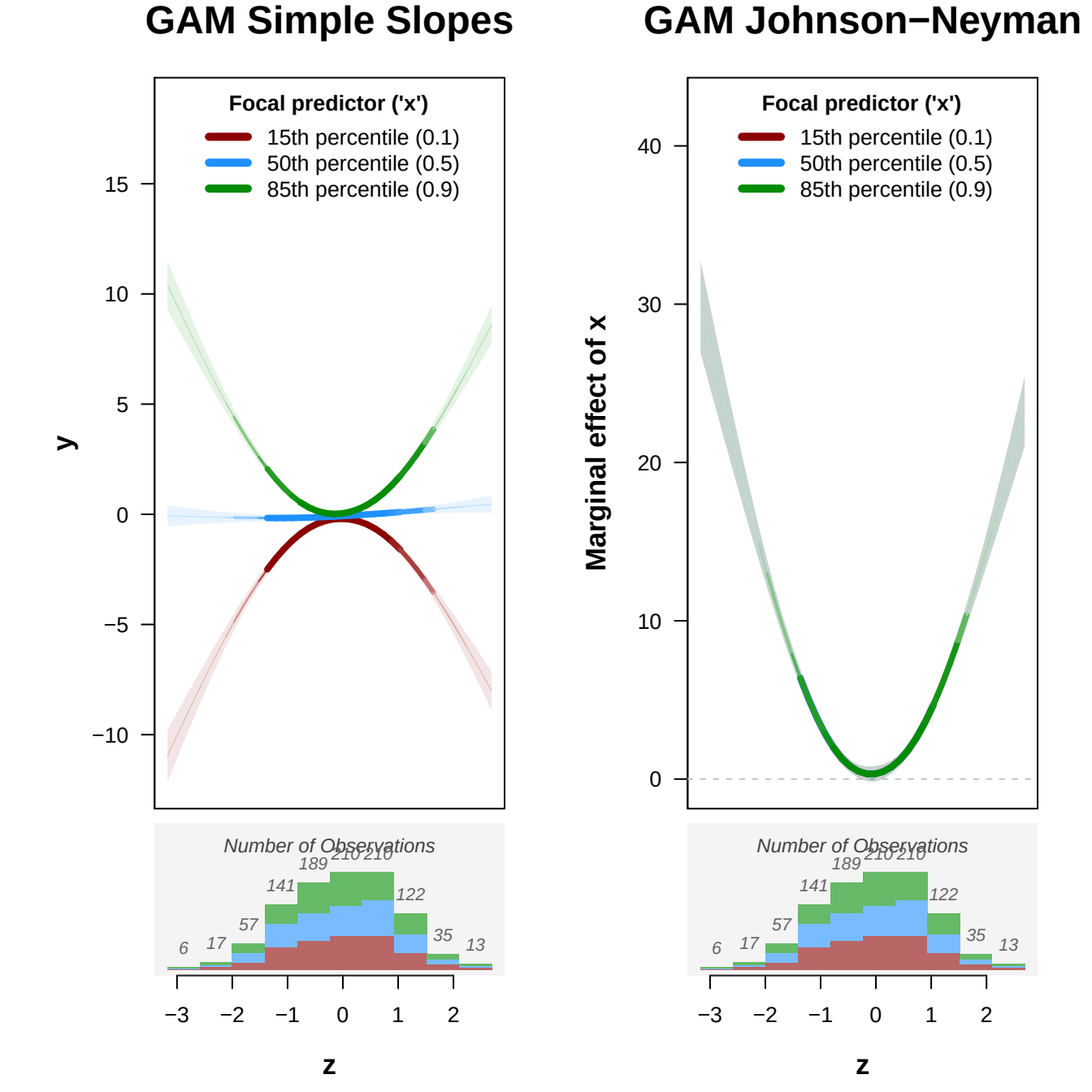

GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
60
50
40
30
20
10
0
Number of Observations
2
0
20
32
97
130
104
73
29
13
−3
−2
−1
0
1
2
z

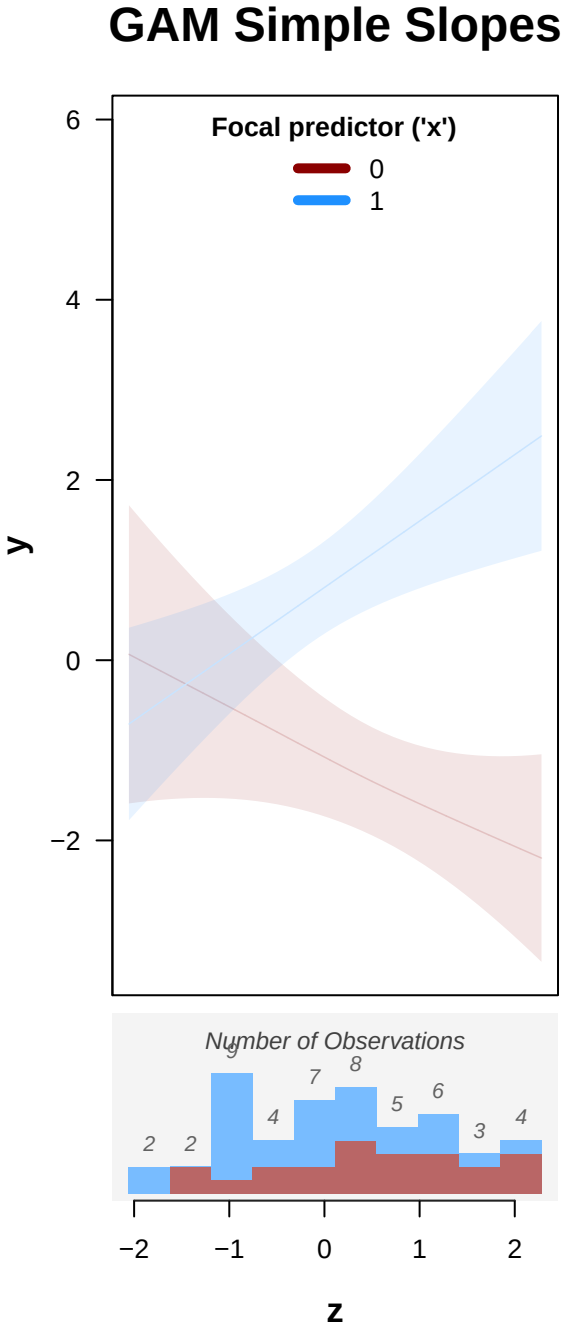

GAM Simple Slopes
Focal predictor ('x')
0
1
y
30
20
10
0
−10
−20
Number of Observations
1
1
10
40
157
257
273
162
76
23
−4
−3
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
50
40
30
20
10
0
Number of Observations
1
1
10
40
157
257
273
162
76
23
−4
−3
−2
−1
0
1
2
z

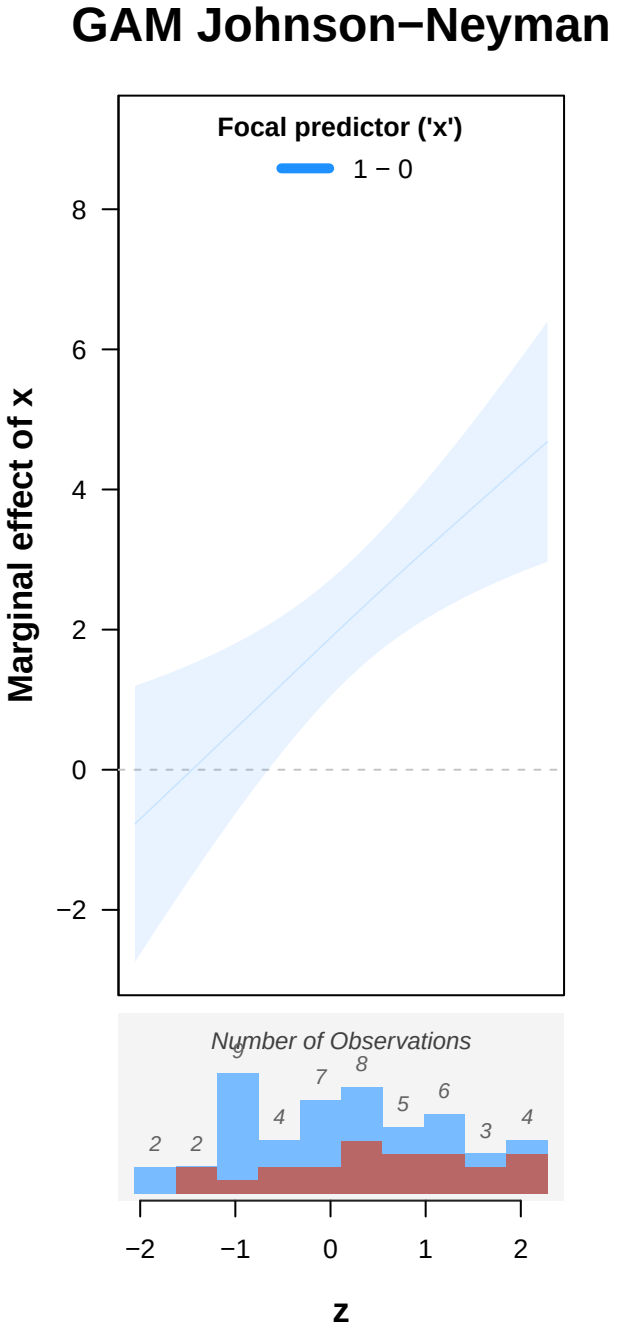

GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
30
20
10
0
−10
Number of Observations
271
210
219
104
114
42
29
3
4
4
−4
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
40
30
20
10
0
Number of Observations
271
210
219
104
114
42
29
3
4
4
−4
−3
−2
−1
0
1
2
3
z

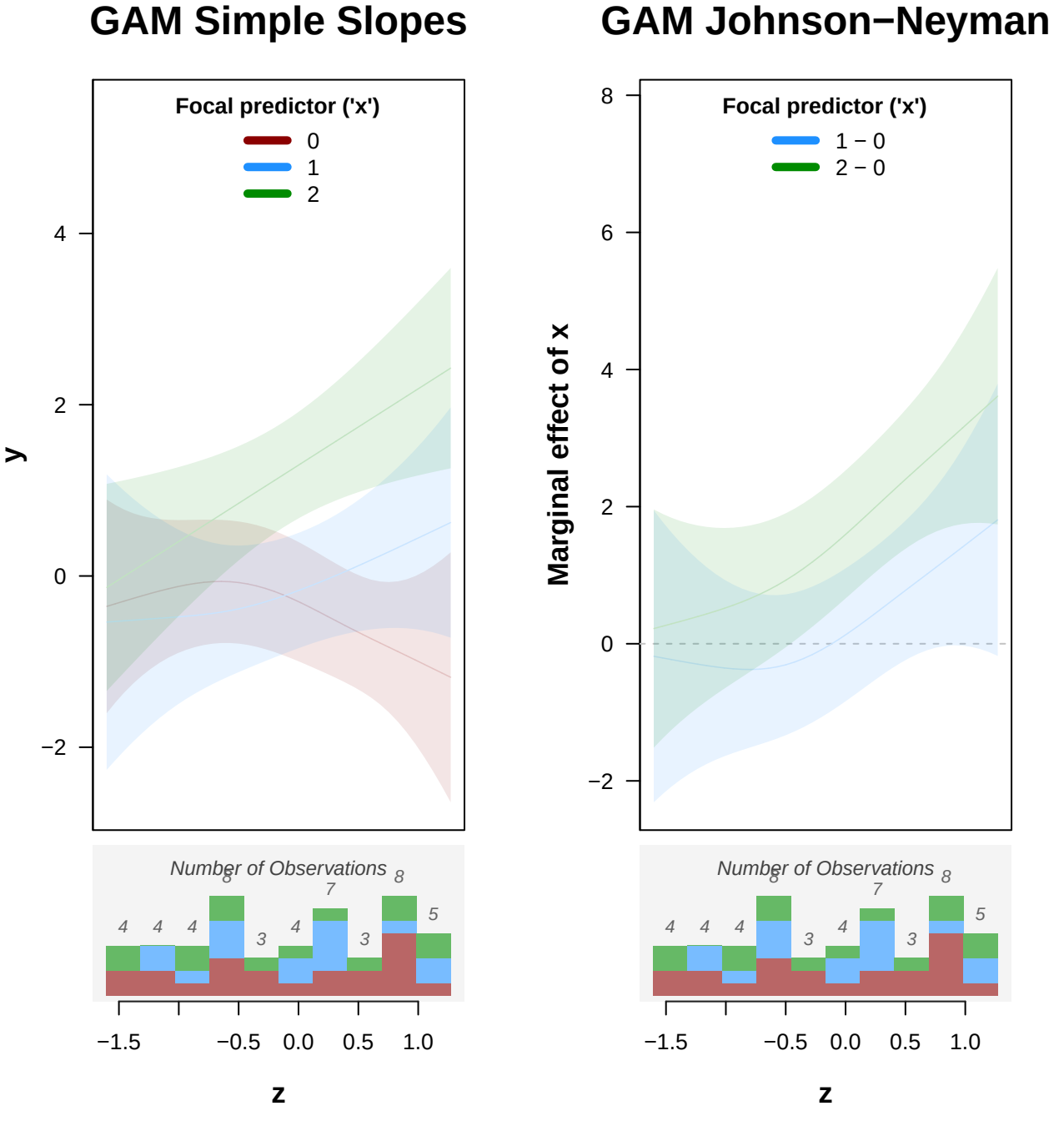

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
y
10
5
0
−5
Number of Observations
6
22
80
181
203
219
163
86
33
7
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
30
25
20
15
10
5
0
Number of Observations
6
22
80
181
203
219
163
86
33
7
−3
−2
−1
0
1
2
3
z

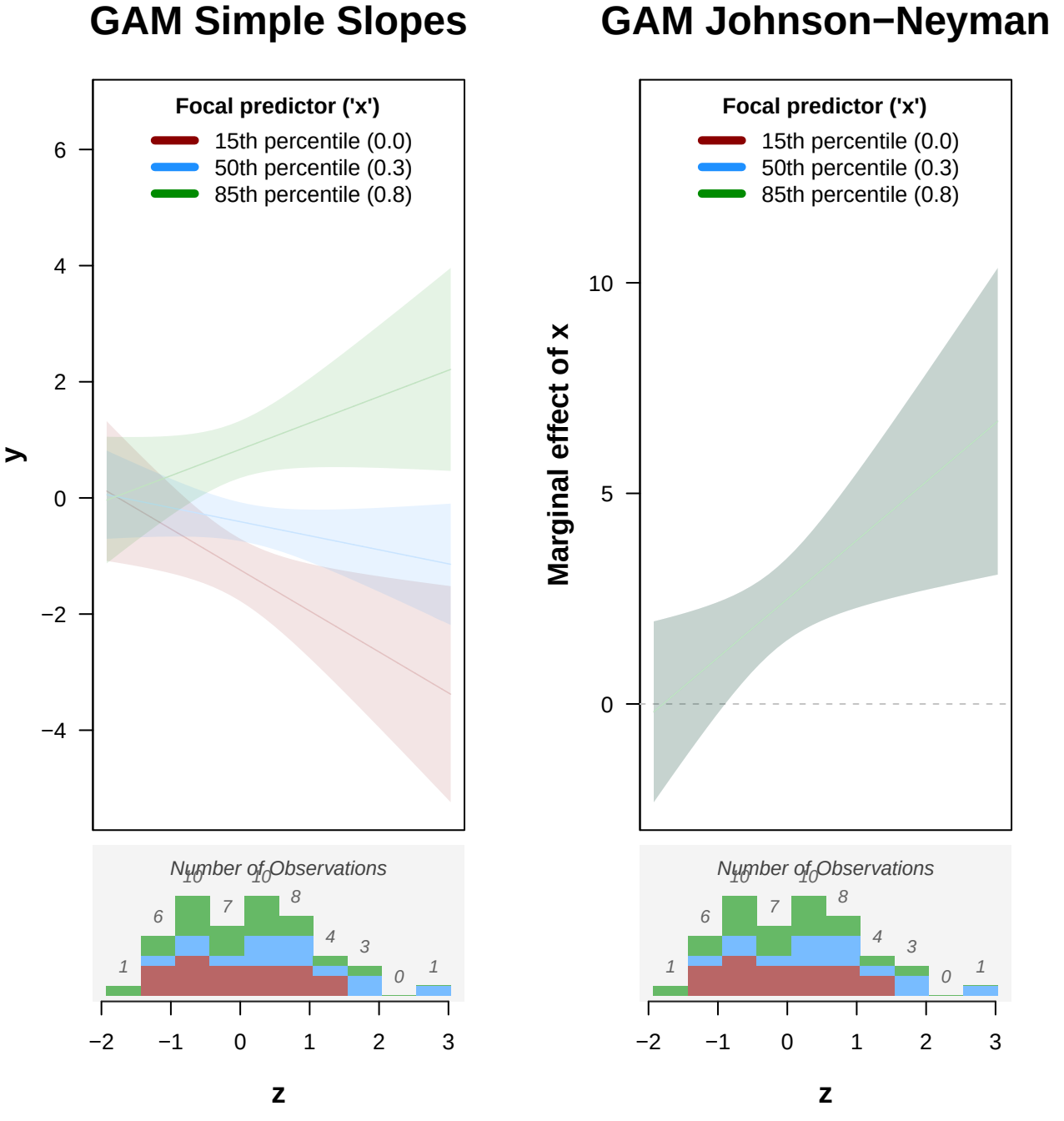

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
y
15
10
5
0
−5
−10
Number of Observations
241
203
179
142
94
58
47
18
13
5
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
40
30
20
10
0
Number of Observations
241
203
179
142
94
58
47
18
13
5
−2
−1
0
1
2
3
z

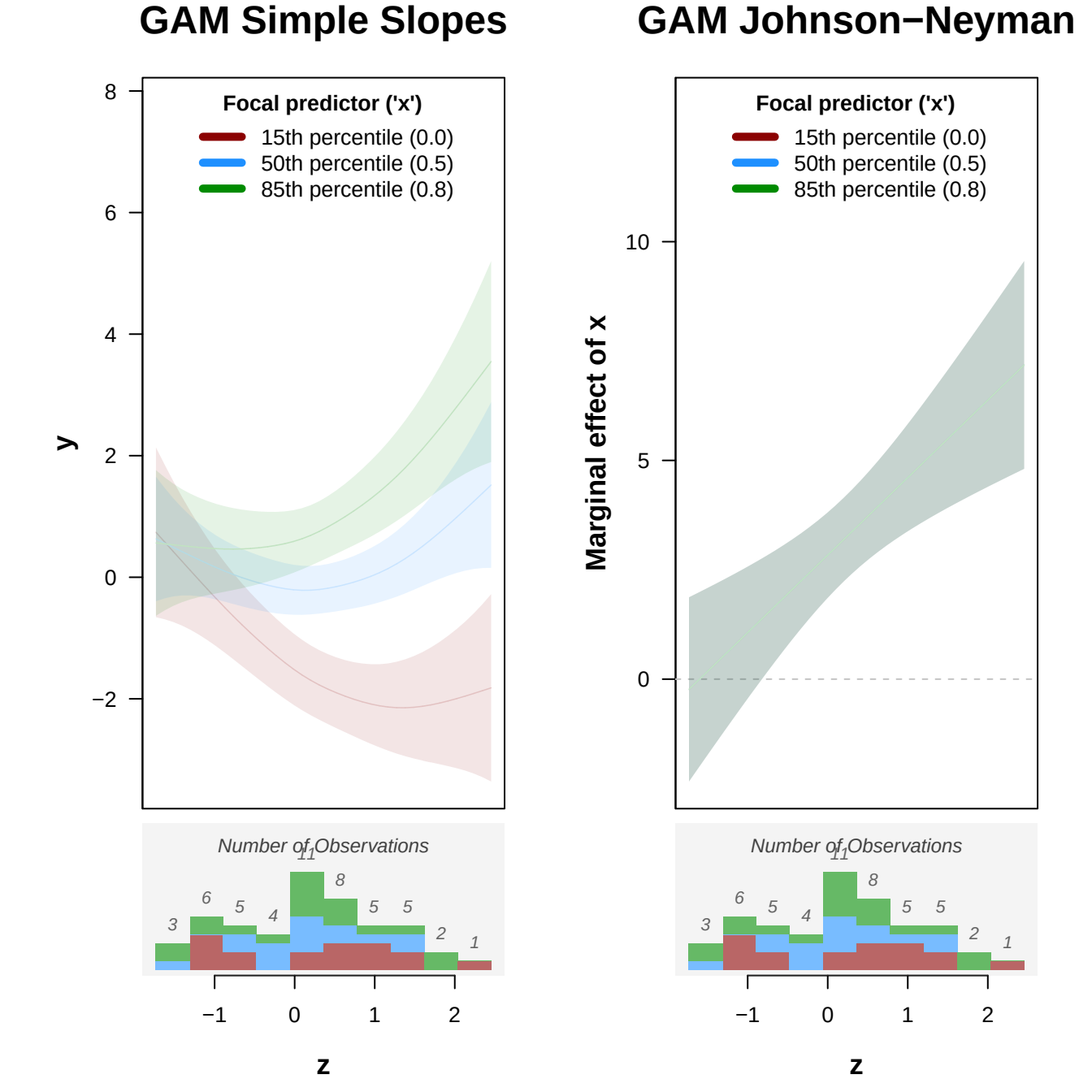

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.9)
y
15
10
5
0
−5
−10
Number of Observations
6
17
57
141
189
210
210
122
35
13
−3
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.9)
Marginal effect of x
40
30
20
10
0
Number of Observations
6
17
57
141
189
210
210
122
35
13
−3
−2
−1
0
1
2
z

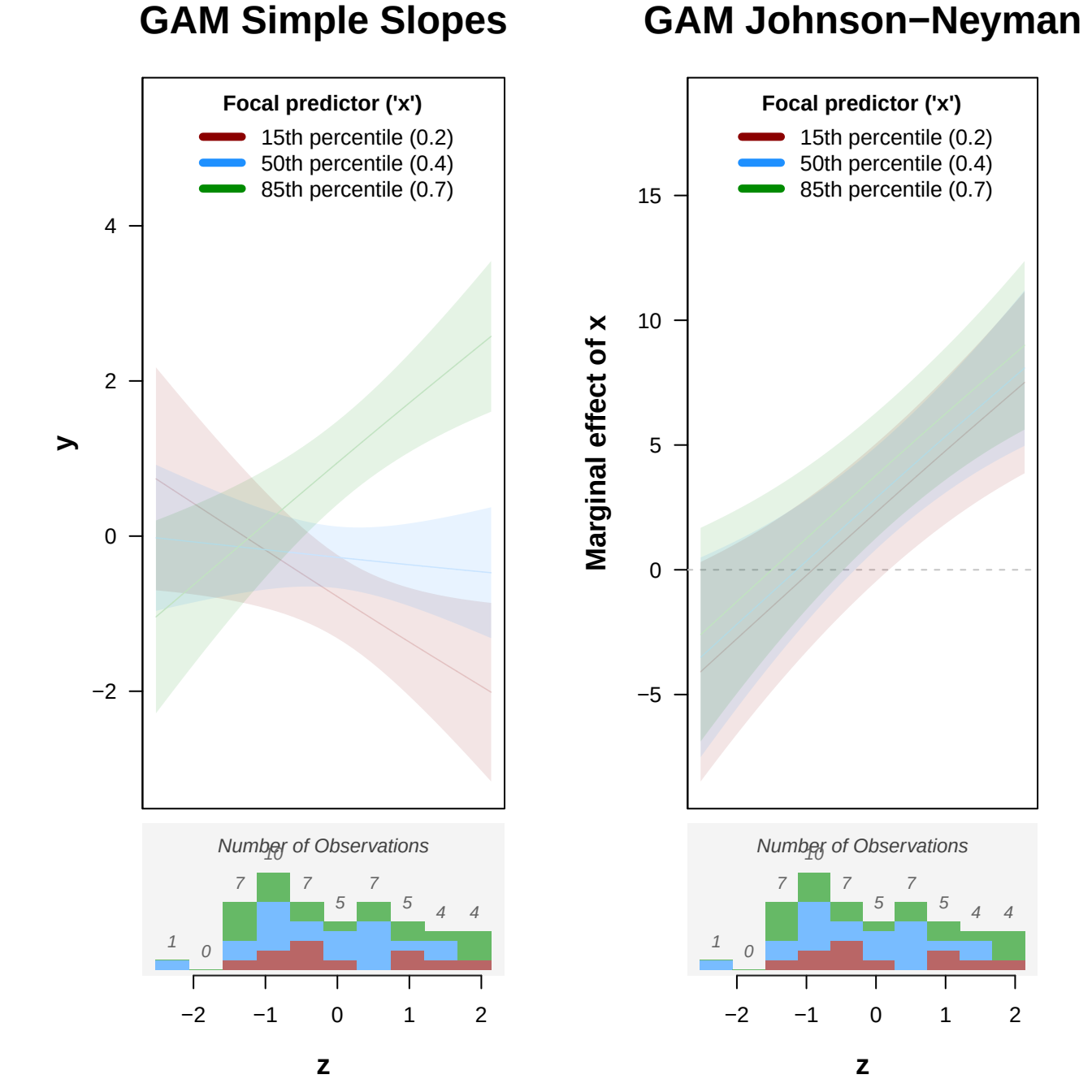

GAM Simple Slopes
Focal predictor ('x')
0
1
y
6
4
2
0
−2
Number of Observations
2 2 4 7 8 5 6 3 4
−2 −1 0 1 2
z

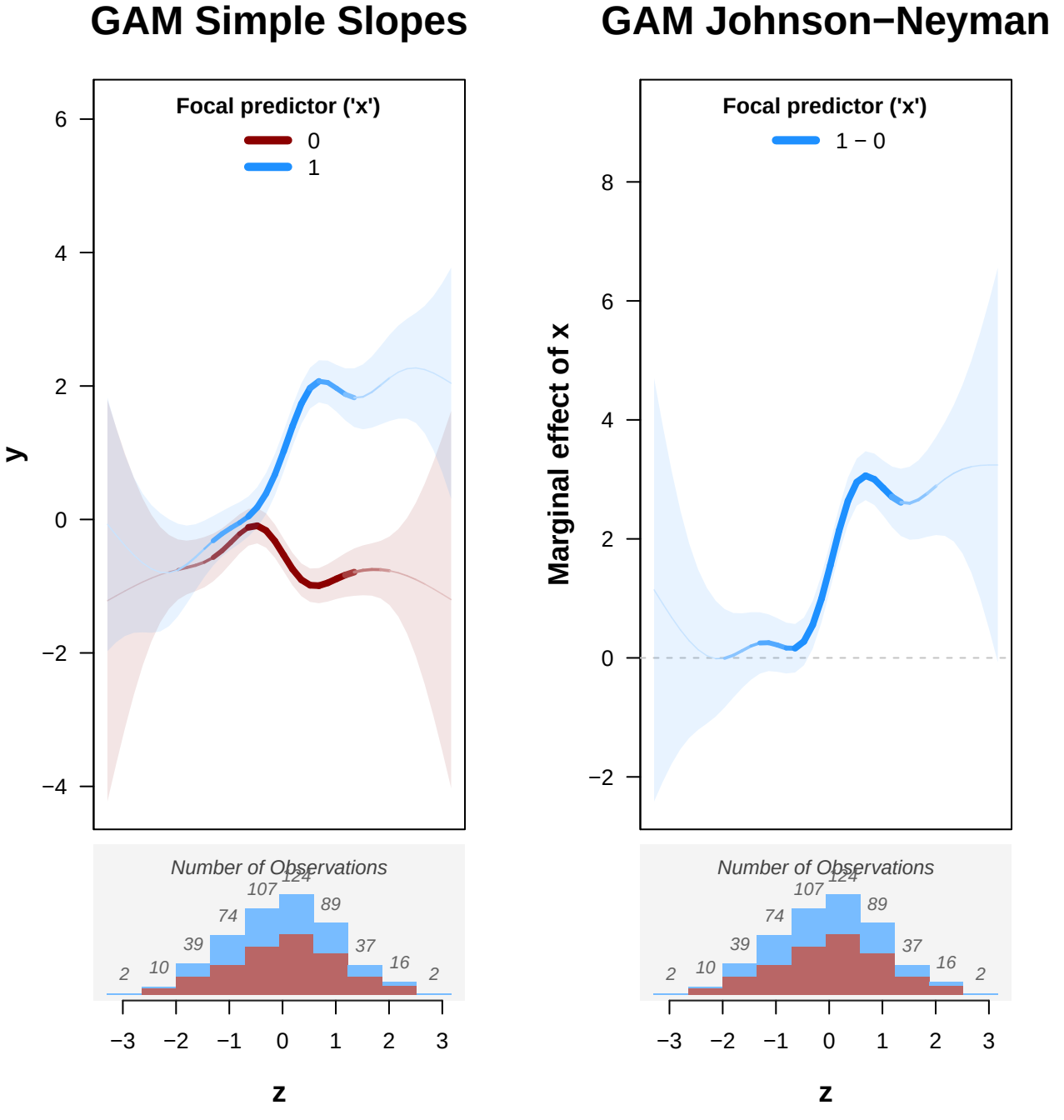

GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
8
6
4
2
0
−2
Number of Observations
2 2 4 7 8 5 6 3 4
−2 −1 0 1 2
z

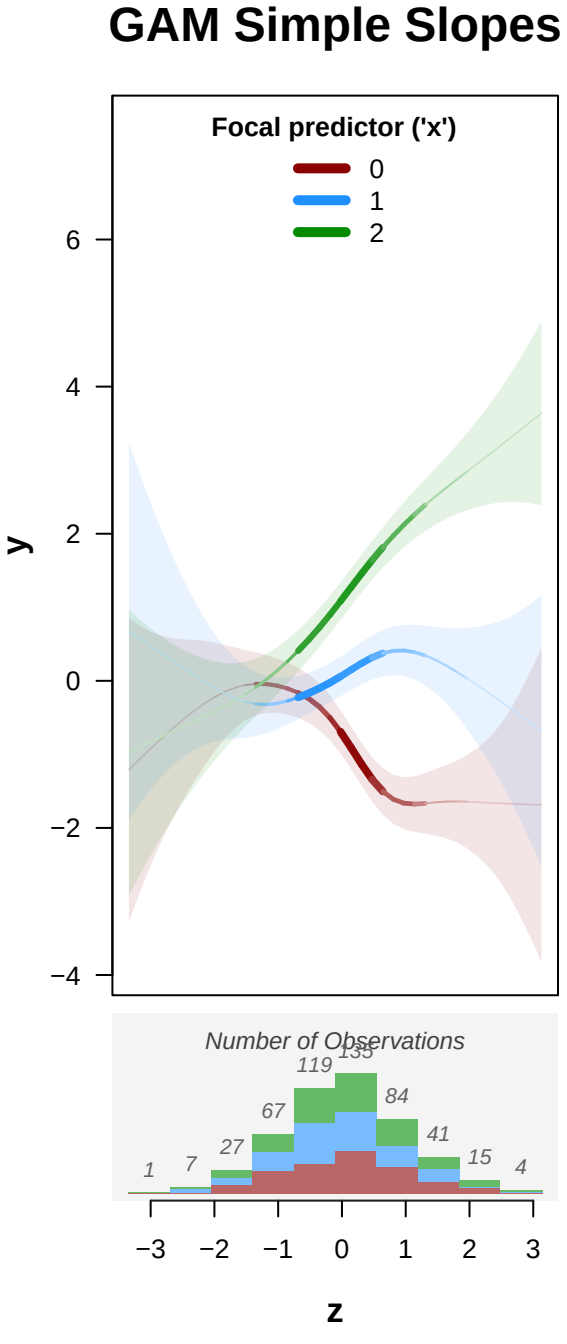

GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
4
2
0
−2
Number of Observations
4 4 4 8 3 4 7 3 8 5
−1.5 −0.5 0.0 0.5 1.0
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
8
6
4
2
0
−2
Number of Observations
4 4 4 8 3 4 7 3 8 5
−1.5 −0.5 0.0 0.5 1.0
z

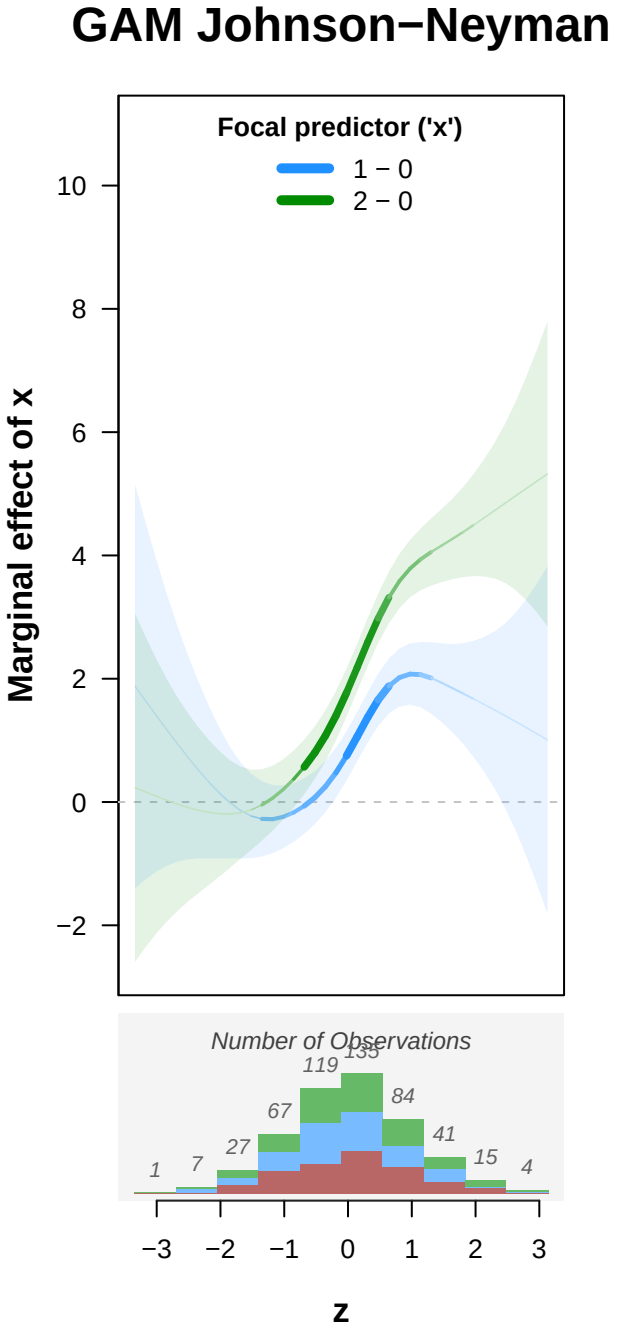

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.3)
85th percentile (0.8)
y
6
4
2
0
−2
−4
Number of Observations
10
10
8
7
6
4
3
1
0
1
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.3)
85th percentile (0.8)
Marginal effect of x
10
5
0
Number of Observations
10
10
8
7
6
4
3
1
0
1
−2
−1
0
1
2
3
z

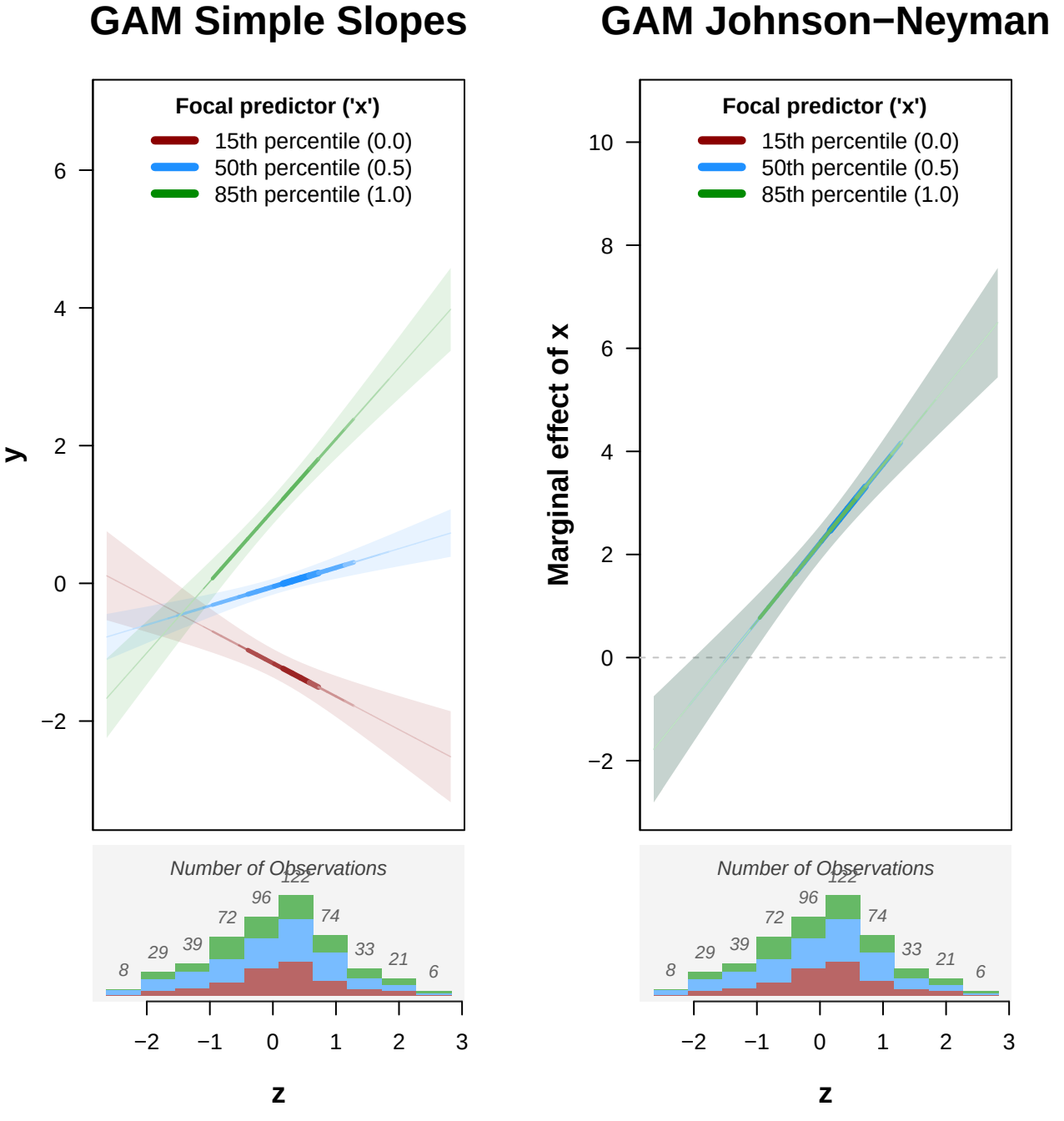

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.5)
85th percentile (0.8)
y
8
6
4
2
0
−2
Number of Observations
3
6
5
4
11
8
5
5
2
1
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
10
5
0
Number of Observations
3
6
5
4
11
8
5
5
2
1
−1
0
1
2
z

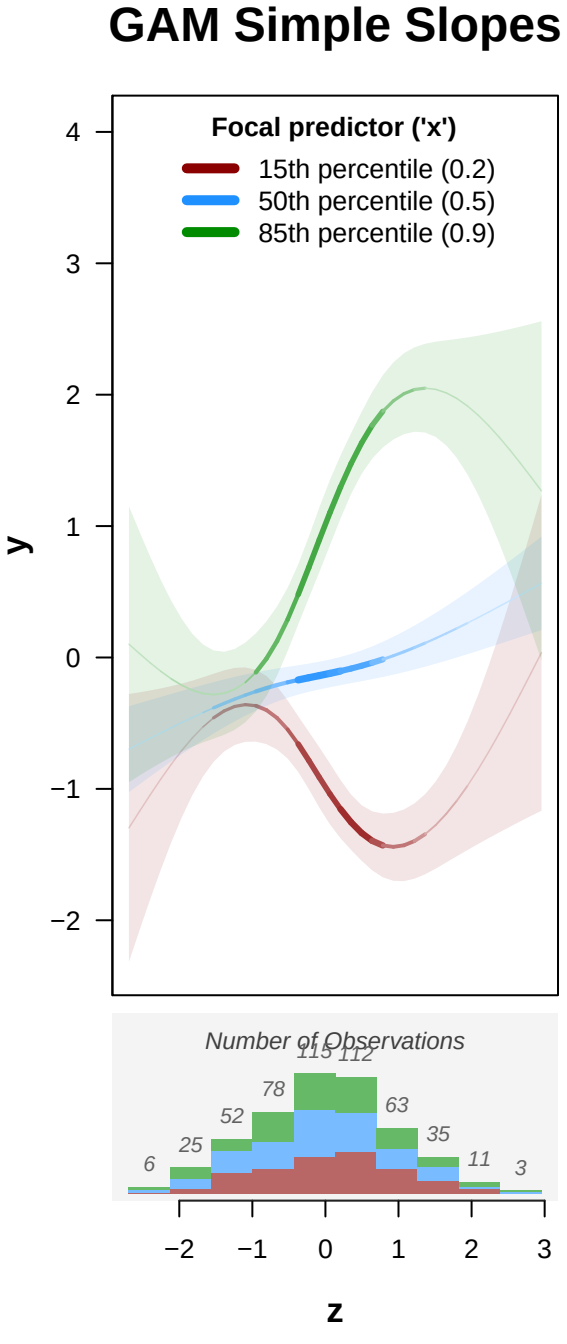

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.4)
85th percentile (0.7)
y
4
2
0
−2
Number of Observations
10
7
7
7
5
5
4
4
1
0
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.4)
85th percentile (0.7)
Marginal effect of x
15
10
5
0
−5
Number of Observations
10
7
7
7
5
5
4
4
1
0
−2
−1
0
1
2
z

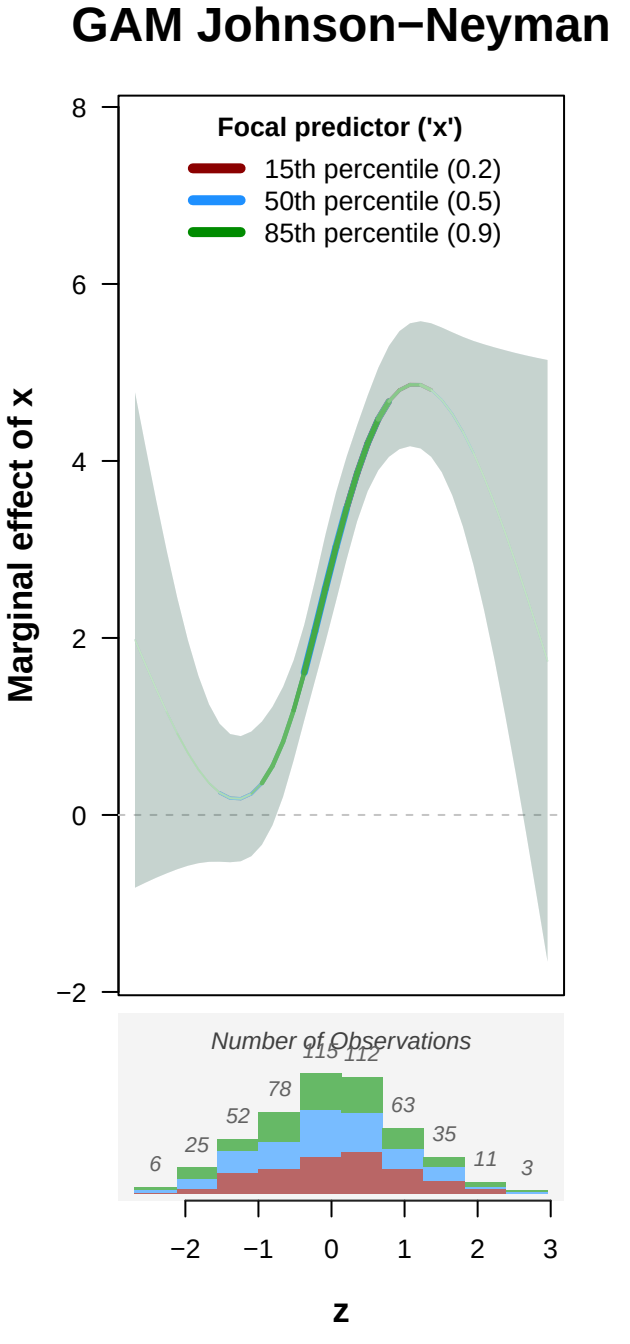

GAM Simple Slopes
Focal predictor ('x')
0
1
y
6
4
2
0
−2
−4
Number of Observations
2
10
39
74
107
124
89
37
16
2
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
8
6
4
2
0
−2
Number of Observations
2
10
39
74
107
124
89
37
16
2
−3
−2
−1
0
1
2
3
z

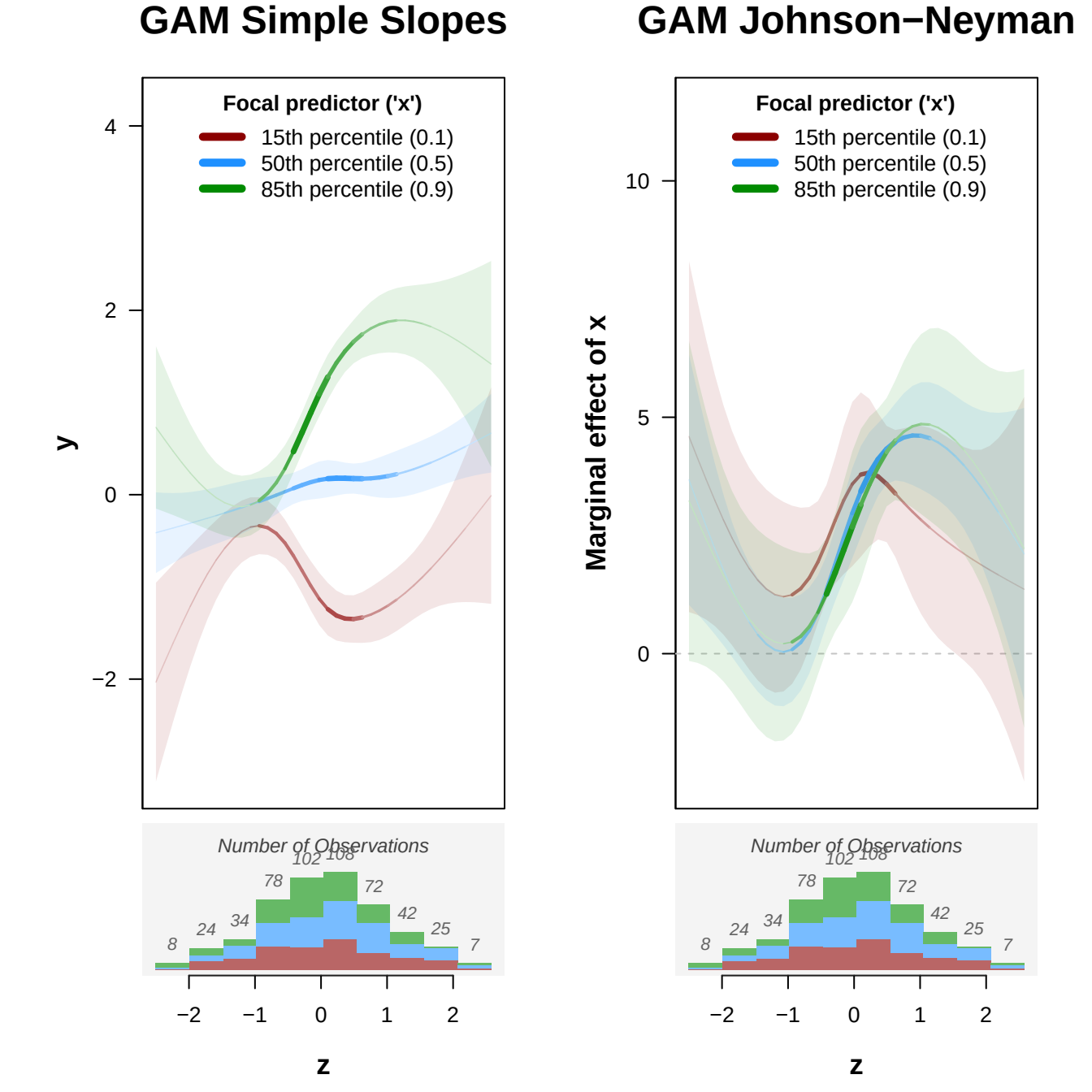

GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
6
4
2
0
−2
−4
Number of Observations
1
7
27
67
119
135
84
41
15
4
−3
−2
−1
0
1
2
3
z

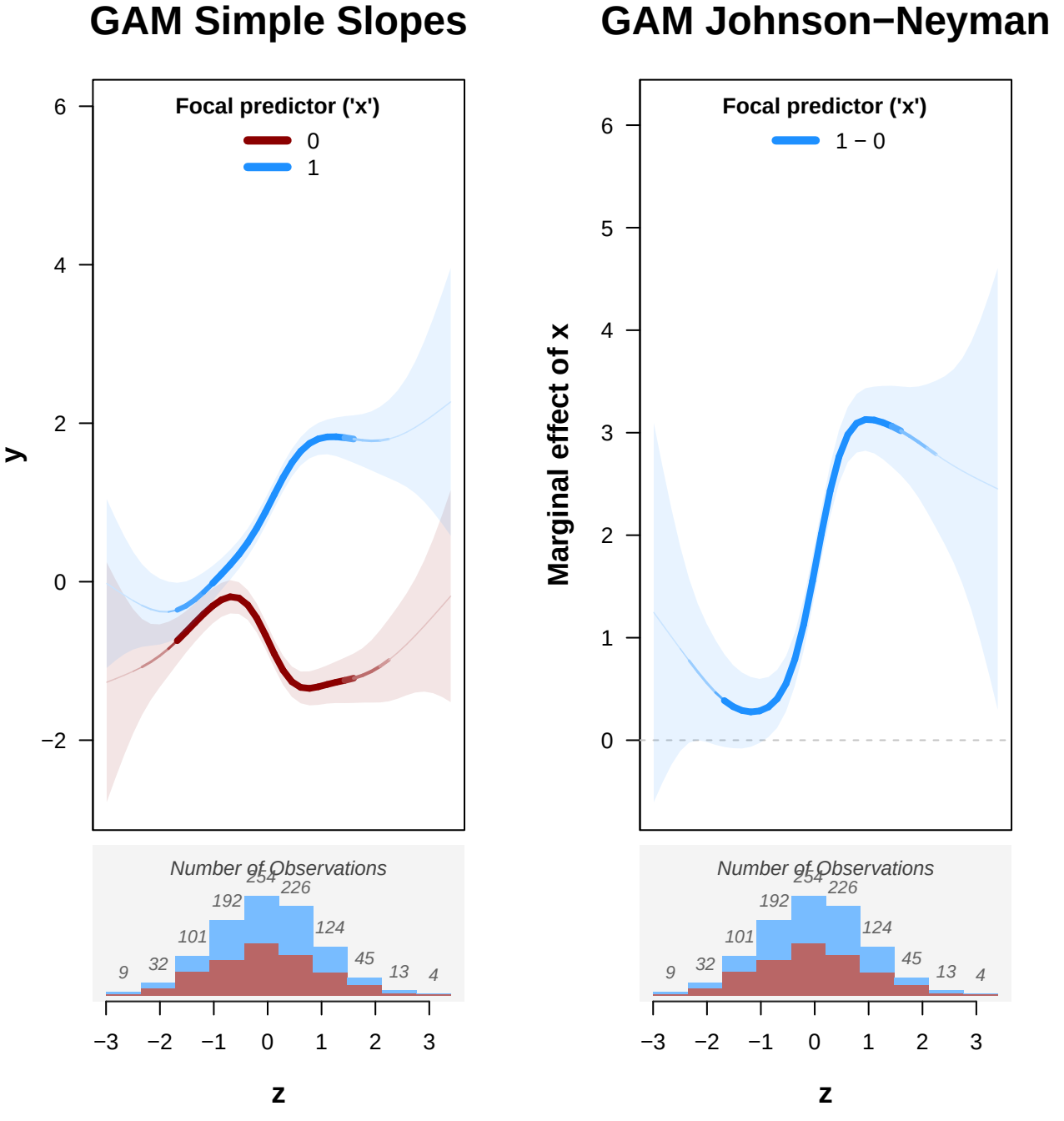

GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
10
8
6
4
2
0
−2
Number of Observations
1
7
27
67
119
135
84
41
15
4
−3
−2
−1
0
1
2
3
z

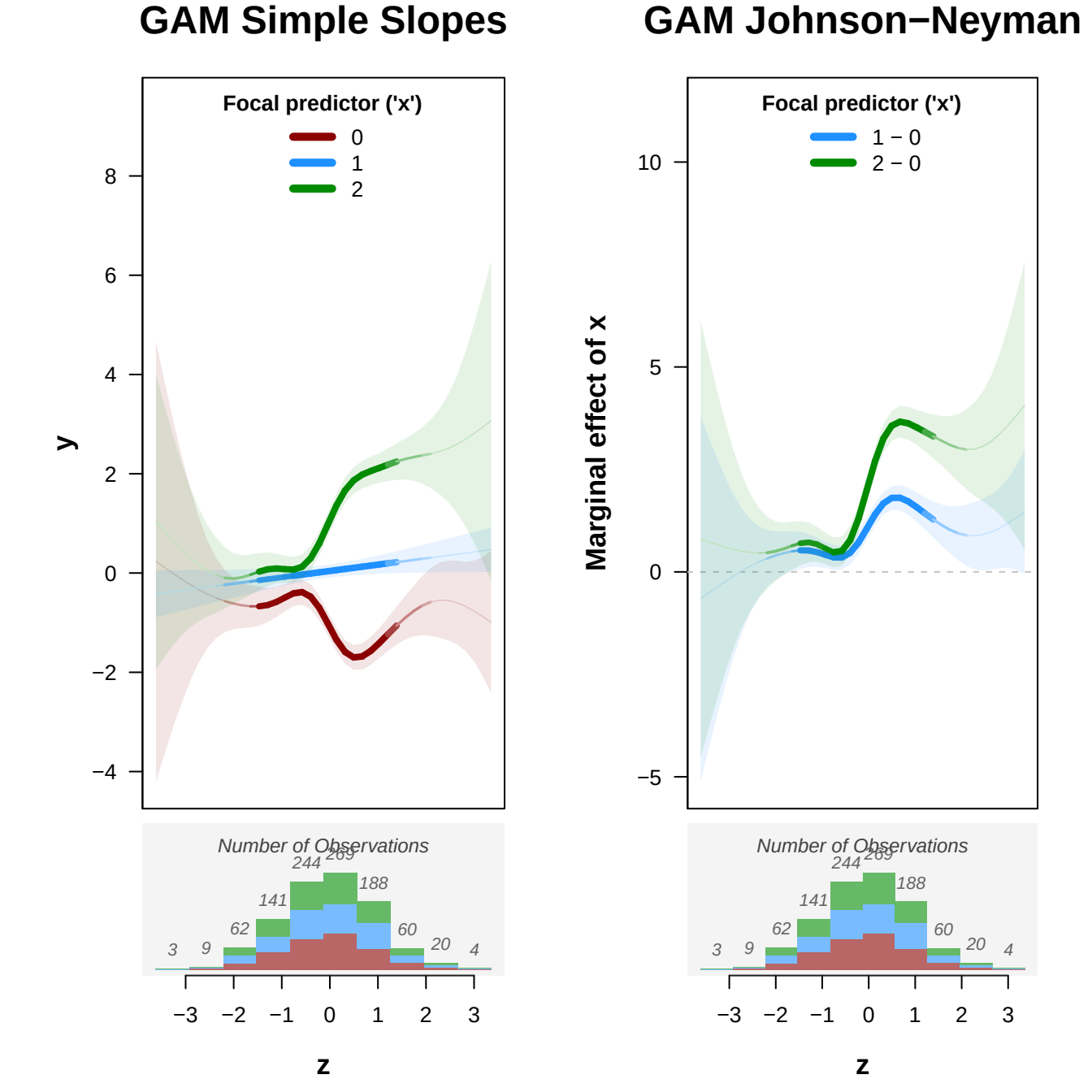

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.5)
85th percentile (1.0)
y
6
4
2
0
−2
Number of Observations
8
29
39
72
96
122
74
33
21
6
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.5)
85th percentile (1.0)
Marginal effect of x
10
8
6
4
2
0
−2
Number of Observations
8
29
39
72
96
122
74
33
21
6
−2
−1
0
1
2
3
z

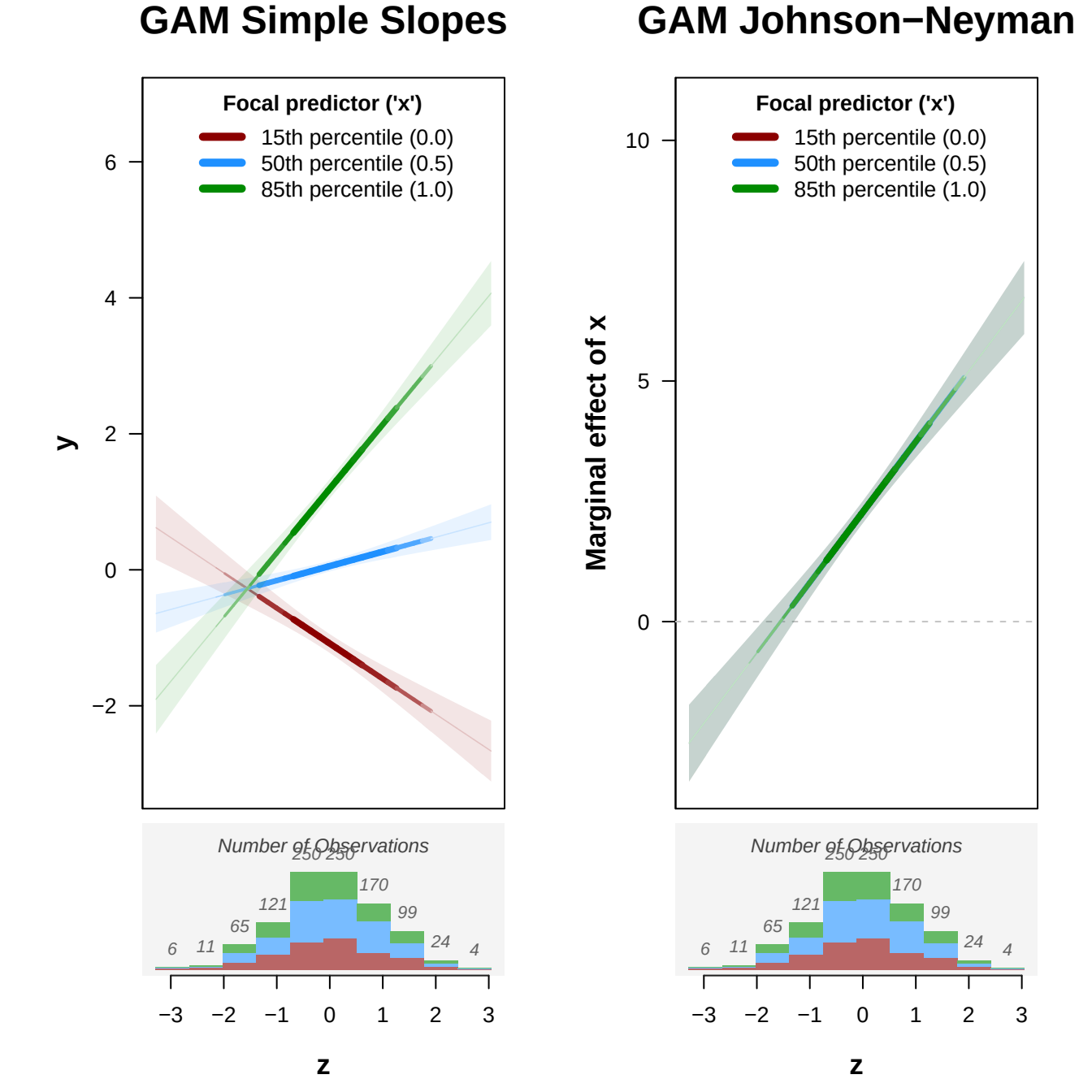

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
y
Number of Observations
6
25
52
78
115
112
63
35
11
3
z

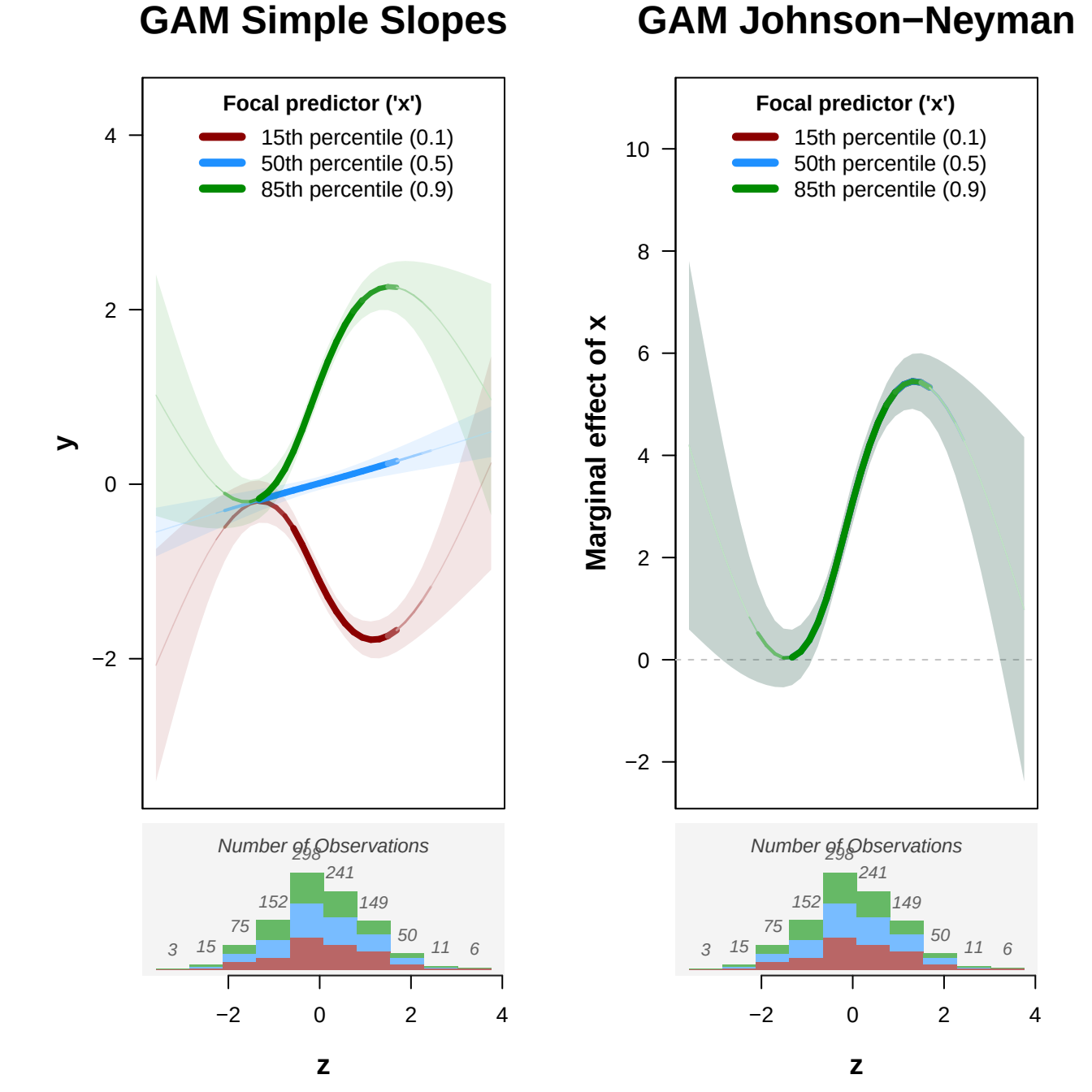

GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
Marginal effect of x
Number of Observations
6
25
52
78
115
112
63
35
11
3
z

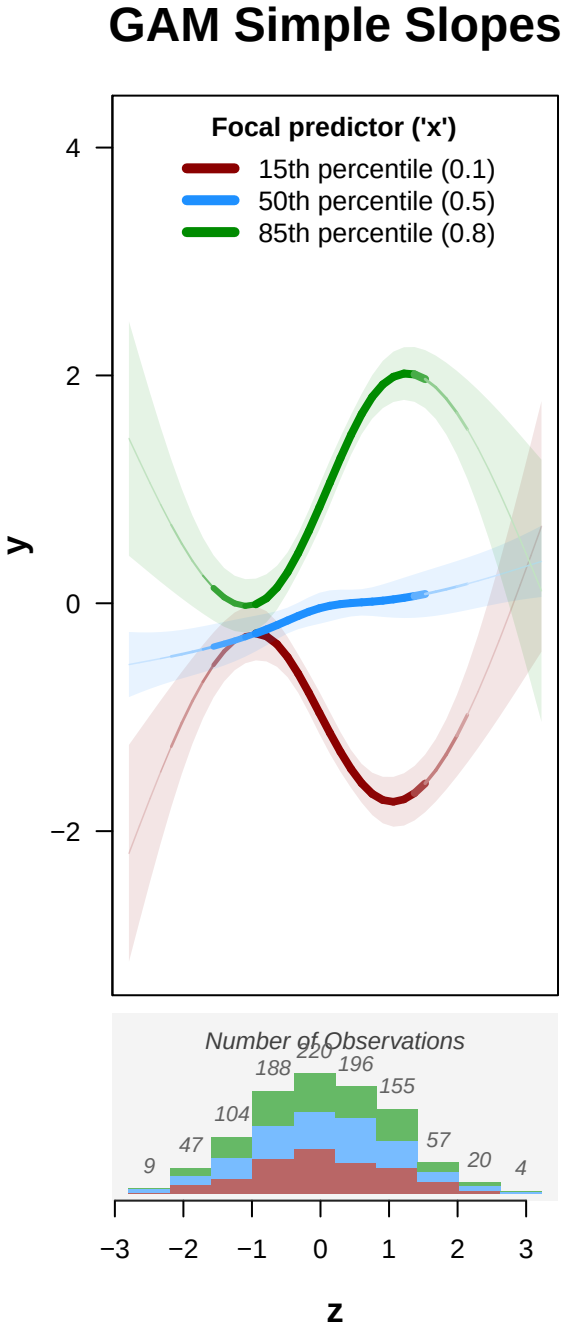

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.9)
y
4
2
0
−2
Number of Observations
8
24
34
78
102
108
72
42
25
7
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.9)
Marginal effect of x
10
5
0
Number of Observations
8
24
34
78
102
108
72
42
25
7
−2
−1
0
1
2
z

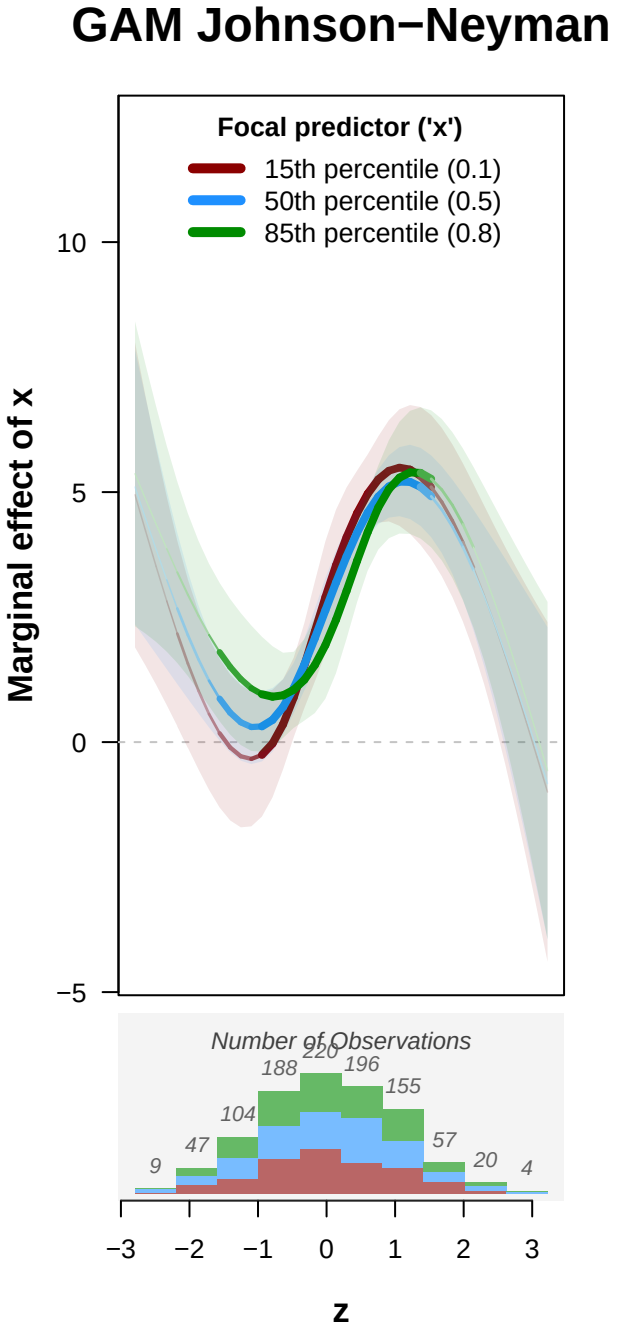

GAM Simple Slopes
Focal predictor ('x')
0
1
y
6
4
2
0
−2
Number of Observations
9
32
101
192
254
226
124
45
13
4
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
6
5
4
3
2
1
0
Number of Observations
9
32
101
192
254
226
124
45
13
4
−3
−2
−1
0
1
2
3
z

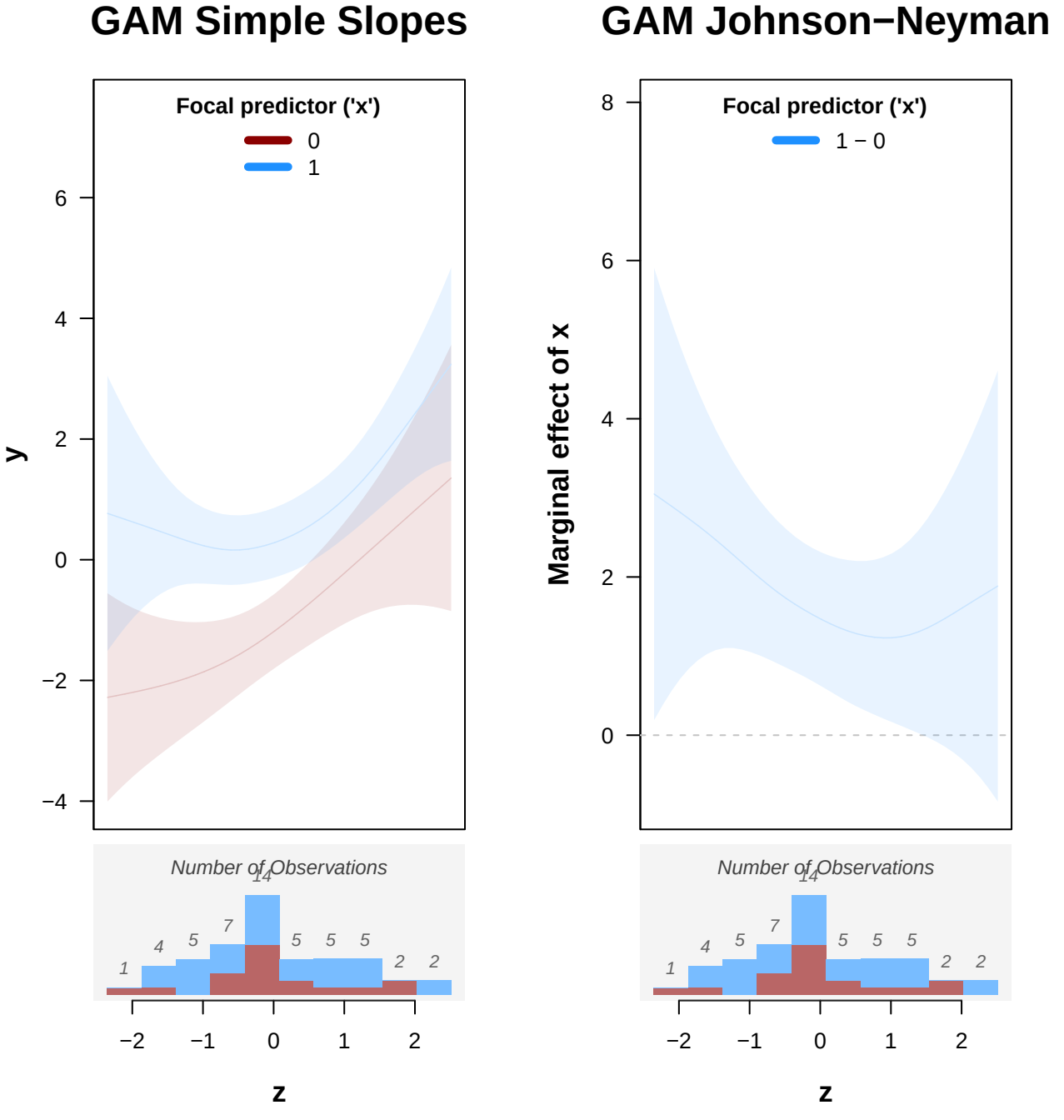

GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
8
6
4
2
0
−2
−4
Number of Observations
3
9
62
141
244
269
188
60
20
4
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
10
5
0
−5
Number of Observations
3
9
62
141
244
269
188
60
20
4
−3
−2
−1
0
1
2
3
z

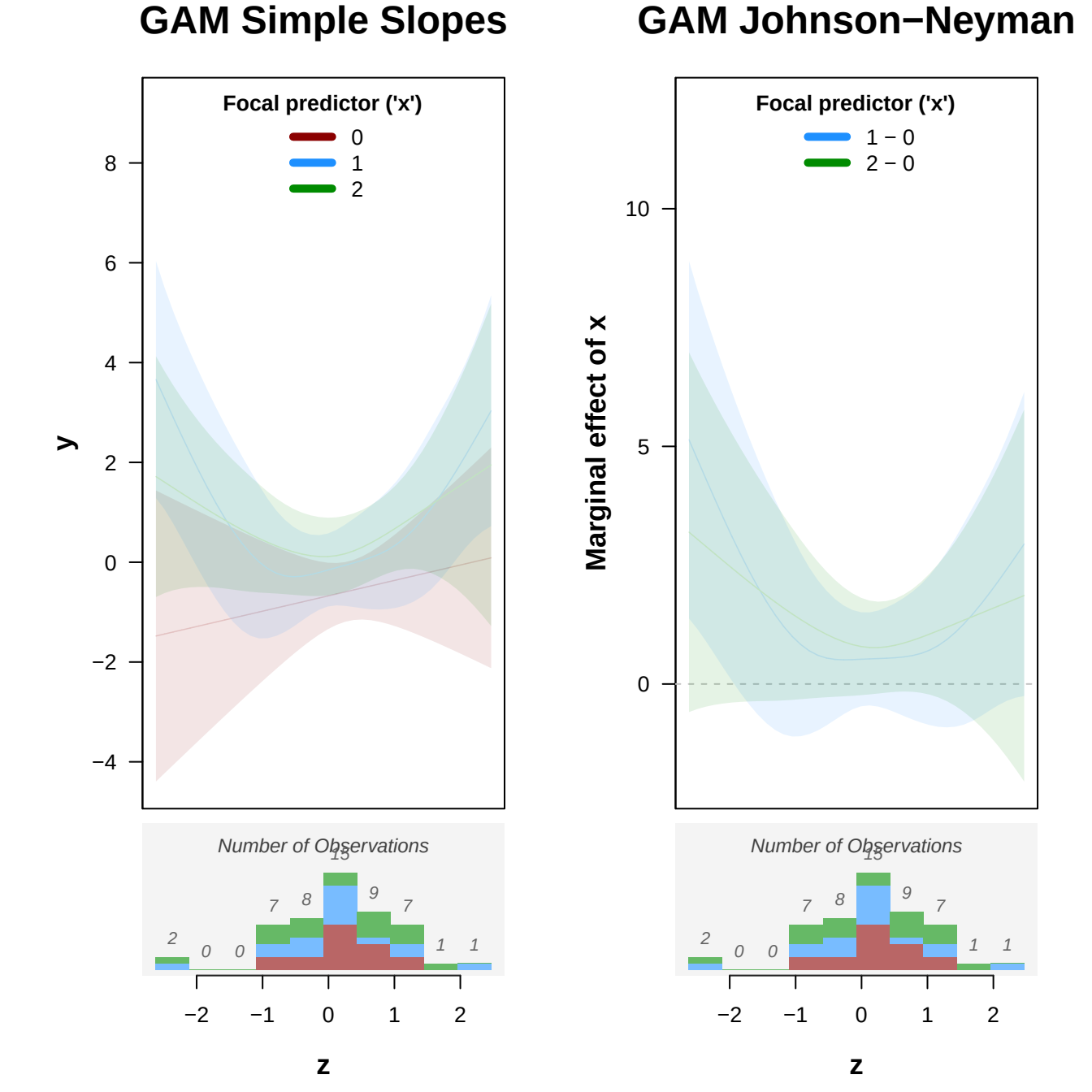

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.5)
85th percentile (1.0)
y
6
4
2
0
−2
Number of Observations
250 250
170
121
99
65
24
11
6
4
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.5)
85th percentile (1.0)
Marginal effect of x
10
5
0
Number of Observations
250 250
170
121
99
65
24
11
6
4
−3
−2
−1
0
1
2
3
z

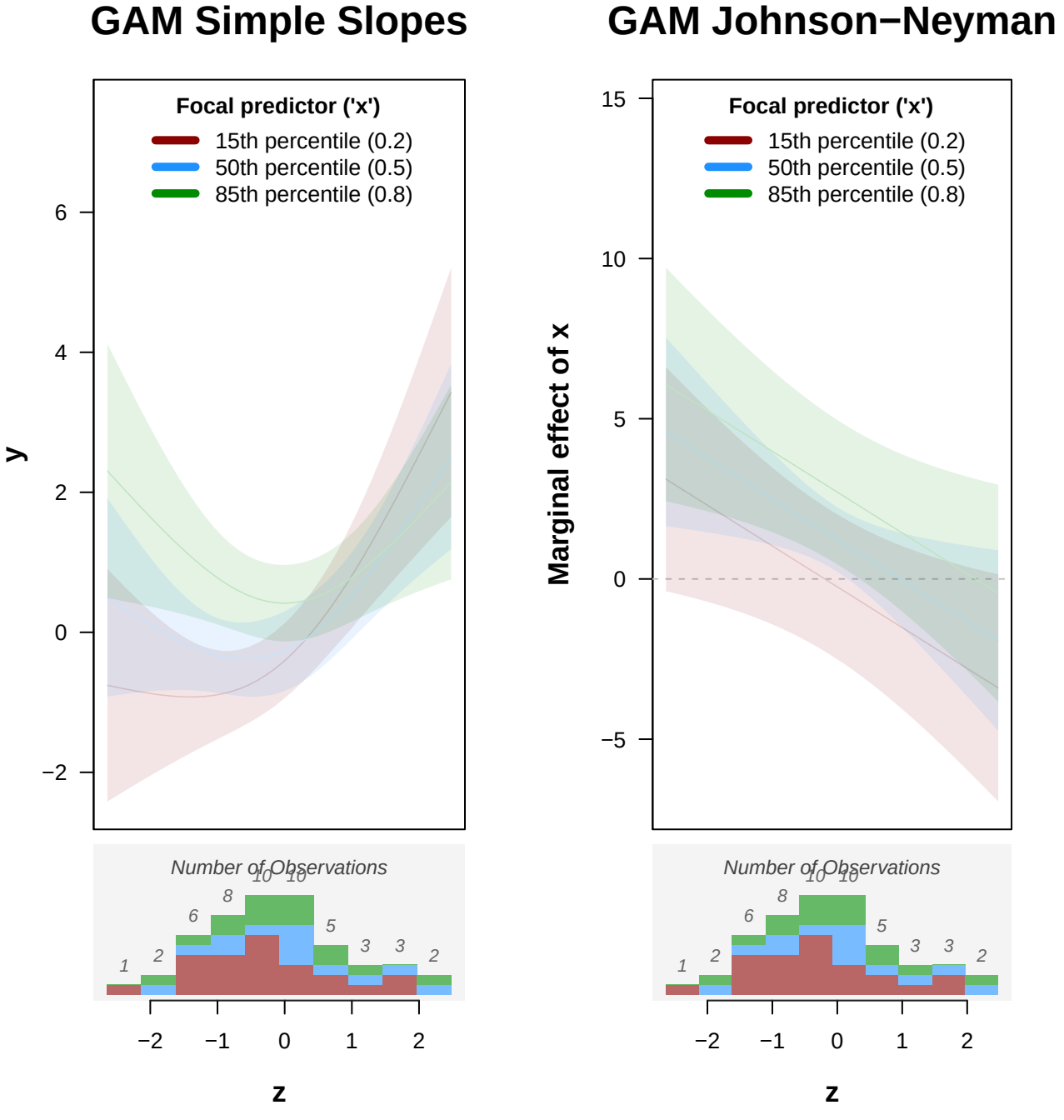

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.9)
y
4
2
0
−2
Number of Observations
298
241
152
149
75
50
11
6
3
15
−2
0
2
4
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.9)
Marginal effect of x
10
8
6
4
2
0
−2
Number of Observations
298
241
152
149
75
50
11
6
3
15
−2
0
2
4
z

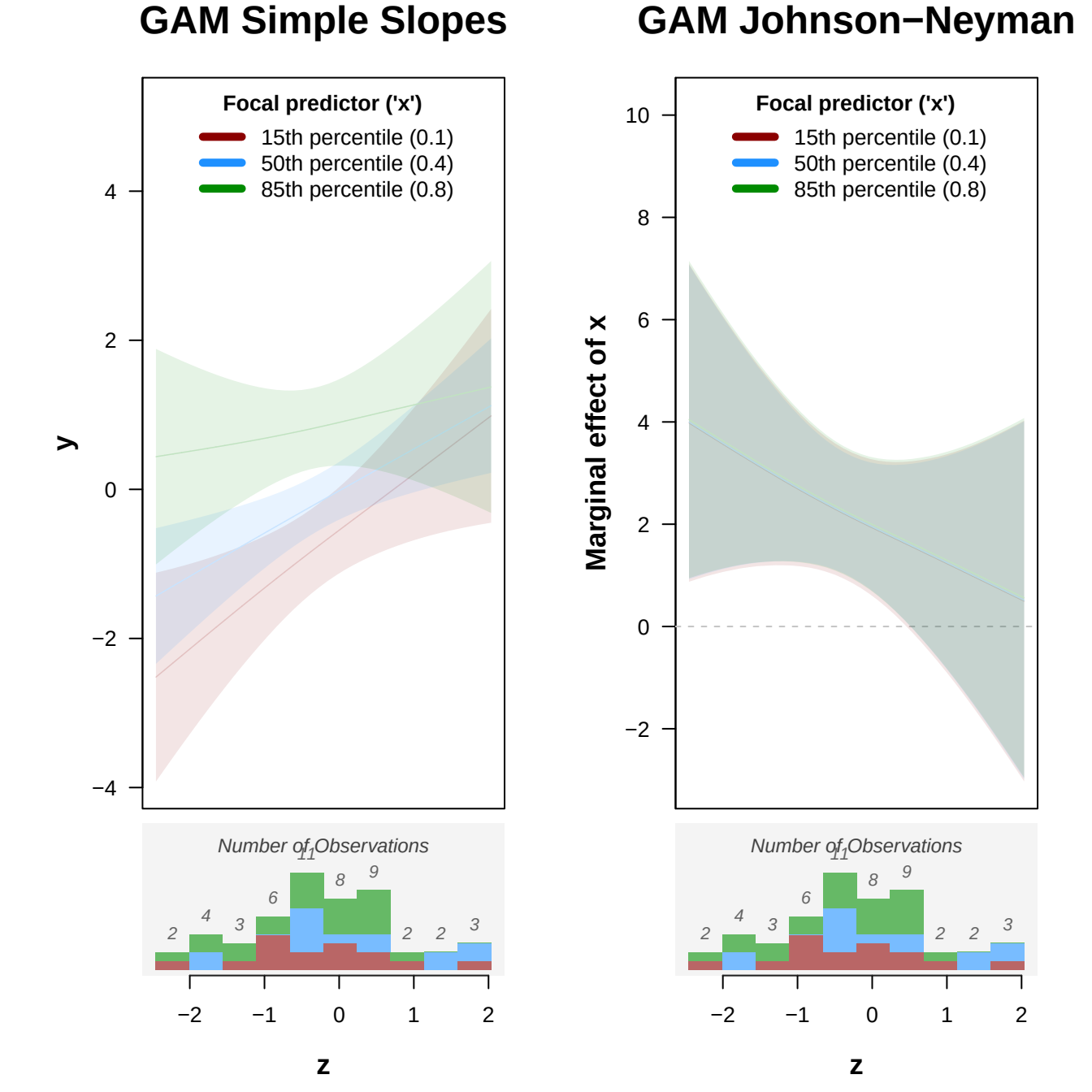

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
y
4
2
0
−2
Number of Observations
9
47
104
188
220
196
155
57
20
4
−3
−2
−1
0
1
2
3
z

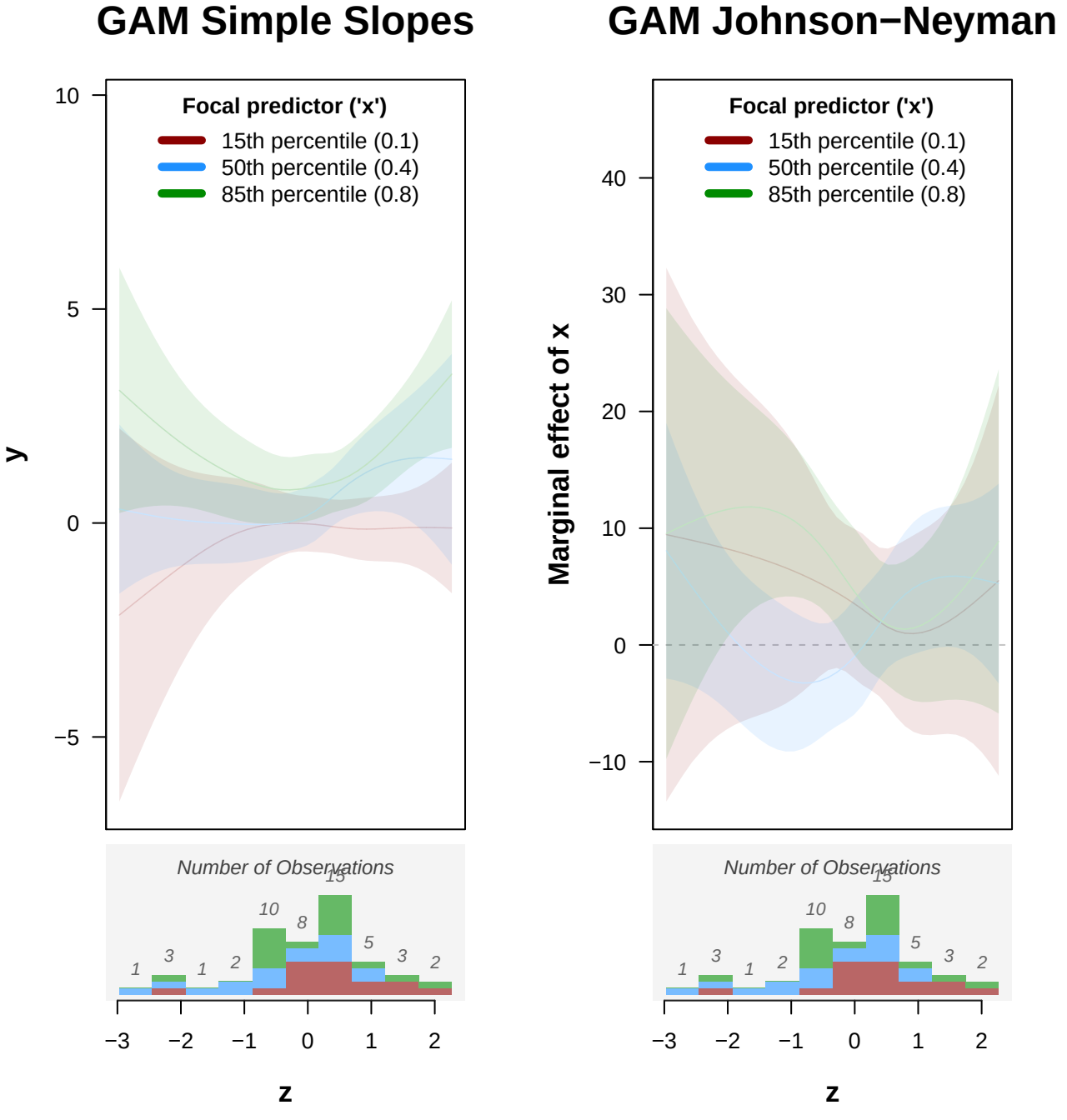

GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
10
5
0
−5
Number of Observations
9
47
104
188
220
196
155
57
20
4
−3
−2
−1
0
1
2
3
z

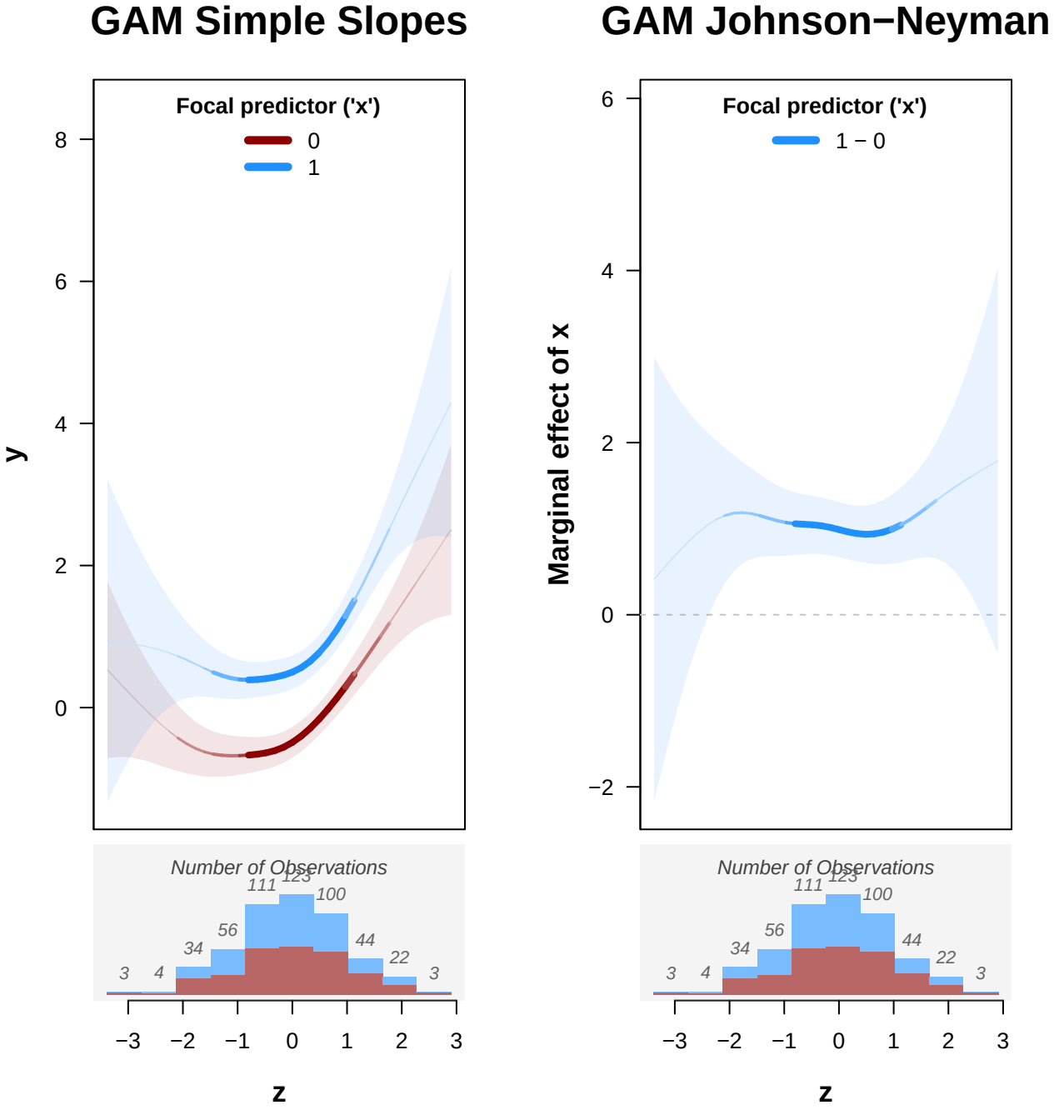

GAM Simple Slopes
Focal predictor ('x')
0
1
y
6
4
2
0
−2
−4
Number of Observations
14
7
5
4
5
5
5
2
2
1
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
8
6
4
2
0
Number of Observations
14
7
5
4
5
5
5
2
2
1
−2
−1
0
1
2
z

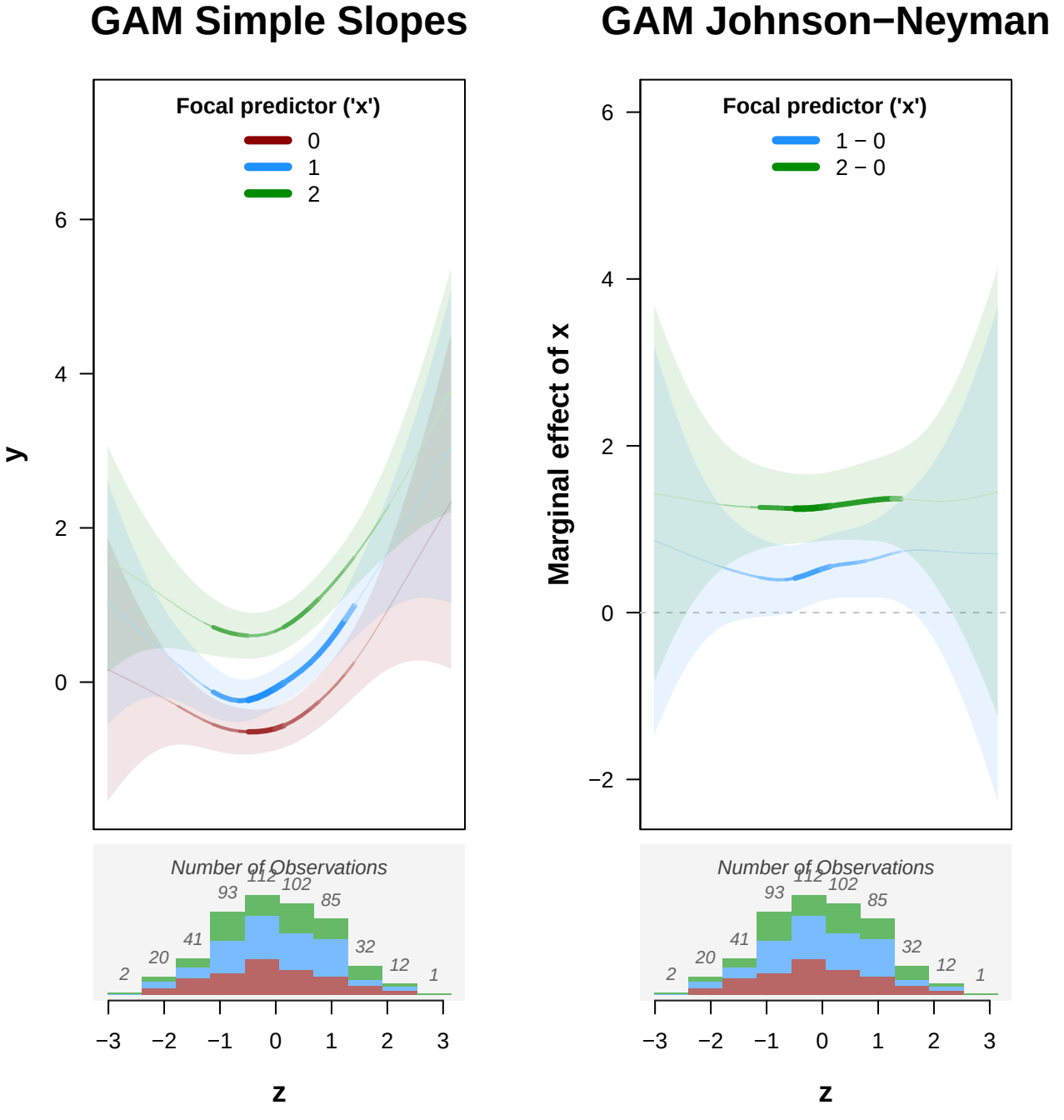

GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
8
6
4
2
0
−2
−4
Number of Observations
2
0
0
7
8
15
9
7
1
1
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
10
5
0
Number of Observations
2
0
0
7
8
15
9
7
1
1
−2
−1
0
1
2
z

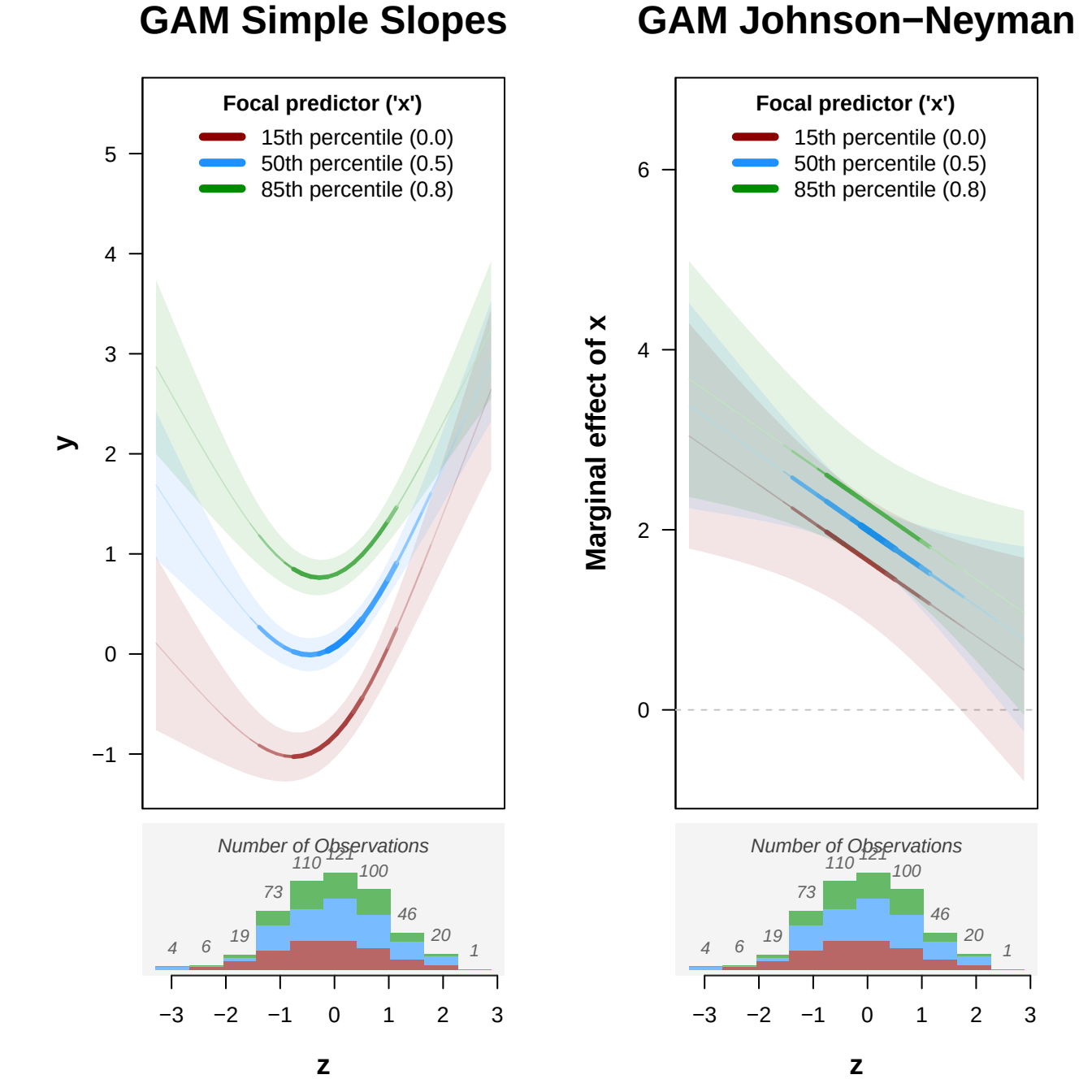

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
y
6
4
2
0
−2
Number of Observations
1
2
6
8
10
10
5
3
3
2
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
15
10
5
0
−5
Number of Observations
1
2
6
8
10
10
5
3
3
2
−2
−1
0
1
2
z

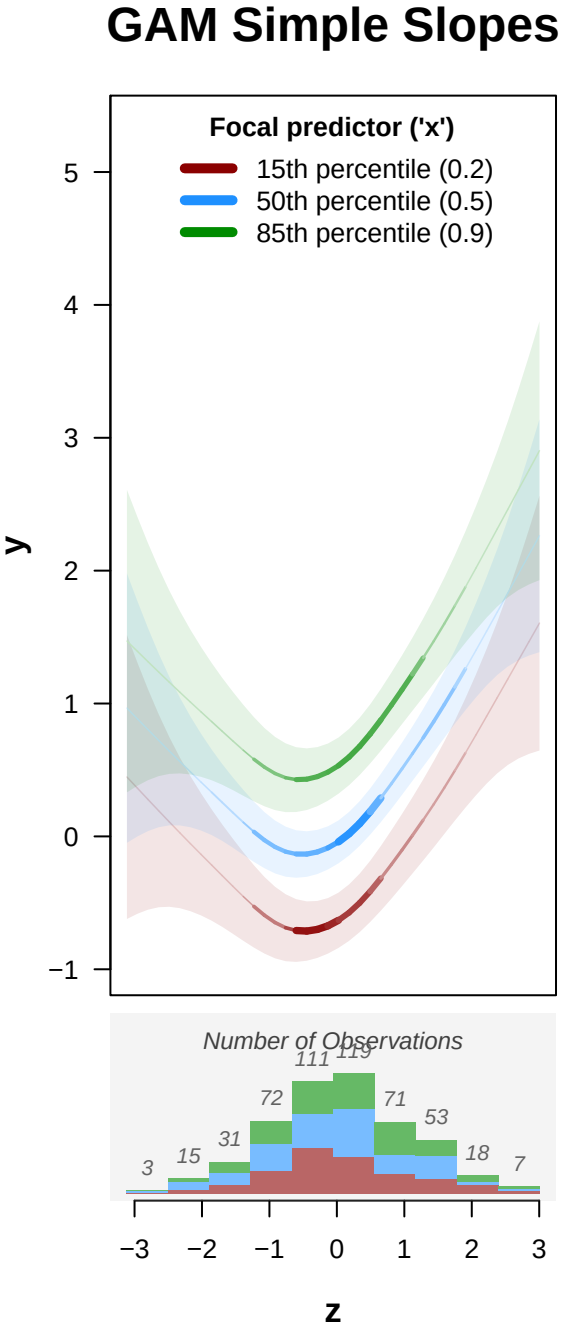

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.4)
85th percentile (0.8)
y
4
2
0
−2
−4
Number of Observations
11
8
9
6
4
3
2
2
2
3
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.4)
85th percentile (0.8)
Marginal effect of x
10
8
6
4
2
0
−2
Number of Observations
11
8
9
6
4
3
2
2
2
3
−2
−1
0
1
2
z

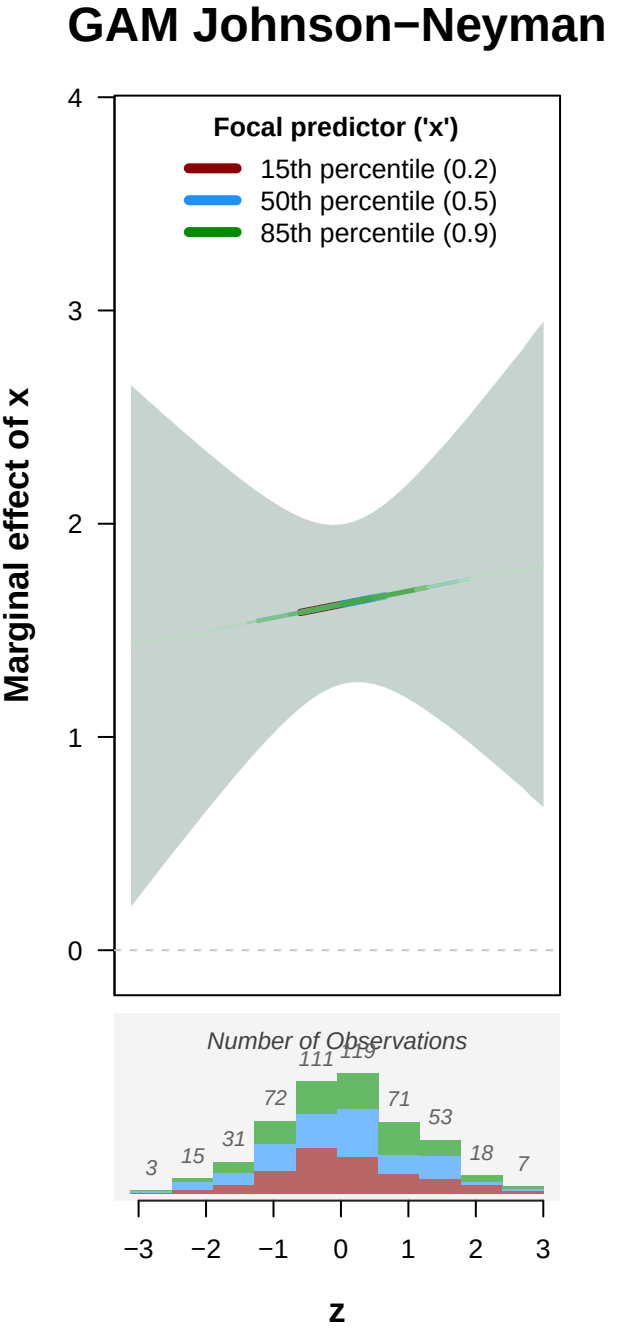

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.4)
85th percentile (0.8)
y
10
5
0
−5
Number of Observations
1
3
1
2
10
8
15
5
3
2
−3
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.4)
85th percentile (0.8)
Marginal effect of x
40
30
20
10
0
−10
Number of Observations
1
3
1
2
10
8
15
5
3
2
−3
−2
−1
0
1
2
z

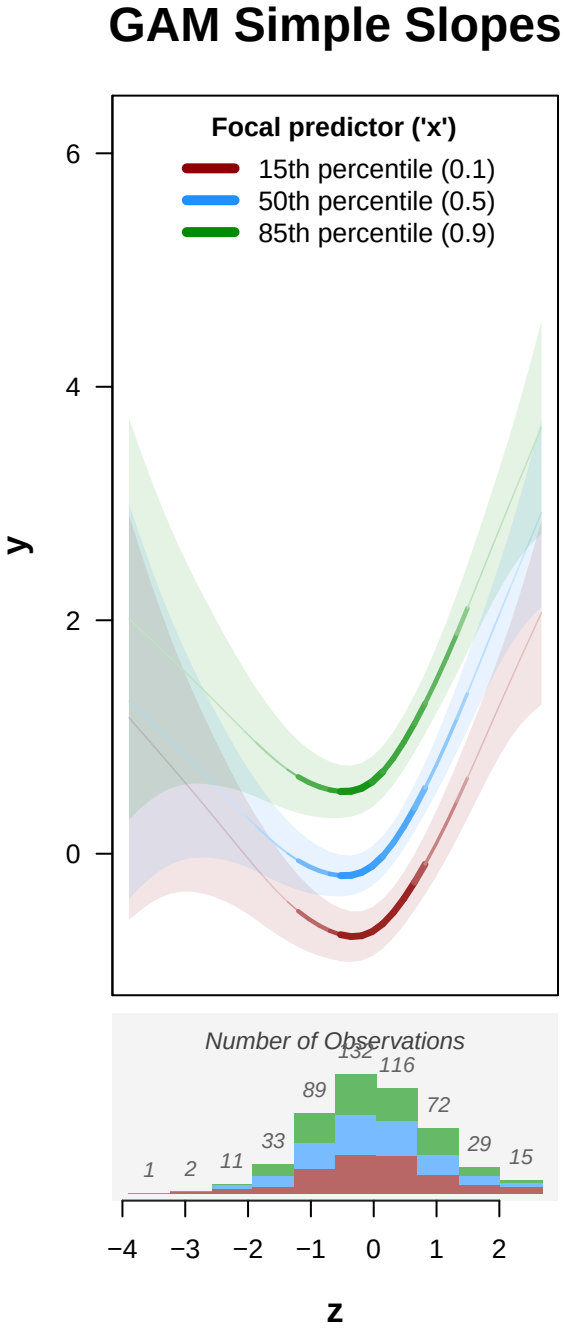

GAM Simple Slopes
Focal predictor ('x')
0
1
y
8
6
4
2
0
Number of Observations
3
4
34
56
111
123
100
44
22
3
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
6
4
2
0
−2
Number of Observations
3
4
34
56
111
123
100
44
22
3
−3
−2
−1
0
1
2
3
z

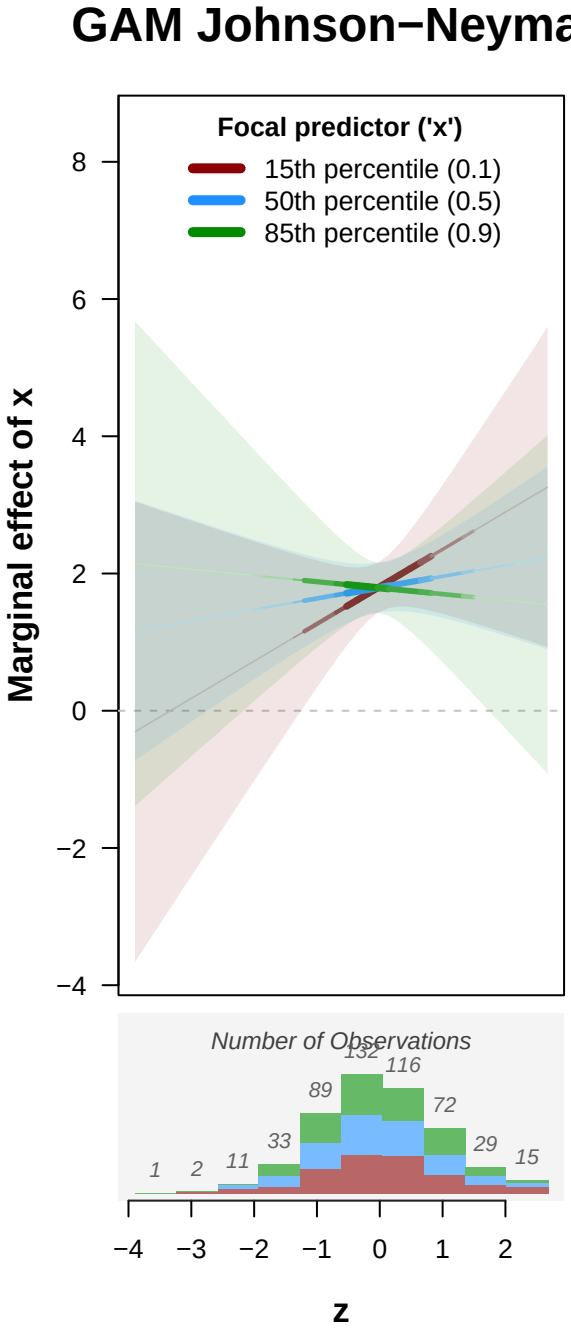

GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
6
4
2
0
Number of Observations
2
20
41
93
112
102
85
32
12
1
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
6
4
2
0
−2
Number of Observations
2
20
41
93
112
102
85
32
12
1
−3
−2
−1
0
1
2
3
z

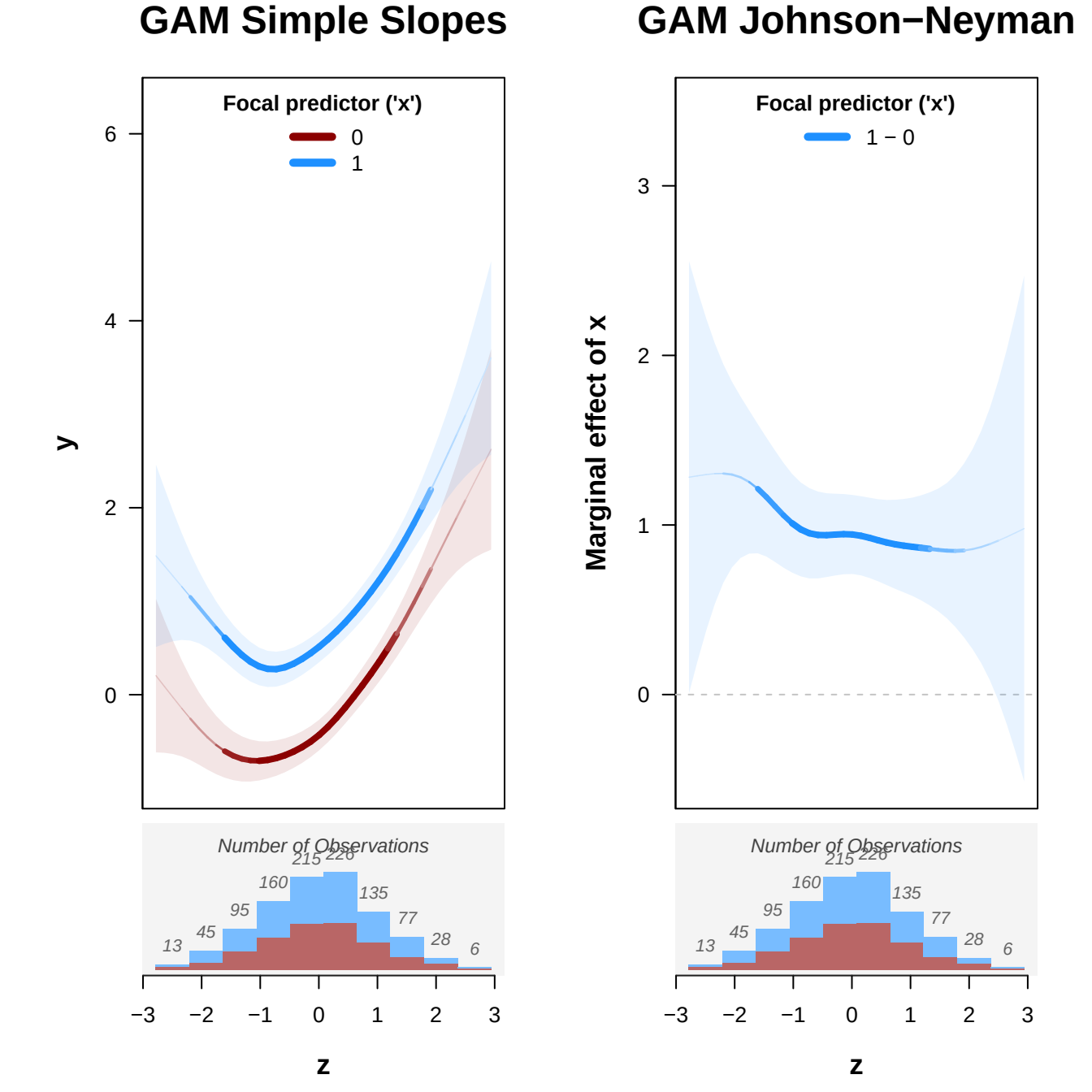

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.5)
85th percentile (0.8)
y
5
4
3
2
1
0
−1
Number of Observations
4
6
19
73
110
121
100
46
20
1
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
6
4
2
0
Number of Observations
4
6
19
73
110
121
100
46
20
1
−3
−2
−1
0
1
2
3
z

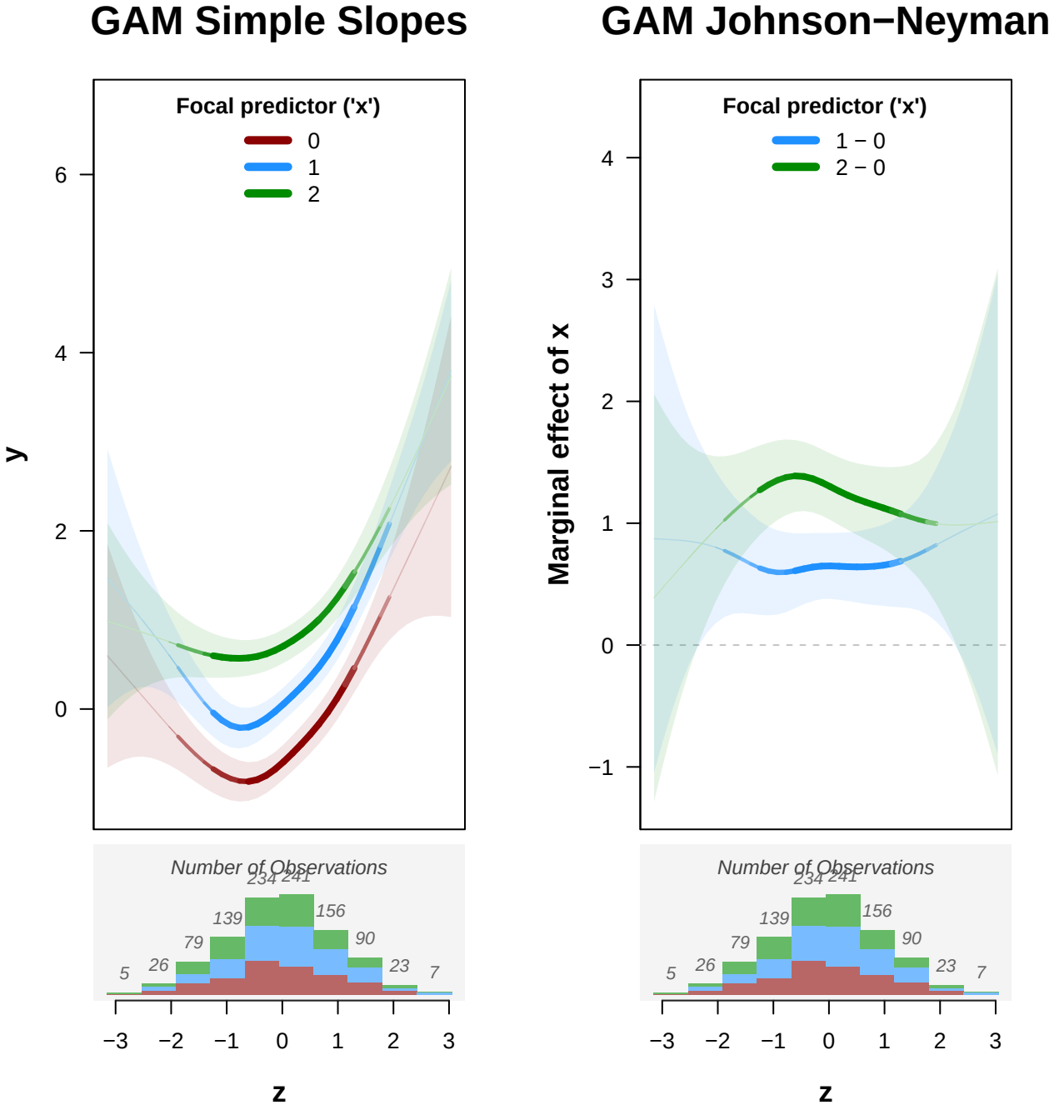

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
y
5
4
3
2
1
0
−1
Number of Observations
3
15
31
72
111
119
71
53
18
7
−3
−2
−1
0
1
2
3
z

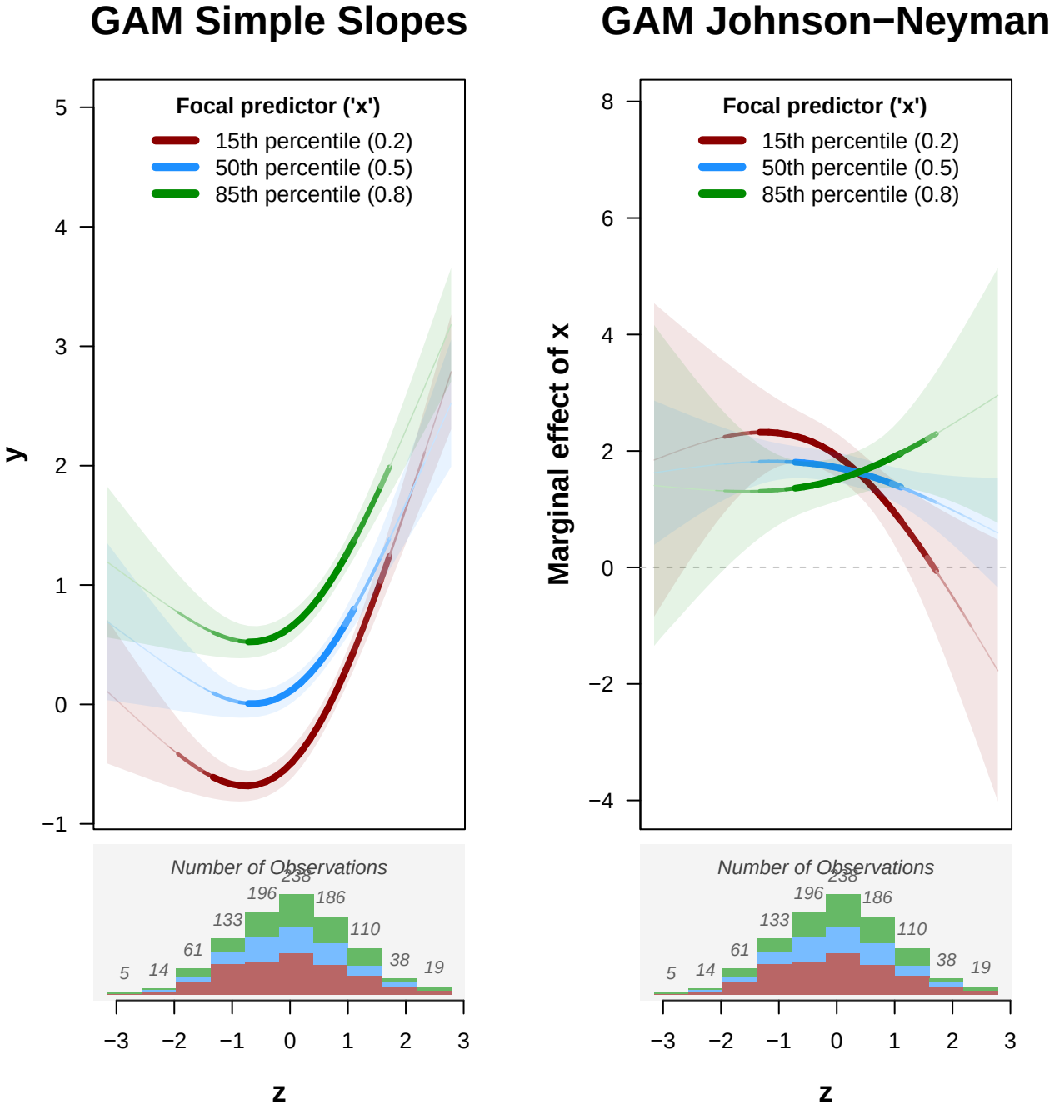

GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
Marginal effect of x
4
3
2
1
0
Number of Observations
3
15
31
72
111
119
71
53
18
7
−3
−2
−1
0
1
2
3
z

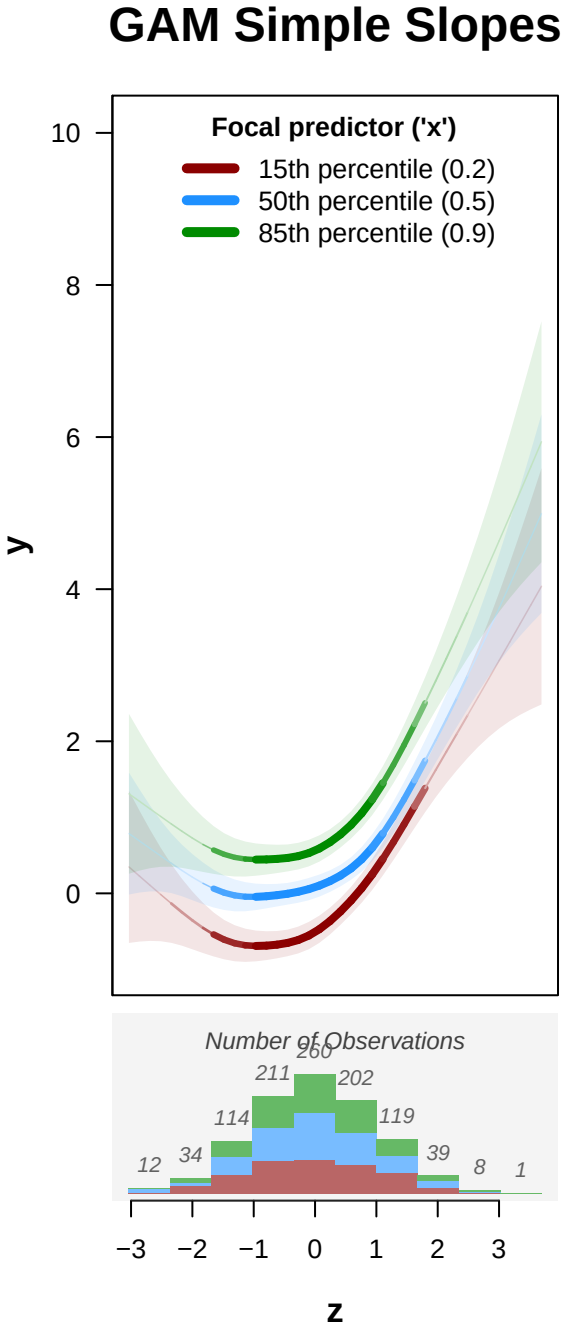

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.9)
y
6
4
2
0
Number of Observations
1
2
11
33
89
132
116
72
29
15
−4
−3
−2
−1
0
1
2
z

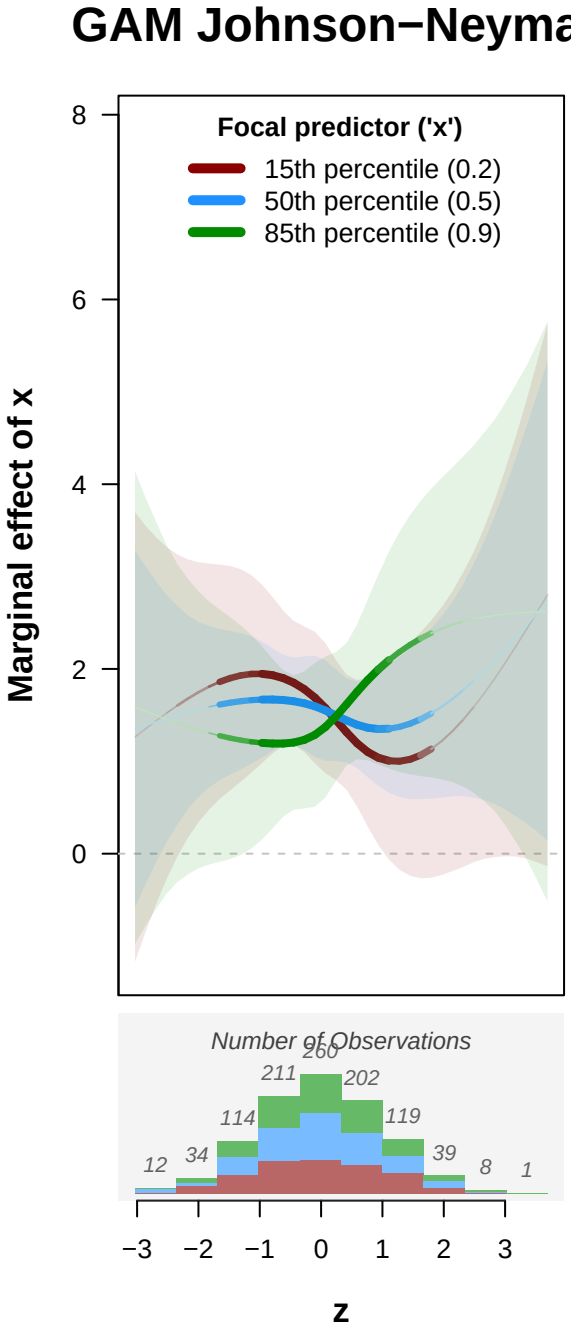

GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.9)
Marginal effect of x
8
6
4
2
0
−2
−4
Number of Observations
1
2
11
33
89
132
116
72
29
15
−4
−3
−2
−1
0
1
2
z

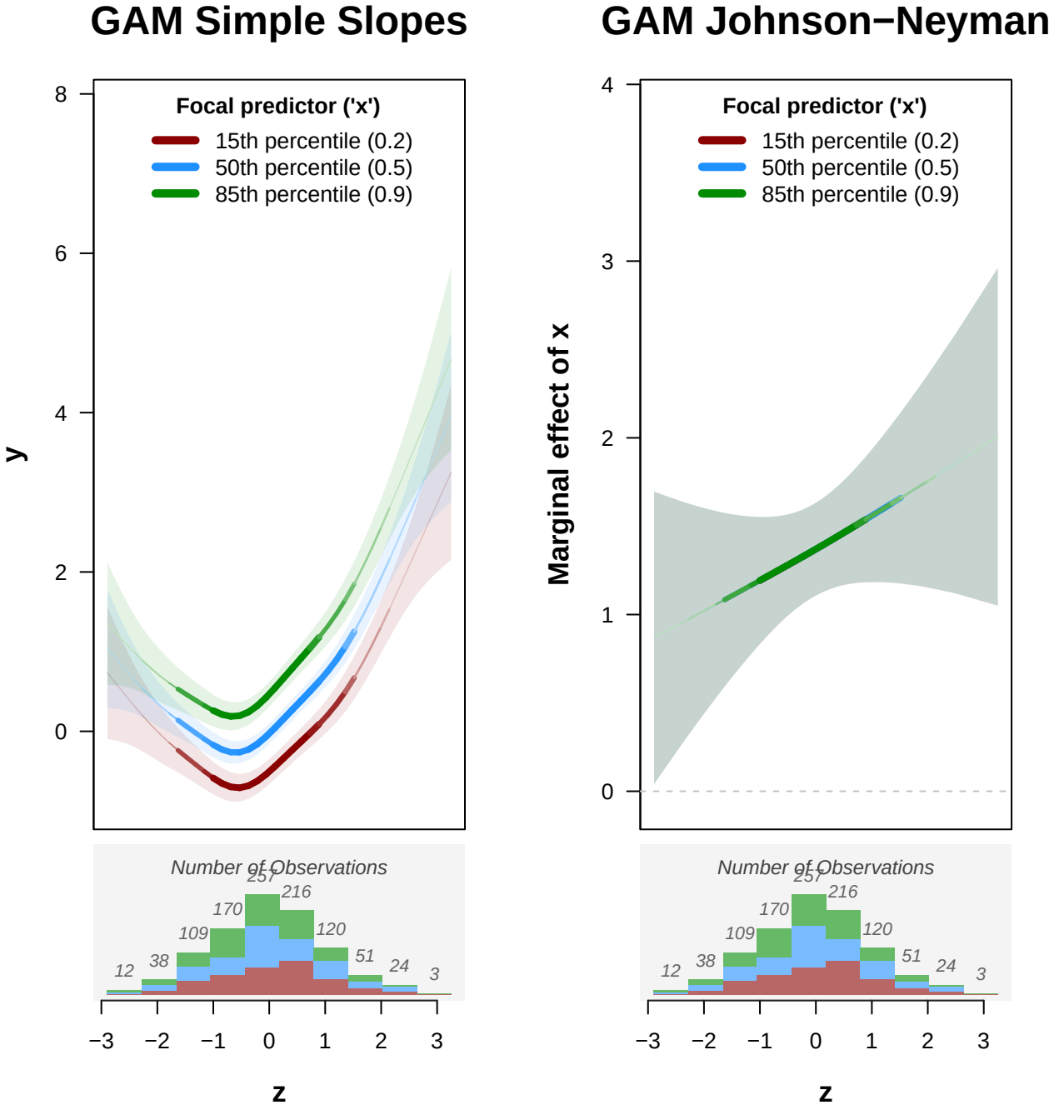

GAM Simple Slopes
Focal predictor ('x')
0
1
y
6
4
2
0
Number of Observations
13
45
95
160
215
226
135
77
28
6
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
3
2
1
0
Number of Observations
13
45
95
160
215
226
135
77
28
6
−3
−2
−1
0
1
2
3
z

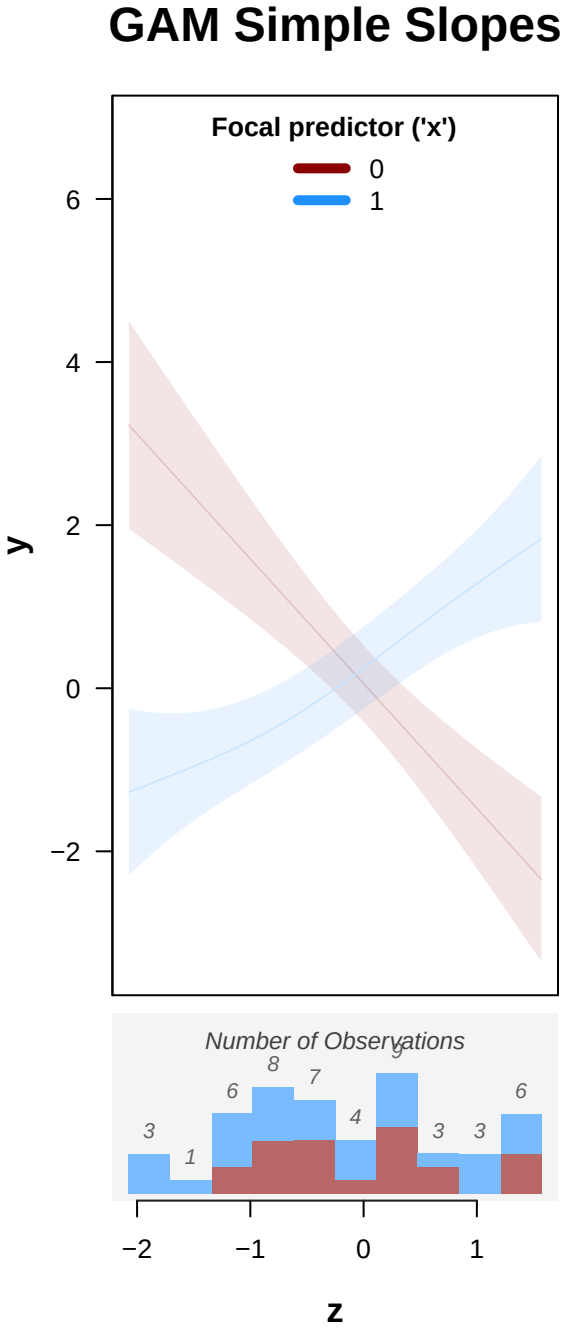

GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
6
4
2
0
Number of Observations
234 241
139
156
79
90
26
23
5
7
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
4
3
2
1
0
−1
Number of Observations
234 241
139
156
79
90
26
23
5
7
−3
−2
−1
0
1
2
3
z

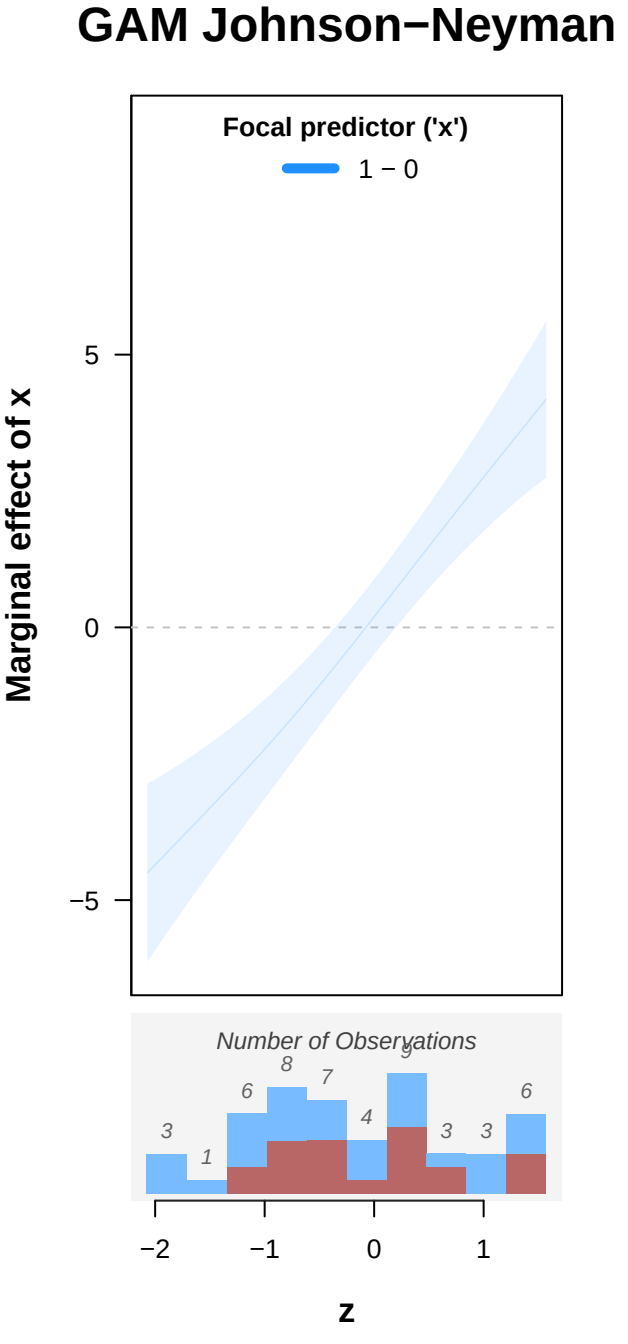

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
y
5
4
3
2
1
0
−1
Number of Observations
5
14
61
133
196
238
186
110
38
19
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
8
6
4
2
0
−2
−4
Number of Observations
5
14
61
133
196
238
186
110
38
19
−3
−2
−1
0
1
2
3
z

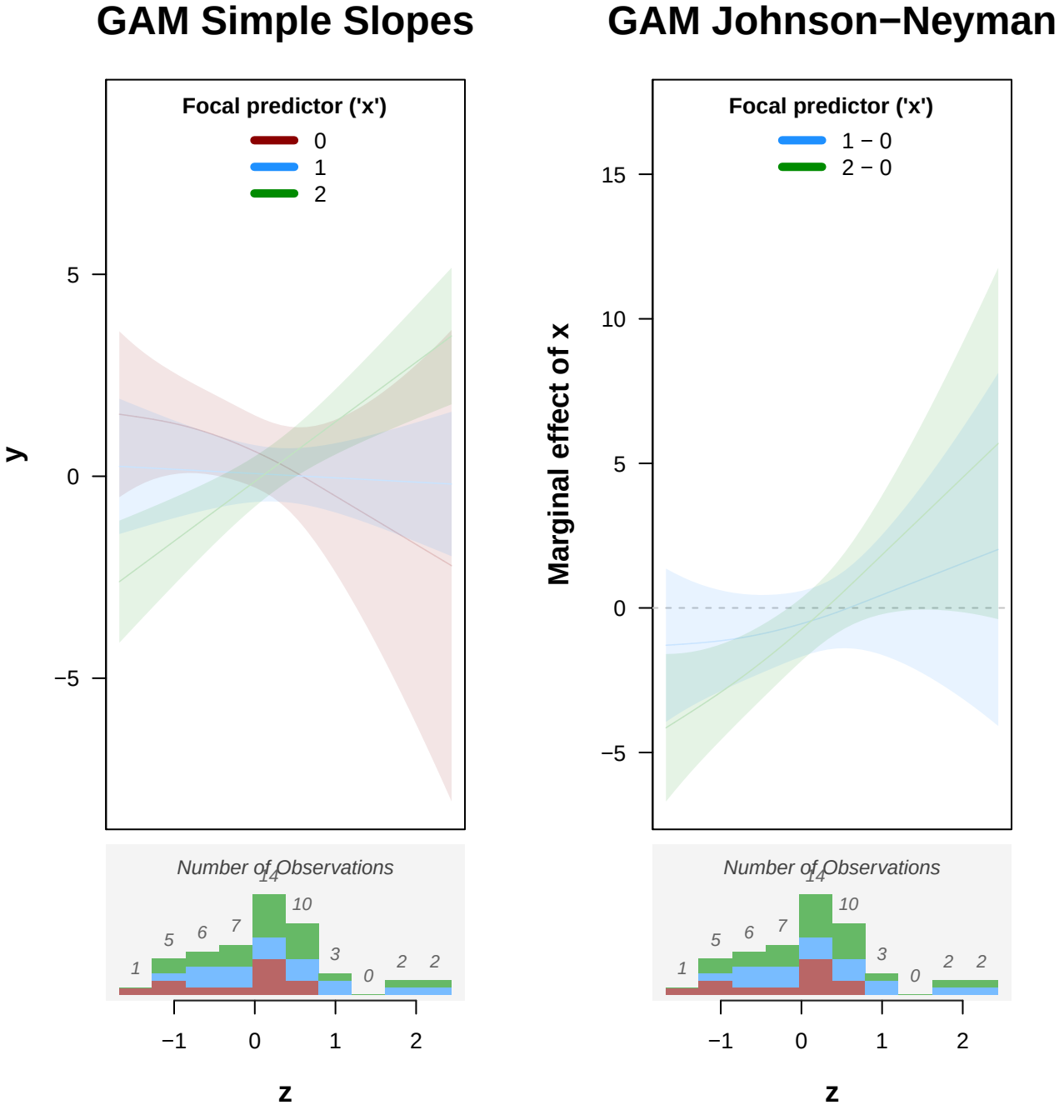

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
y
10
8
6
4
2
0
Number of Observations
260
211
202
114
119
39
8
1
12
34
−3
−2
−1
0
1
2
3
z

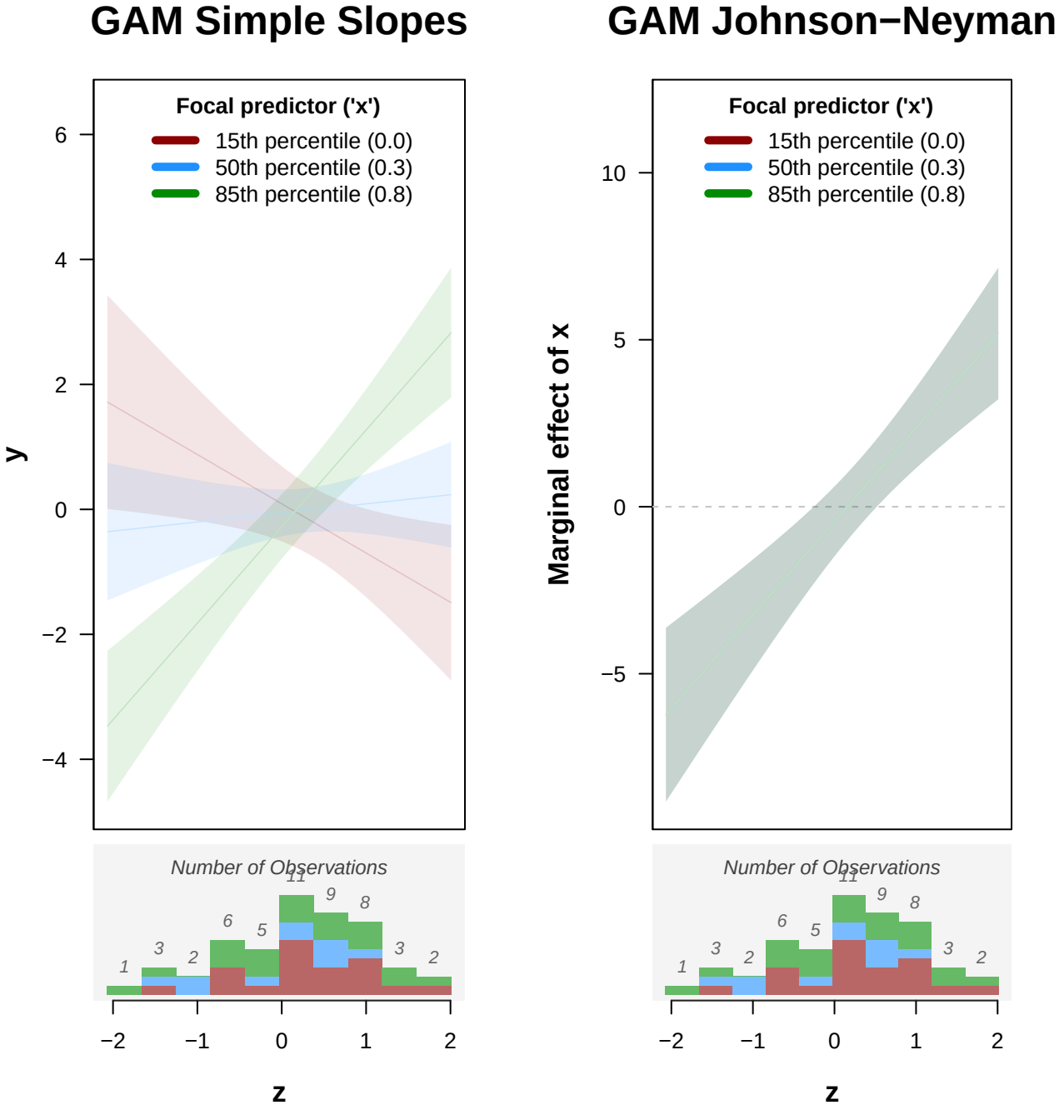

GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
Marginal effect of x
8
6
4
2
0
Number of Observations
260
211
202
114
119
39
8
1
12
34
−3
−2
−1
0
1
2
3
z

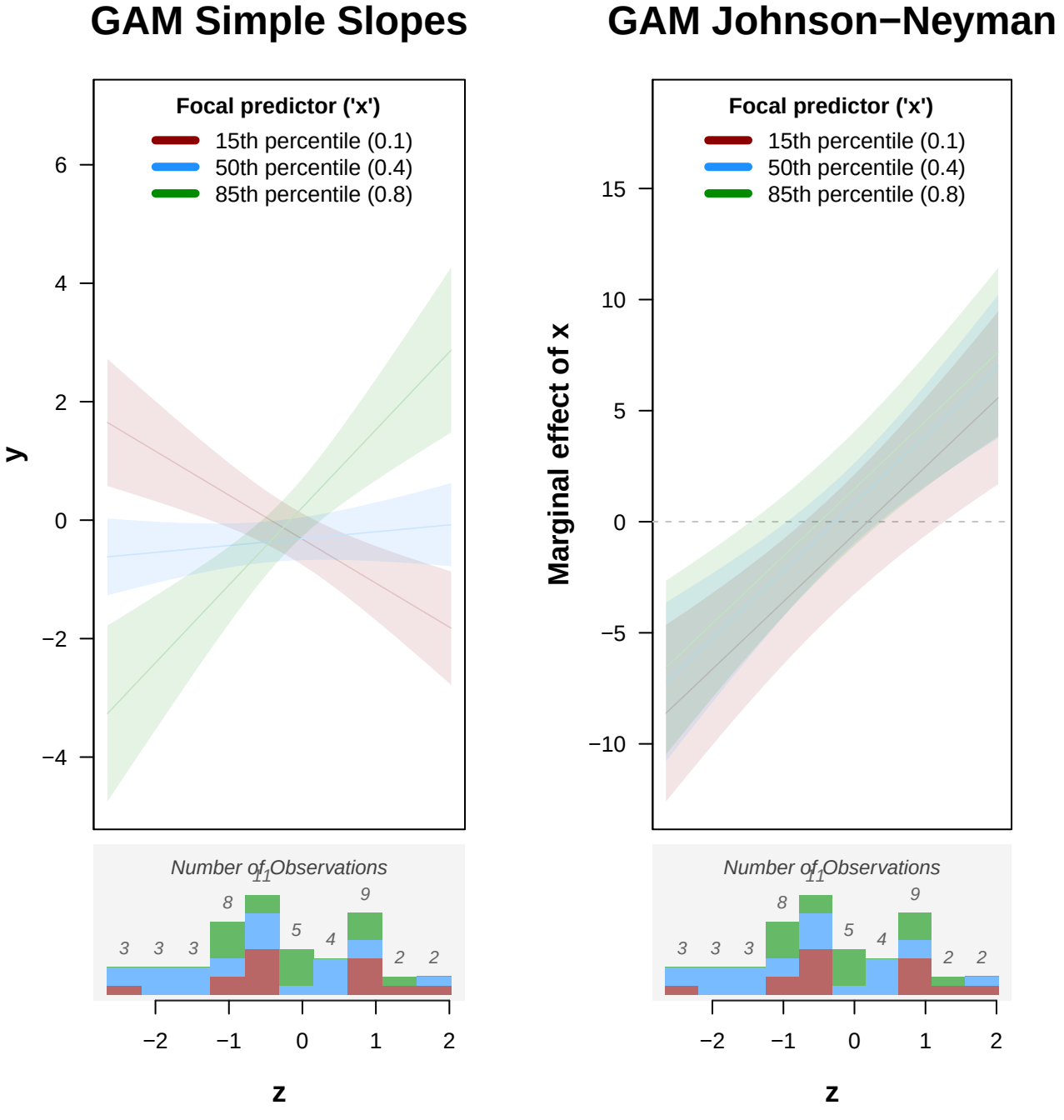

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
y
8
6
4
2
0
Number of Observations
257
216
170
120
109
51
38
24
12
3
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
Marginal effect of x
4
3
2
1
0
Number of Observations
257
216
170
120
109
51
38
24
12
3
−3
−2
−1
0
1
2
3
z

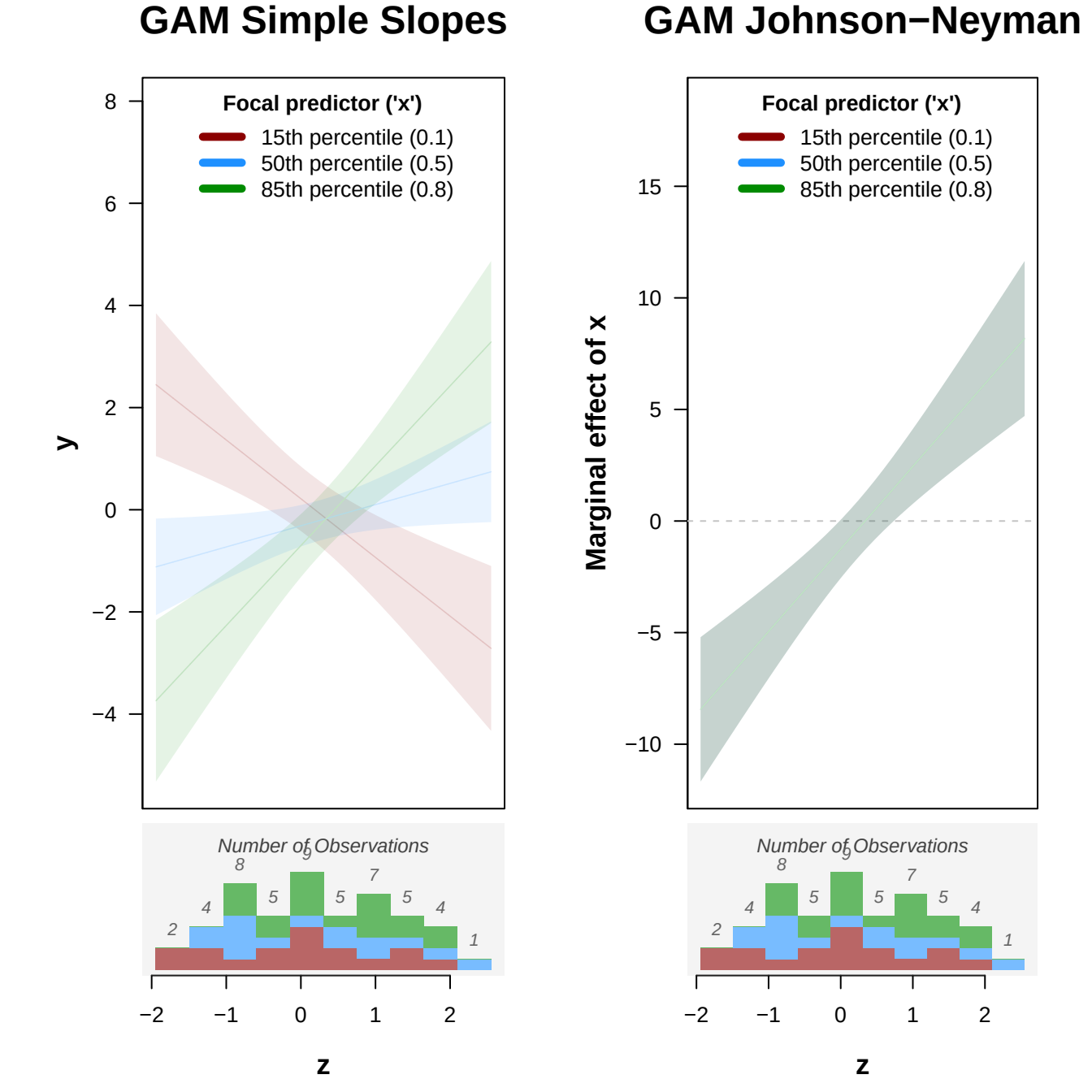

GAM Simple Slopes
Focal predictor ('x')
0
1
y
6
4
2
0
−2
Number of Observations
3
1
6
8
7
4
9
3
3
6
−2
−1
0
1
z

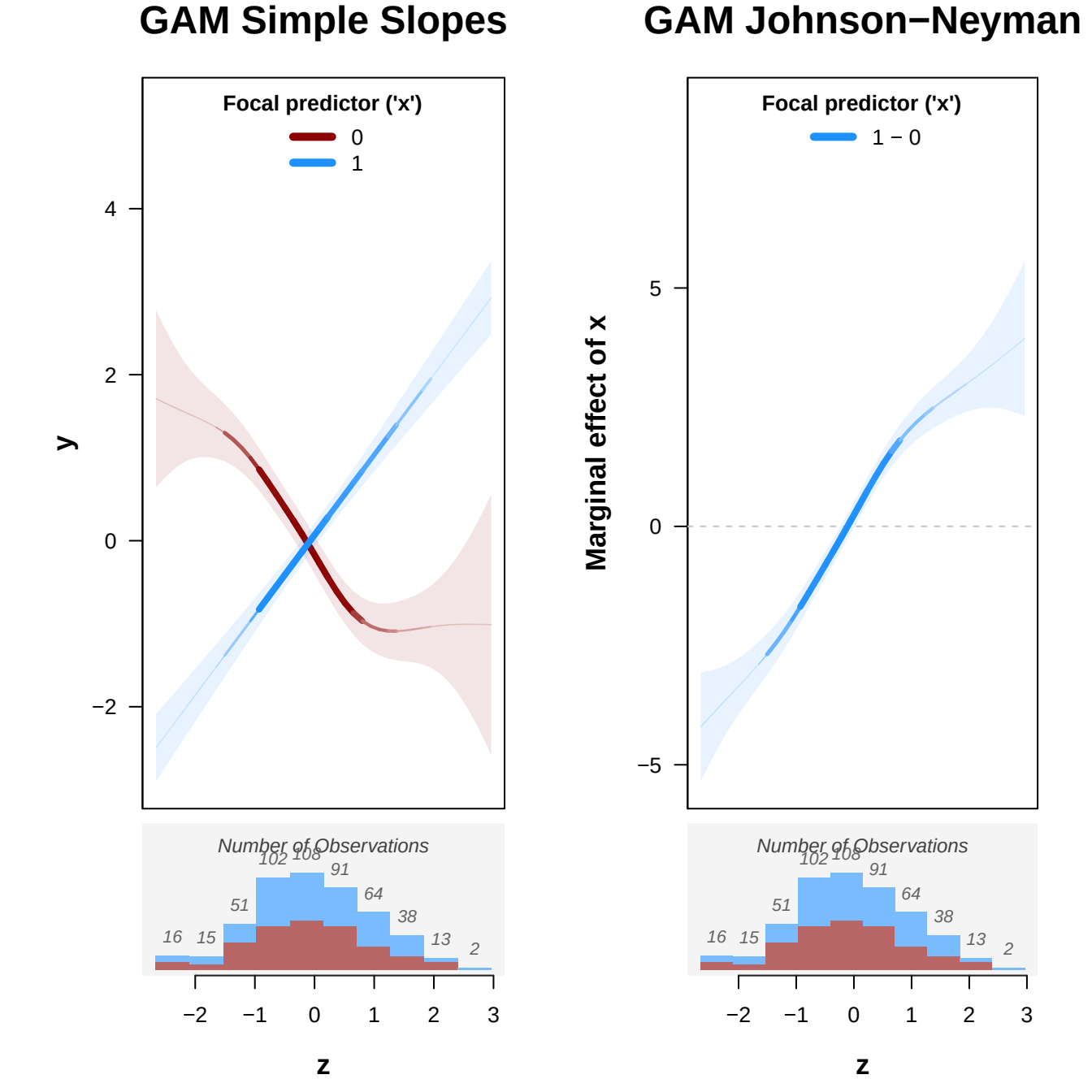

GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
5
0
−5
Number of Observations
3
1
6
8
7
4
9
3
3
6
−2
−1
0
1
z

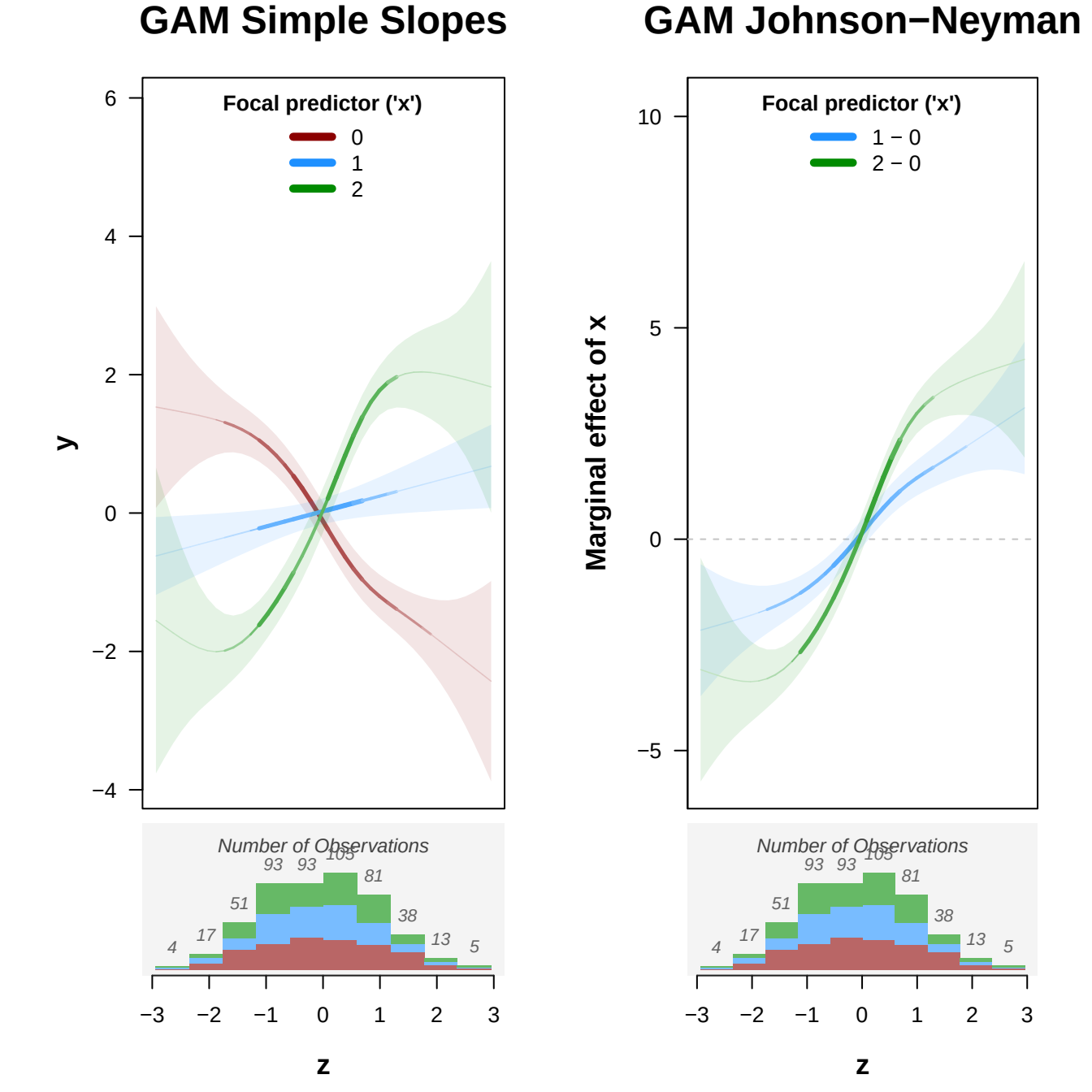

GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
5
0
−5
Number of Observations
14
10
7
6
5
1
3
0
2
2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
15
10
5
0
−5
Number of Observations
14
10
7
6
5
1
3
0
2
2
−1
0
1
2
z

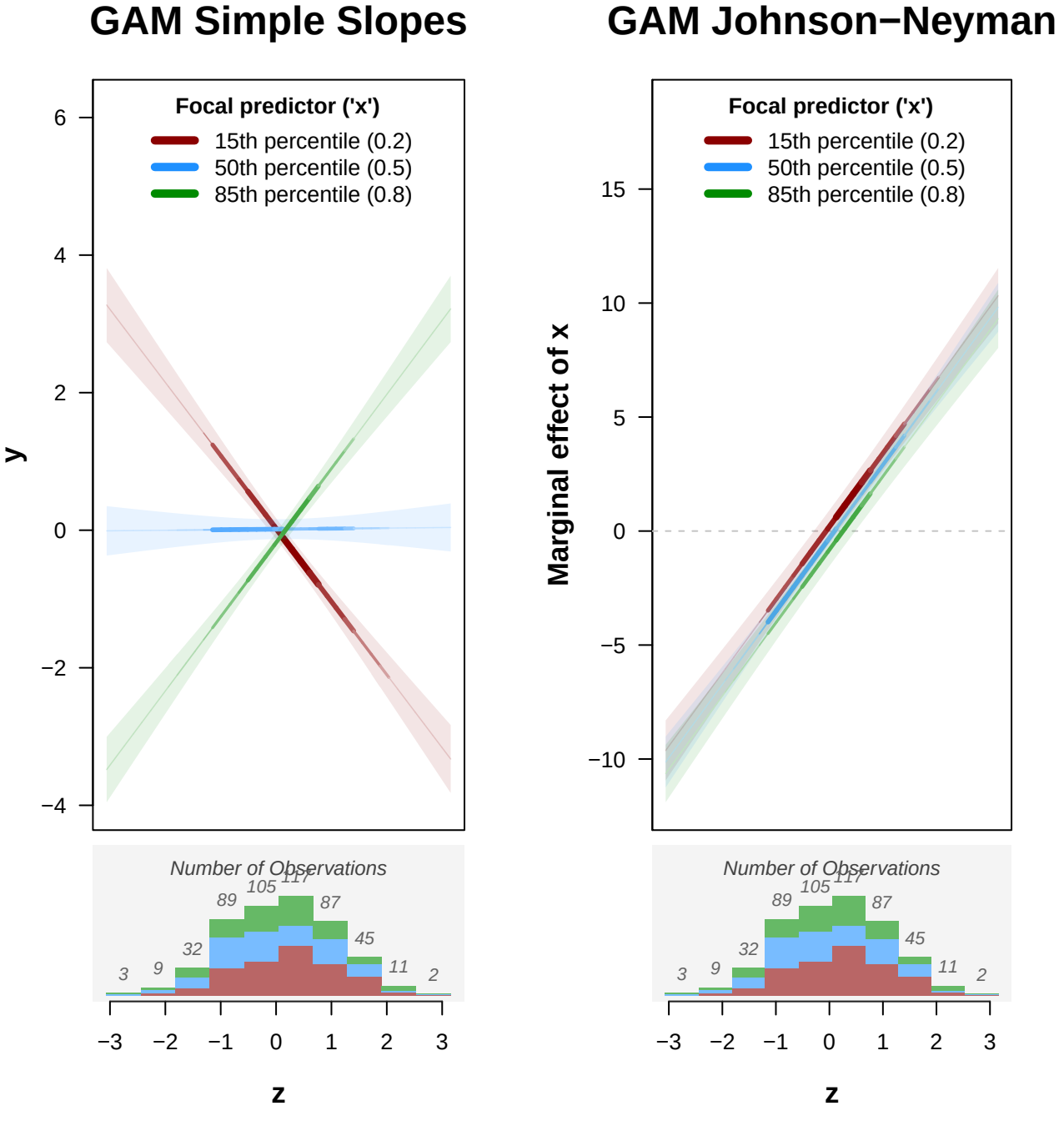

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.3)
85th percentile (0.8)
y
6
4
2
0
−2
−4
Number of Observations
1
3
2
6
5
11
9
8
3
2
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.0)
50th percentile (0.3)
85th percentile (0.8)
Marginal effect of x
10
5
0
−5
Number of Observations
1
3
2
6
5
11
9
8
3
2
−2
−1
0
1
2
z

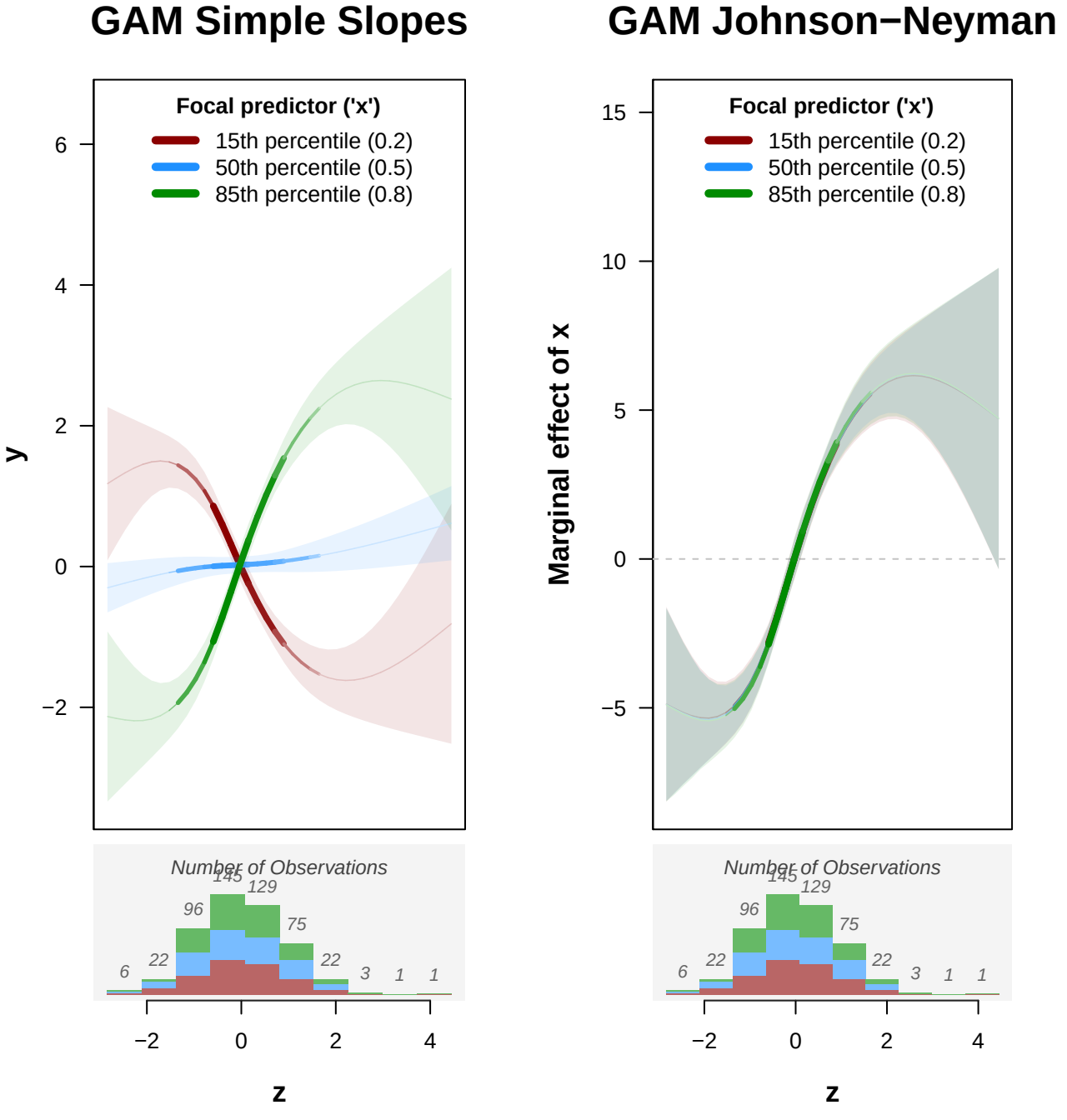

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.4)
85th percentile (0.8)
y
6
4
2
0
−2
−4
Number of Observations
3 3 3 8 11 5 4 9 2 2
−2 −1 0 1 2
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.4)
85th percentile (0.8)
Marginal effect of x
15
10
5
0
−5
−10
Number of Observations
3 3 3 8 11 5 4 9 2 2
−2 −1 0 1 2
z

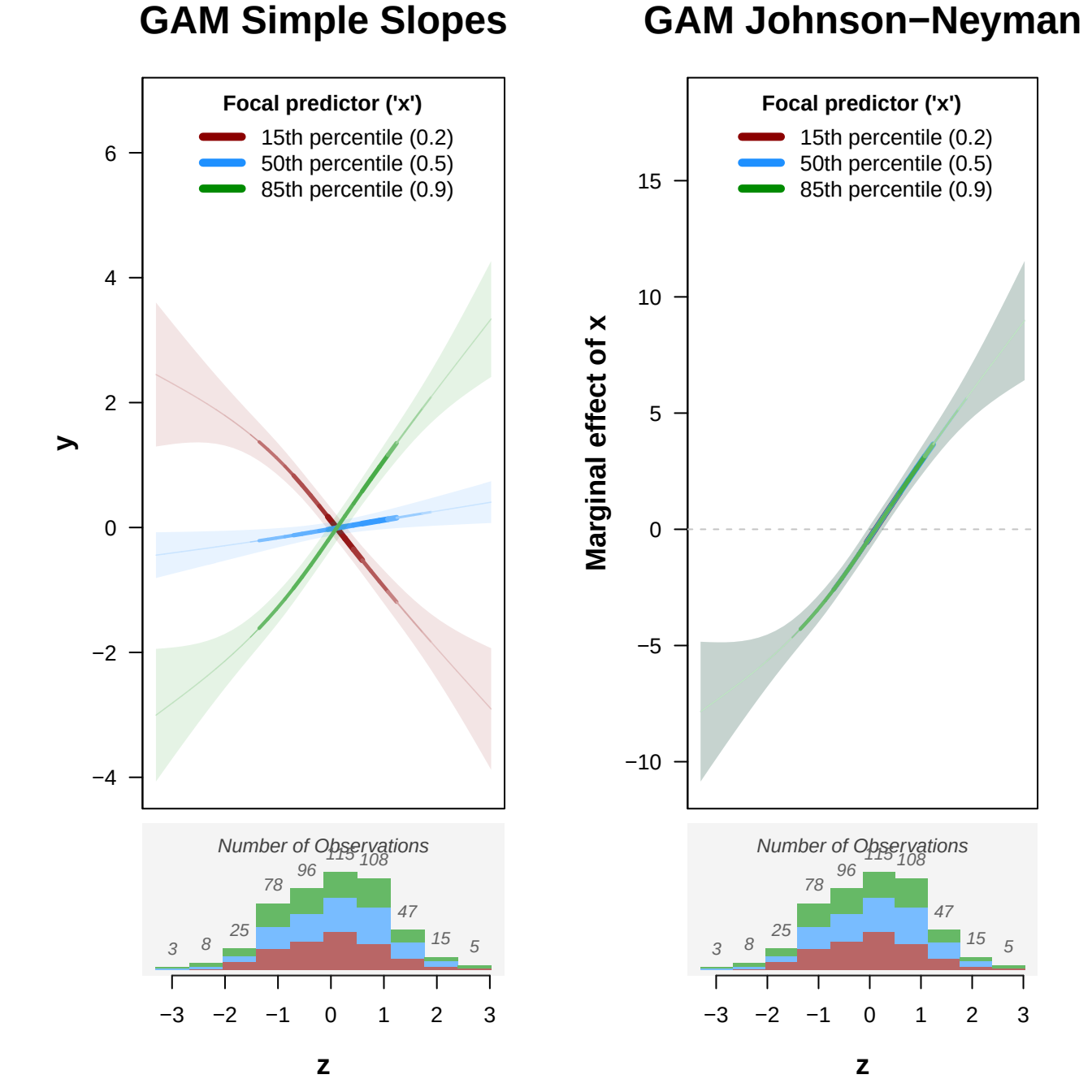

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
y
8
6
4
2
0
−2
−4
Number of Observations
9
8
7
5
5
5
4
4
2
1
−2
−1
0
1
2
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
15
10
5
0
−5
−10
Number of Observations
9
8
7
5
5
5
4
4
2
1
−2
−1
0
1
2
z

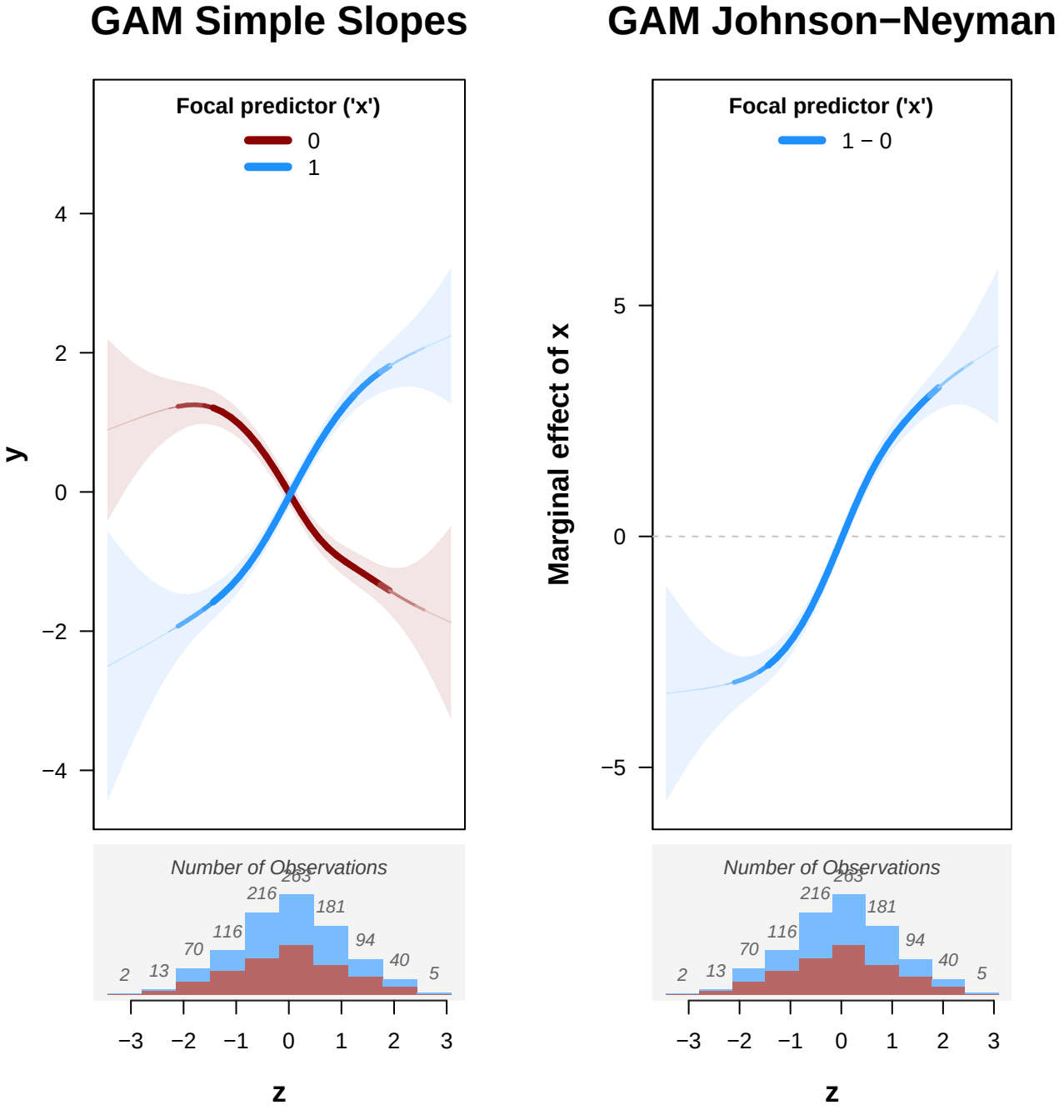

GAM Simple Slopes
Focal predictor ('x')
0
1
y
4
2
0
−2
Number of Observations
16
15
51
102
108
91
64
38
13
2
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
5
0
−5
Number of Observations
16
15
51
102
108
91
64
38
13
2
−2
−1
0
1
2
3
z

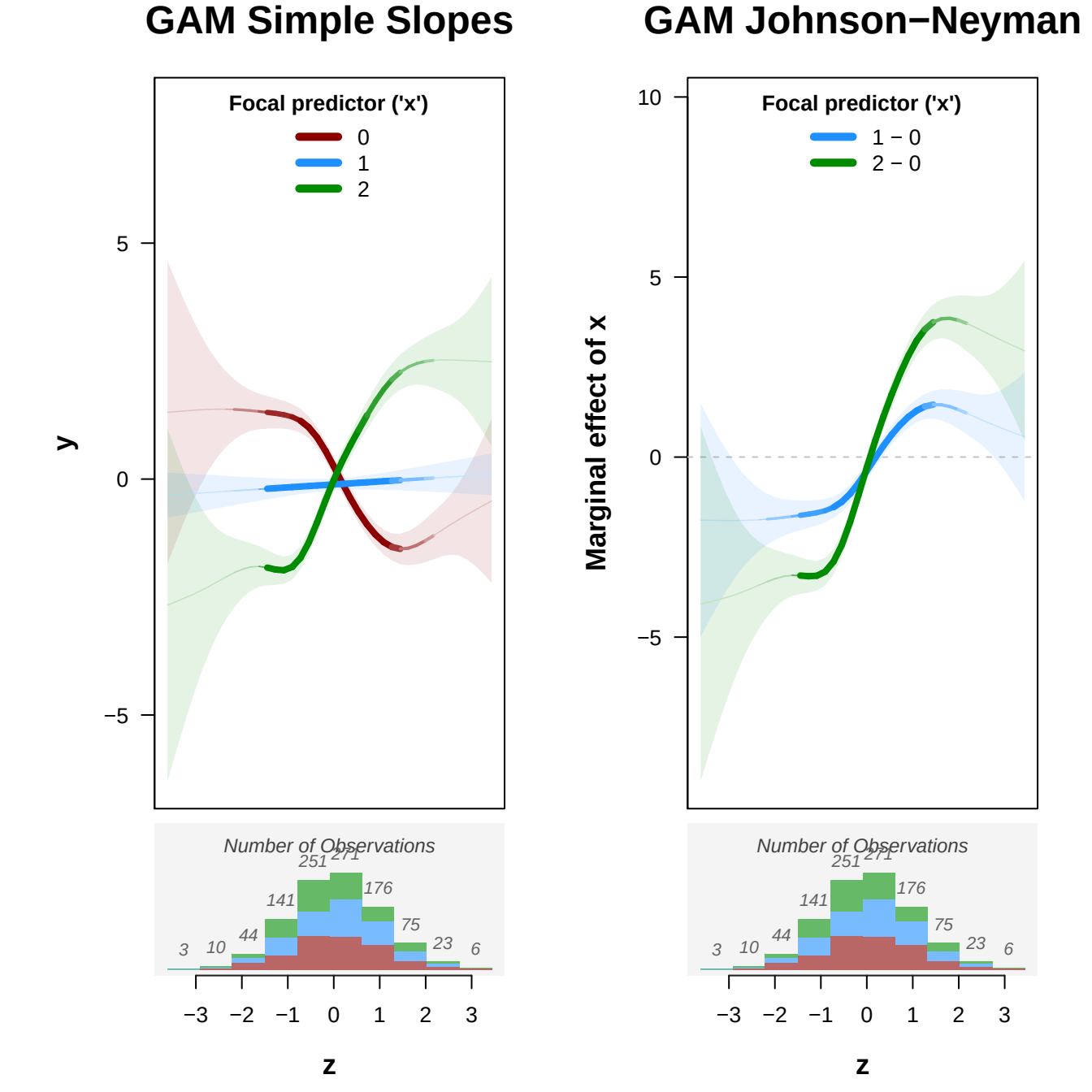

GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
6
4
2
0
−2
−4
Number of Observations
4
17
51
93
93
105
81
38
13
5
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
10
5
0
−5
Number of Observations
4
17
51
93
93
105
81
38
13
5
−3
−2
−1
0
1
2
3
z

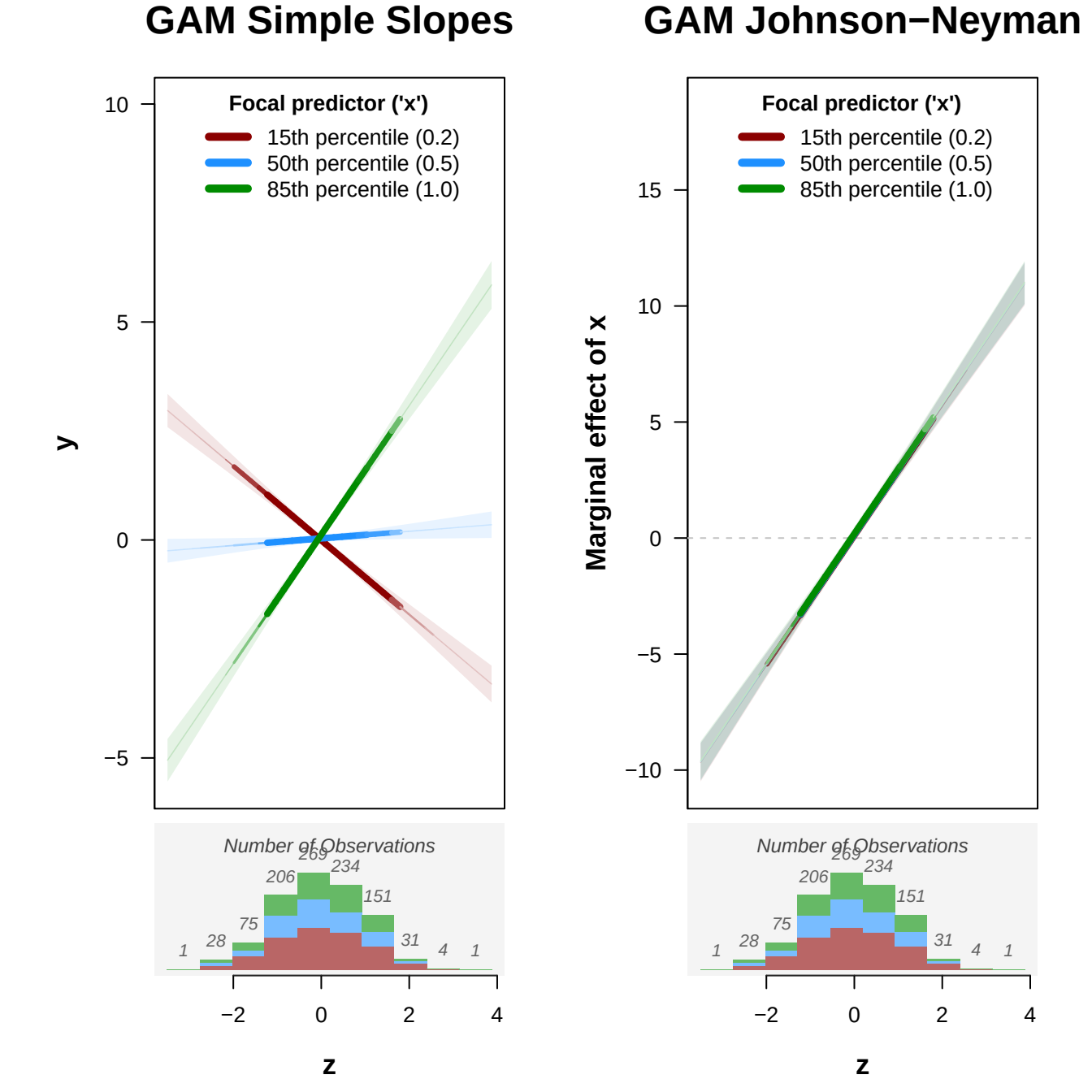

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
y
6
4
2
0
−2
−4
Number of Observations
3
9
32
89
105
117
87
45
11
2
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
15
10
5
0
−5
−10
Number of Observations
3
9
32
89
105
117
87
45
11
2
−3
−2
−1
0
1
2
3
z

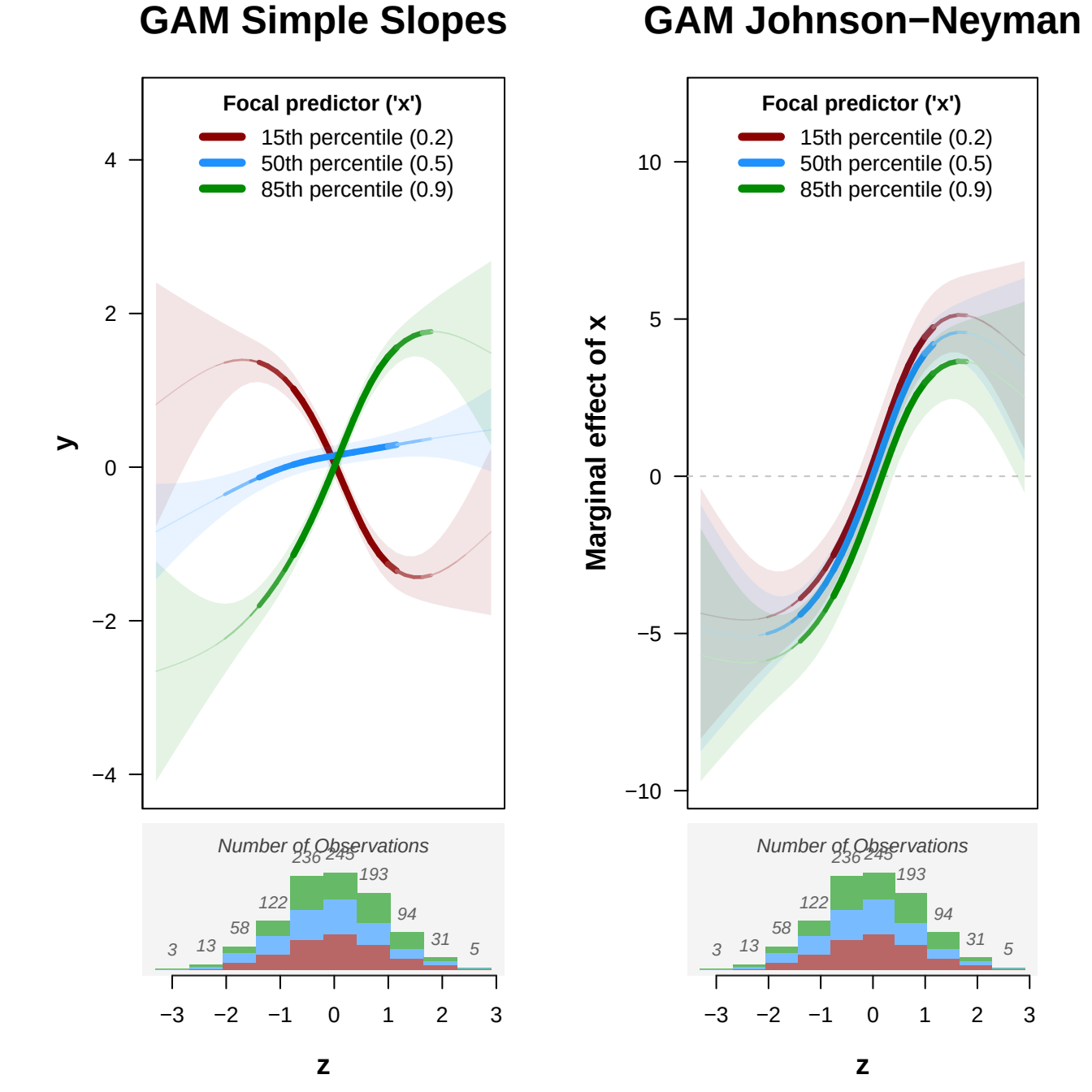

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
y
6
4
2
0
−2
Number of Observations
145
129
96
75
22
22
3
1
1
6
−2
0
2
4
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
15
10
5
0
−5
Number of Observations
145
129
96
75
22
22
3
1
1
6
−2
0
2
4
z

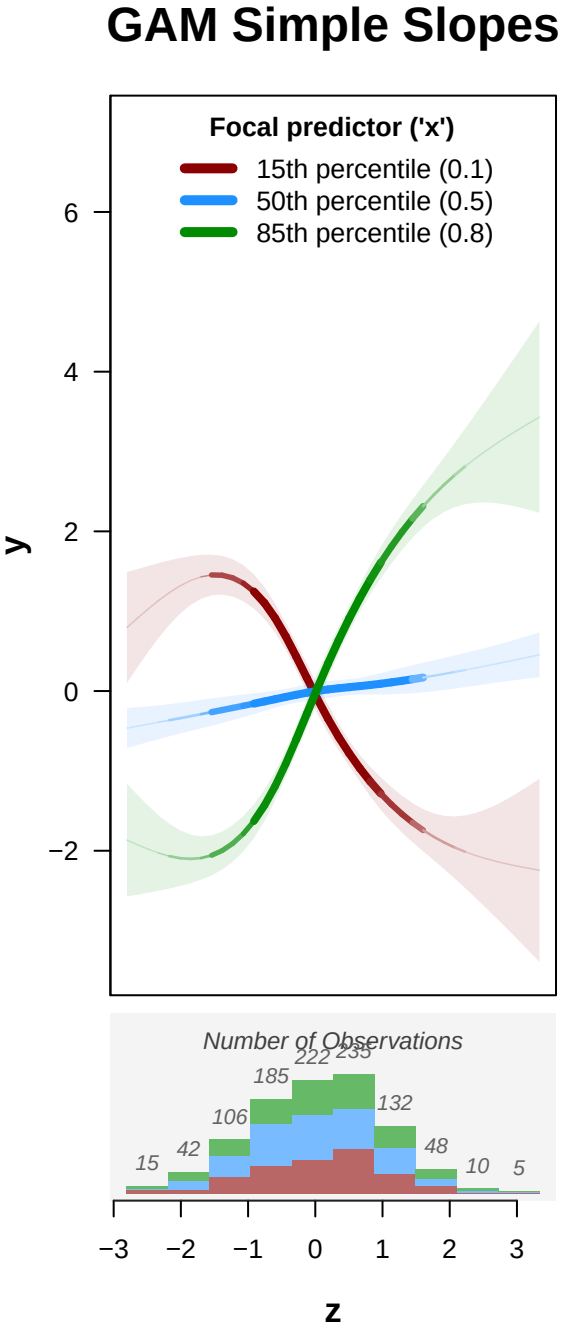

GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
y
6
4
2
0
−2
−4
Number of Observations
3
8
25
78
96
115
108
47
15
5
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
Marginal effect of x
15
10
5
0
−5
−10
Number of Observations
3
8
25
78
96
115
108
47
15
5
−3
−2
−1
0
1
2
3
z

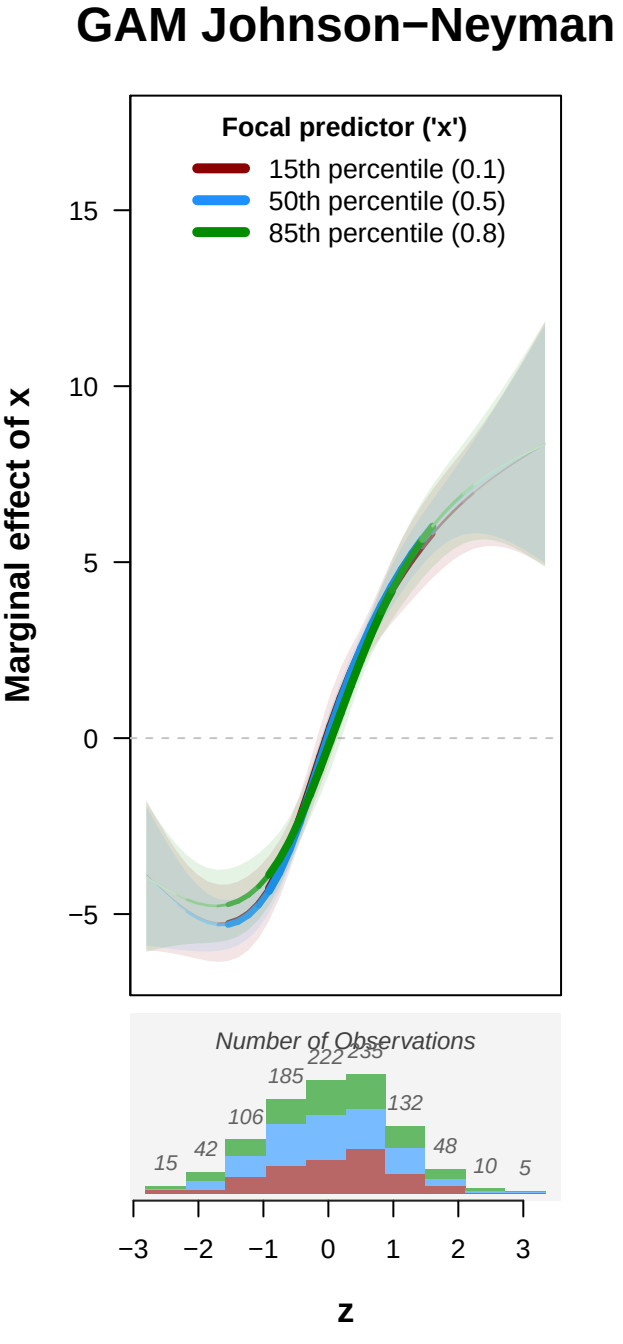

GAM Simple Slopes
Focal predictor ('x')
0
1
y
4
2
0
−2
−4
Number of Observations
263
216
181
116
94
70
40
13
2
5
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
Marginal effect of x
5
0
−5
Number of Observations
263
216
181
116
94
70
40
13
2
5
−3
−2
−1
0
1
2
3
z


GAM Simple Slopes
Focal predictor ('x')
0
1
2
y
5
0
−5
Number of Observations
3
10
44
141
251
271
176
75
23
6
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
1 − 0
2 − 0
Marginal effect of x
10
5
0
−5
Number of Observations
3
10
44
141
251
271
176
75
23
6
−3
−2
−1
0
1
2
3
z


GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (1.0)
y
10
5
0
−5
Number of Observations
1
28
75
206
269
234
151
31
4
1
−2
0
2
4
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (1.0)
Marginal effect of x
15
10
5
0
−5
−10
Number of Observations
1
28
75
206
269
234
151
31
4
1
−2
0
2
4
z


GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
y
4
2
0
−2
−4
Number of Observations
3
13
58
122
236
245
193
94
31
5
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.2)
50th percentile (0.5)
85th percentile (0.9)
Marginal effect of x
10
5
0
−5
−10
Number of Observations
3
13
58
122
236
245
193
94
31
5
−3
−2
−1
0
1
2
3
z


GAM Simple Slopes
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
y
6
4
2
0
−2
Number of Observations
15
42
106
185
222
235
132
48
10
5
−3
−2
−1
0
1
2
3
z


GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (0.1)
50th percentile (0.5)
85th percentile (0.8)
Marginal effect of x
15
10
5
0
−5
Number of Observations
15
42
106
185
222
235
132
48
10
5
−3
−2
−1
0
1
2
3
z